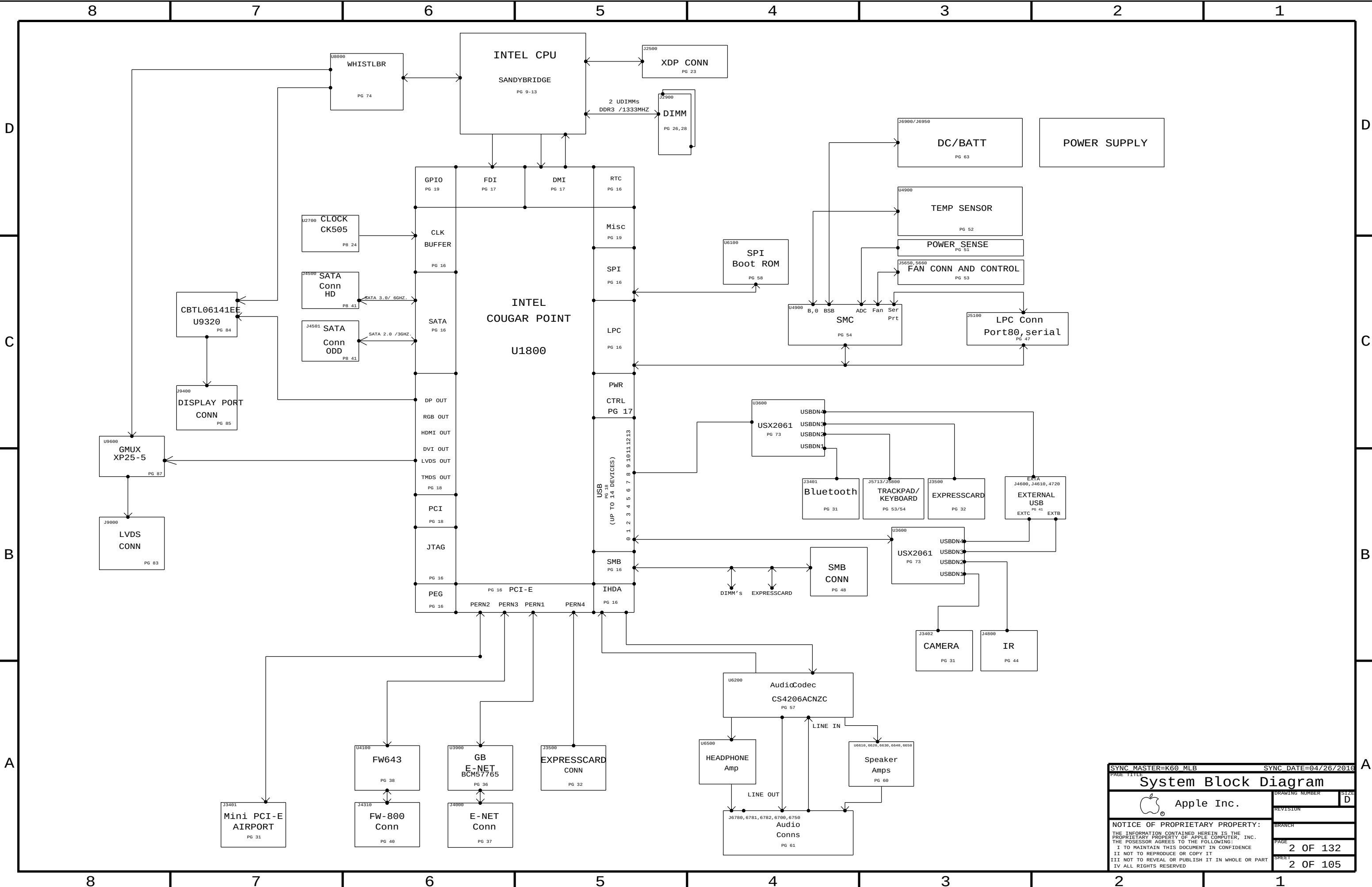




8		7		6		5		4		3		2		1	
D															
C															
B															
A															
PROT02/EVT 11/11/10 rev3.0 for board 820-2914-05.brd release PROT02/EVT 11/15/10 rev3.6 for board 820-2914-06.brd release PROT02/EVT 11/19/10 rev3.7 for board 820-2914-07.brd release EVT 11/22/10 rev3.9 for board 820-2914-07.brd release															
D															
C															
B															
A															
8		7		6		5		4		3		2		1	

SYNC MASTER=K17 MLB

SYNC DATE=04/26/2010

PAGE TITLE

Revision History

 Apple Inc.

NOTICE OF PROPRIETARY PROPERTY:
THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE COMPUTER, INC.
THE POSSESSOR AGREES TO THE FOLLOWING:
I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE
II NOT TO REPRODUCE OR COPY IT
III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART
IV ALL RIGHTS RESERVED

DRAWING NUMBER

REVISION

BRANCH

PAGE

SHEET

SIZE

D

4 OF 132

4 OF 105

PROT02/EVT 11/11/10 rev3.0 for board 820-2914-05.brd release
PROT02/EVT 11/15/10 rev3.6 for board 820-2914-06.brd release
PROT02/EVT 11/19/10 rev3.7 for board 820-2914-07.brd release
EVT 11/22/10 rev3.9 for board 820-2914-07.brd release

SYNC MASTER=K17_MLB

SYNC DATE=04/26/2010

PAGE TITLE		Revision History	
 Apple Inc.		DRAWING NUMBER	SIZE
		REVISION	D
NOTICE OF PROPRIETARY PROPERTY: THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE COMPUTER, INC. THE POSSESSOR AGREES TO THE FOLLOWING: I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE II NOT TO REPRODUCE OR COPY IT III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART IV ALL RIGHTS RESERVED		BRANCH	
		PAGE	4 OF 132
		SHEET	4 OF 105

BOM Variants

BOM NUMBER	BOM NAME	BOM OPTIONS
639-1303	PCBA, MLB, K92	K92_COMMON, CPU:2_2GHZ, FB_1G_SAMSUNG, EEEE_DG5Y
639-1464	PCBA, MLB, CF62, K92	K92_COMMON, CPU:2_2GHZ, FB_1G_HYNIX, VRAM_HYNIX, EEEE_DG60
639-1466	PCBA, MLB, CF63, K92	K92_COMMON, CPU:2_3GHZ, FB_1G_SAMSUNG, EEEE_DG62
639-1465	PCBA, MLB, CF64, K92	K92_COMMON, CPU:2_3GHZ, FB_1G_HYNIX, VRAM_HYNIX, EEEE_DG61
085-1898	K92 MLB DEVELOPMENT BOM	K92_DEVEL:ENG

K92 BOM GROUPS

BOM GROUP	BOM OPTIONS
K92_COMMON	ALTERNATE, COMMON, K92_COMMON1, K92_COMMON2, K92_PROGPARTS
K92_COMMON1	CPUMEM_S0, EXT_HP_AMP, SMC_DEBUG_YES, USBHUB_2514B
K92_COMMON2	GPUVID_1P11V, HUB1_2NONREM, HUB2_2NONREM, KB_BL, ENET:B0, T29BST:Y, T29:YES, T29_DP_HPD:ALL_OR
K92_DEVEL:ENG	SNB_CPT_XDP, DEBUG_ADC, LPCPLUS:YES, VREFMRGN, GMUX_3TAG_CONN, S0PG00D_ISL, BMON:ENG, CPURIPPLE_ENG, 1MVP15NS_ENG, SDRV12C:MCU
K92_DEVEL:PVT	SNB_CPT_XDP, LPCPLUS:YES, VREFMRGN_NOT
K92_PROGPARTS	SMC_PROG:EVT, BOOTROM_PROG:EVT, ENETROM_PROG:B0_NOSD, TPAD_PROG:EVT, T29ROM:PROG, GMUX_PROG, T29MCU:PROG
K92_PVT	VREFMRGN_NOT, XDP, XDP_CPU_BPM, BMON:PROD
SNB_CPT_XDP	XDP, XDP_CONN, XDP_CPU_BPM, XDP_PCH

Bar Code Labels / EEEE #'s

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
826-4393	1	LBL, P/N LABEL, PCB, 28MM X 6 MM	[EEEE_DG5Y]	CRITICAL	EEEE_DG5Y
826-4393	1	LBL, P/N LABEL, PCB, 28MM X 6 MM	[EEEE_DG60]	CRITICAL	EEEE_DG60
826-4393	1	LBL, P/N LABEL, PCB, 28MM X 6 MM	[EEEE_DG61]	CRITICAL	EEEE_DG61
826-4393	1	LBL, P/N LABEL, PCB, 28MM X 6 MM	[EEEE_DG62]	CRITICAL	EEEE_DG62

Module Parts

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
337S4032	1	IC, CPU, SNB, SR00W, PRQ, D2, 2:2, 45W, 4+2, 1.30, 6M, BGA	U1000	CRITICAL	CPU:2_2GHZ
337S4033	1	IC, CPU, SNB, SR00U, PRQ, D2, 2:3, 45W, 4+2, 1.30, 8M, BGA	U1000	CRITICAL	CPU:2_3GHZ
337S4029	1	IC, PCH, COUGARPOINT SLH90, PRQ, B0B2HM63	U1800	CRITICAL	
343S0534	1	IC, ASIC, GBIT ETHNETASD CTRLR, 686 QFN 8X8	U3900	CRITICAL	ENET:B0
343S0494	1	IC, ASIC, GBIT ETHNETASD CTRLR, 686 QFN 8X8	U3900	CRITICAL	ENET:A0
338S0753	1	IC, FM643-E, 1394B PHY/DMCI LINK/PCI-E, 12	U4100	CRITICAL	
333S0543	4	IC, SGRAM, GDDR5, 32MX32, 1.25GHZ, E-DIE, HF	UB400, UB450, UB500, UB550	CRITICAL	FB_512_SAMSUNG
333S0564	4	IC, SGRAM, GDDR5, 32MX32, 1.25GHZ, A-DIE1.35V	UB400, UB450, UB500, UB550	CRITICAL	FB_512_HYNIX
333S0571	4	IC, SGRAM, GDDR5, 64MX32, 3.6GBPS, C-DIE, HF	UB400, UB450, UB500, UB550	CRITICAL	FB_1G_SAMSUNG
333S0572	4	IC, SGRAM, GDDR5, 64MX32, 3.6GBPS, M-DIE, HF	UB400, UB450, UB500, UB550	CRITICAL	FB_1G_HYNIX
337S3936	1	IC, GPU, AMD, WHISTLER, 962PCBGA, 40NP, ES	U8000	CRITICAL	
338S0945	1	Light Ridge, S LHAJ, FCBGA, 15x15mm	U3600	CRITICAL	T29:YES
353S3055	1	IC, P13VEDP212, x2 DISPLAYPORT 2:1 MUX, QFN	U9390	CRITICAL	

Programmed Parts-All Builds

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
335S0777	1	IC, EEPROM, SERIAL, 8KB, SOIC	U3690	CRITICAL	T29ROM:BLANK
341S2899	1	IC, T29 EEPROM, K92	U3690	CRITICAL	T29ROM:PROG
341S2384	1	IR, ENCORE II, CY7C63833-LFXC	U4800	CRITICAL	
335S0724	1	IC, GPU ROM, K91/F, K92	U8701	CRITICAL	GPUROM:BLANK
341S2957	1	IC, GPU ROM, K91/F, K92	U8701	CRITICAL	GPUROM:PROG
336S0042	1	IC, PLD, LATTICE, LFXP2-5E-5, 132 BALL CSBGA	U9600	CRITICAL	GMUX_BLANK
341S2996	1	IC, GMUX, K92	U9600	CRITICAL	GMUX_PROG
337S3997	1	IC, MCU, 32B, LPC1112A, 16KB/2KB, HVQFN25	U9330	CRITICAL	T29MCU:BLANK
341S2939	1	IC, PROGRAMMED MCU, 32B, LPC1112A, 16KB/2KB, HVQFN25	U9330	CRITICAL	T29MCU:PROG

GPUROM will NOSUFFED @EVT

Alternate Parts

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
157S0058	157S0055		ALL	Delta alt to TDK Magnetics
152S0896	152S0518		ALL	MAG LAYERS ALT TO CYNTEC
152S0915	152S0796		ALL	MAG LAYERS ALT TO CYNTEC
155S0457	155S0329		ALL	MAG LAYERS ALT TO MURATA
516S0805	516S0806		ALL	FOXCONN ALT TO MOLEX
353S2805	353S2603		ALL	Fairchild B' alt to S'wafer
127S0111	127S0060		ALL	Rohm alt to Kemet
353S3085	353S1658		ALL	ST Micro alt to LT
152S0905	152S1307		ALL	Cyntec (used on K90i) as alt
353S3055	353S3151		ALL	Pericom alt to NXP DP Mux
376S0855	376S0613		ALL	radar8515240 Toshiba FET
870-1939	870-1698		ALL	Silver alt to gold tall pogo pins
870-2015	870-1699		ALL	Silver alt to gold short pogo pins
376S0972	376S0612		ALL	add Rohm part as 2nd source
138S0676	138S0691		ALL	add Murata part as 2nd source
128S0327	128S0264		ALL	add NEC part as 2nd source
376S0977	376S0859		ALL	add new part as 2nd source
138S0681	138S0638		ALL	add new part as 2nd source

(U5850)
(L7630)
(U9390)
(Q3200, etc)

(Q3888, Q9430)

SMC

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
338S0895	1	IC, SMC, HS0/2117, 9MMx9MM, TLP	U4900	CRITICAL	SMC_BLANK
341S2855	1	IC, SMC, DEVELOPMENT-PROTO, K92	U4900	CRITICAL	SMC_PROG:PROTO0
341S2855	1	IC, SMC, DEVELOPMENT-PROTO1, K92	U4900	CRITICAL	SMC_PROG:PROTO1
341S2995	1	IC, SMC, DEVELOPMENT-PROTO2, K92	U4900	CRITICAL	SMC_PROG:PROTO2
341S2862	1	IC, SMC, DEVELOPMENT-EVT, K92	U4900	CRITICAL	SMC_PROG:EVT
341S2865	1	IC, SMC, DEVELOPMENT-DVT, K92	U4900	CRITICAL	SMC_PROG:DVT
341S2868	1	IC, SMC, DEVELOPMENT-PVT, K92	U4900	CRITICAL	SMC_PROG:PVT

EFI

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
335S0740	1	64 MBIT SPI SERIAL DUAL I/O FLASH	U6100	CRITICAL	BOOTROM_BLANK
341S2893	1	IC, EFI, ROM, PROTO, K90/K901/K91/K91F/K92	U6100	CRITICAL	BOOTROM_PROG:PROTO0
341S2934	1	IC, EFI, ROM, PROTO1, K90/K901/K91/K91F/K92	U6100	CRITICAL	BOOTROM_PROG:PROTO1
341S2991	1	IC, EFI, ROM, PROTO1, K90/K901/K91/K91F/K92	U6100	CRITICAL	BOOTROM_PROG:PROTO2
341S2894	1	IC, EFI, ROM, EVT, K90/K901/K91/K91F/K92	U6100	CRITICAL	BOOTROM_PROG:EVT
341S2895	1	IC, EFI, ROM, DVT, K90/K901/K91/K91F/K92	U6100	CRITICAL	BOOTROM_PROG:DVT
341S2896	1	IC, EFI, ROM, PVT, K90/K901/K91/K91F/K92	U6100	CRITICAL	BOOTROM_PROG:PVT

Ethernet

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
335S0539	1	IC, FLASH, SERIAL, SPI, 1MBIT, 2V7, 8P, SOIC	U3990	CRITICAL	ENETROM_BLANK
341S2685	1	IC, ENET ROM, PROTO1, K92	U3990	CRITICAL	ENETROM_PROG:1A0_SD
341S3027	1	IC, ENET ROM, PROTO2, EVT, DVT, PVT, K92	U3990	CRITICAL	ENETROM_PROG:B0_NOSD

PSOC

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
341S2902	1	IC, TP PSOC, PROTO, K90, K901, K91, K91F, K92	U5701	CRITICAL	TPAD_PROG:PROTO1
341S3024	1	IC, TP PSOC, proto2, K90, K901, K91, K91F, K92T	U5701	CRITICAL	TPAD_PROG:PROTO2
341S3024	1	IC, TP PSOC, proto1, EVT, K90, K901, K91, K91F, K92T	U5701	CRITICAL	TPAD_PROG:EVT
341S3024	1	IC, TP PSOC, proto1, DVT, PVT, K90, K901, K91, K91F, K92T	U5701	CRITICAL	TPAD_PROG:DVTPTVT

SYNC MASTER=K17 MLB

SYNC DATE=04/26/2010

PAGE TITLE

BOM Configuration

 Apple Inc.

NOTICE OF PROPRIETARY PROPERTY:
THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE COMPUTER, INC. THE POSSESSOR AGREES TO THE FOLLOWING:
I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE
II NOT TO REPRODUCE OR COPY IT
III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART
IV ALL RIGHTS RESERVED

DRAWING NUMBER

REVISION

BRANCH

PAGE

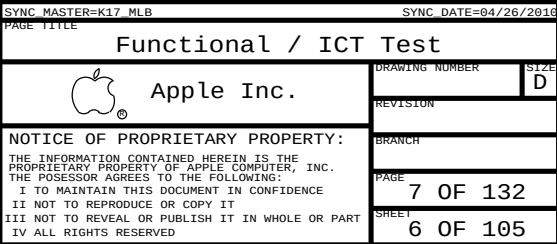
SHEET

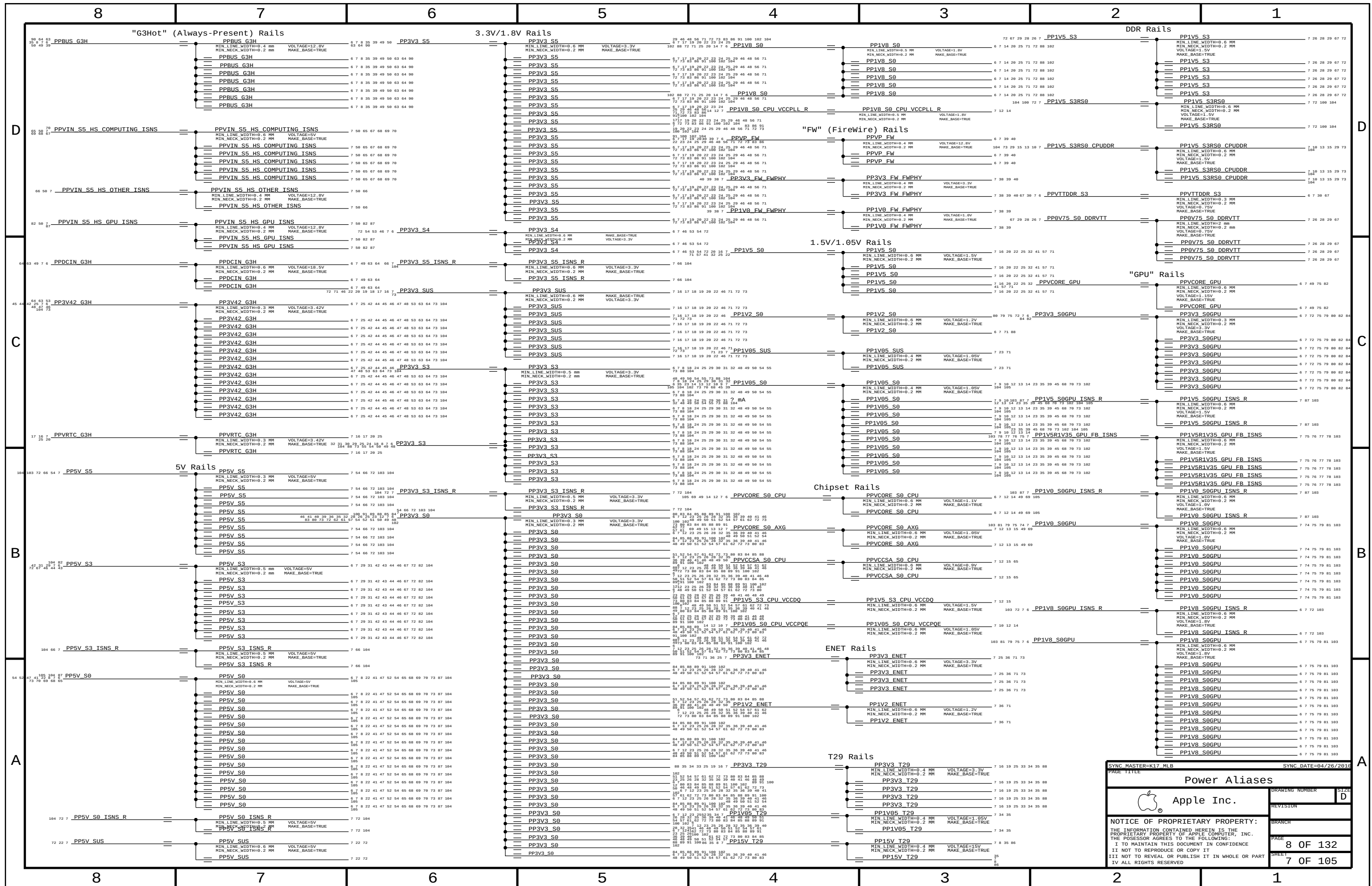
SIZE

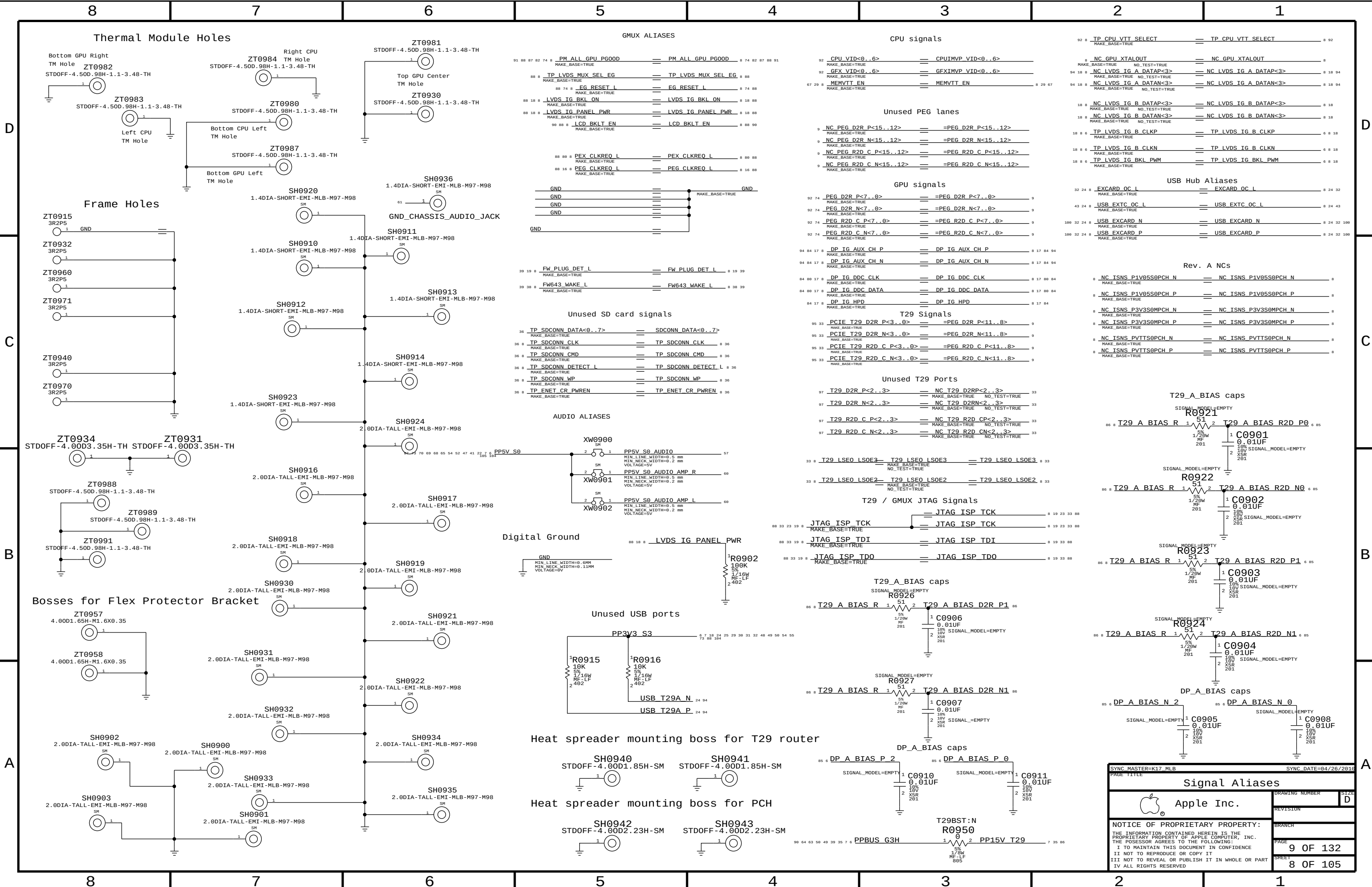
D

5 OF 132

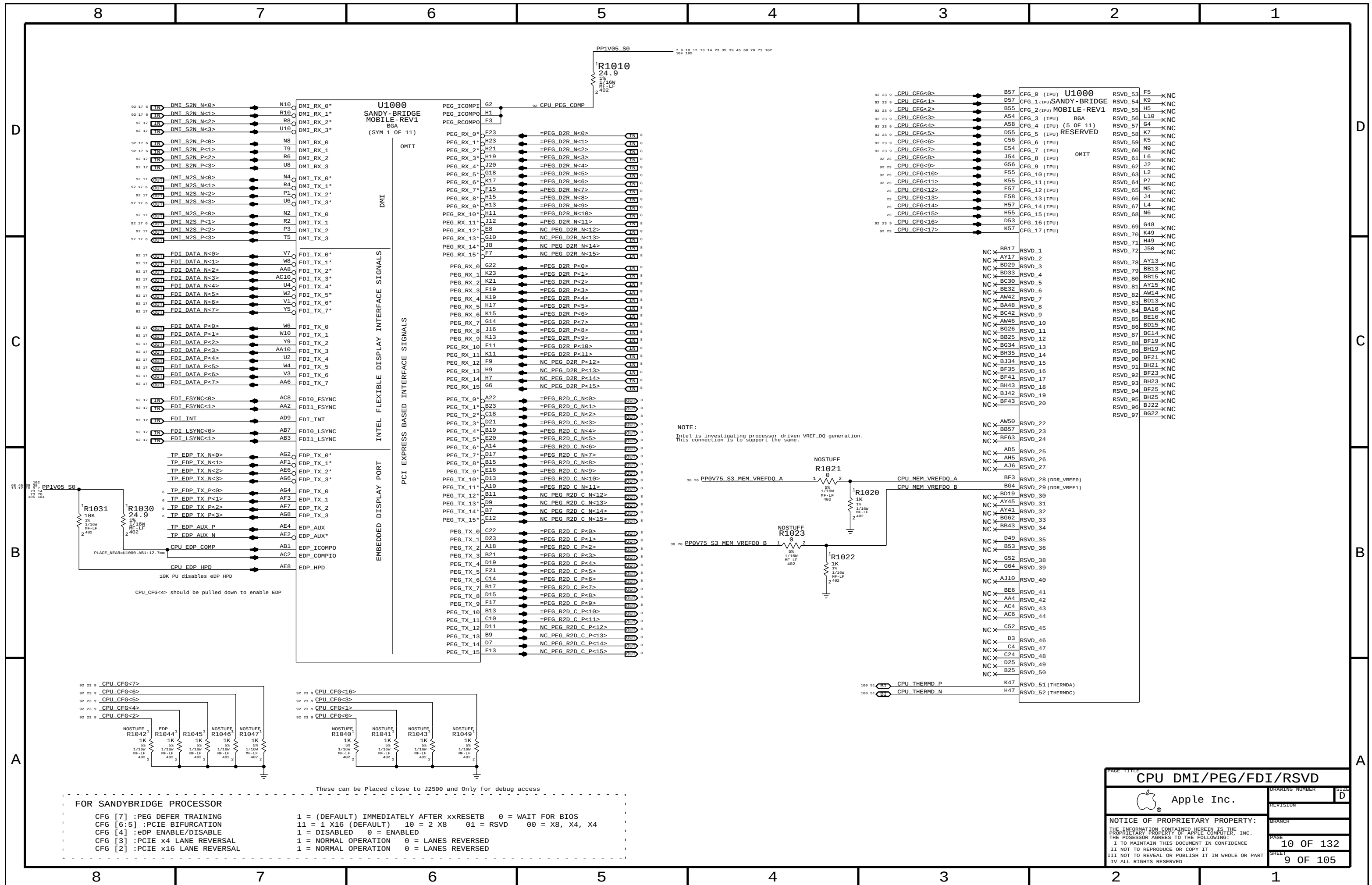
5 OF 105

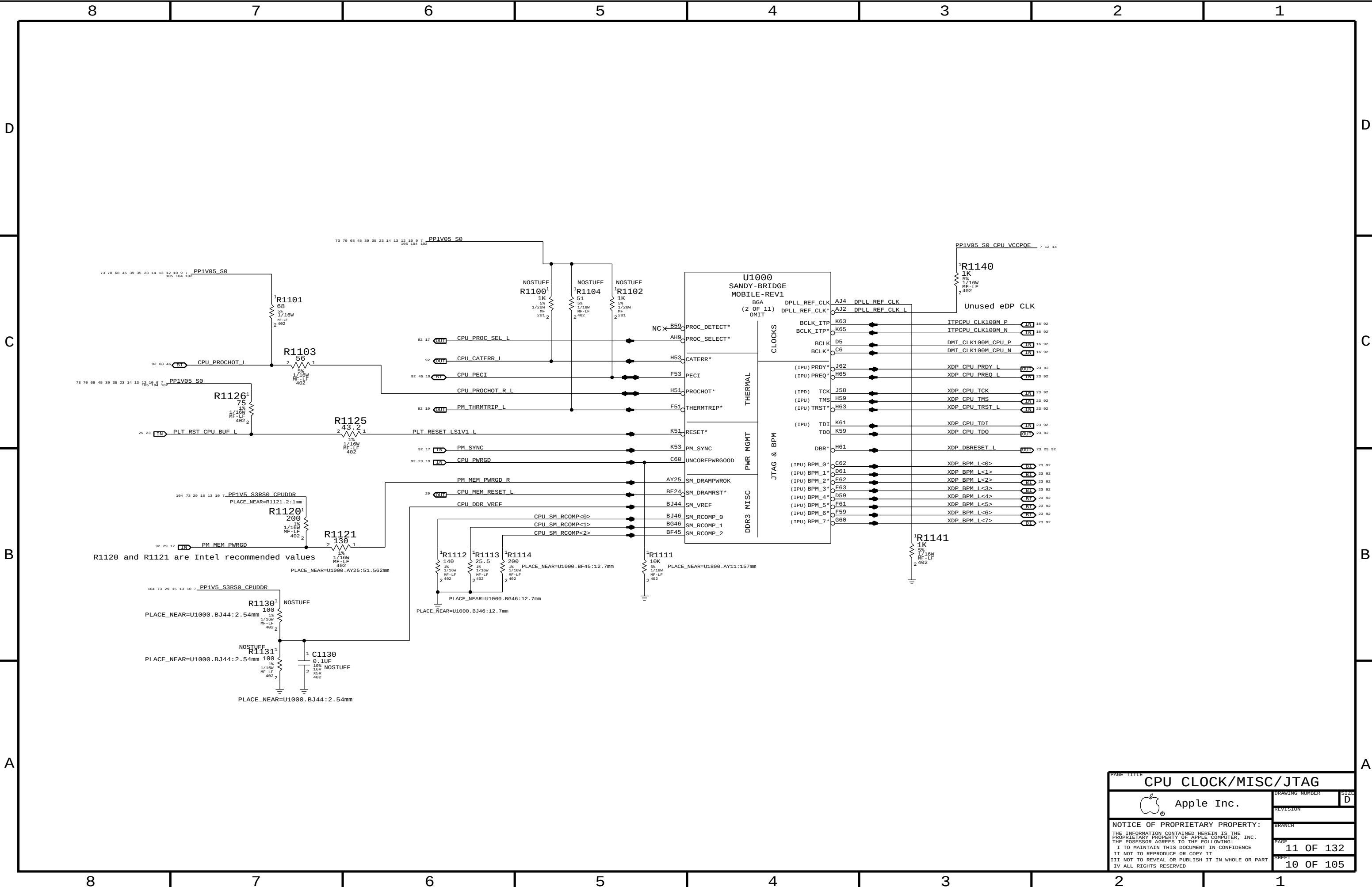




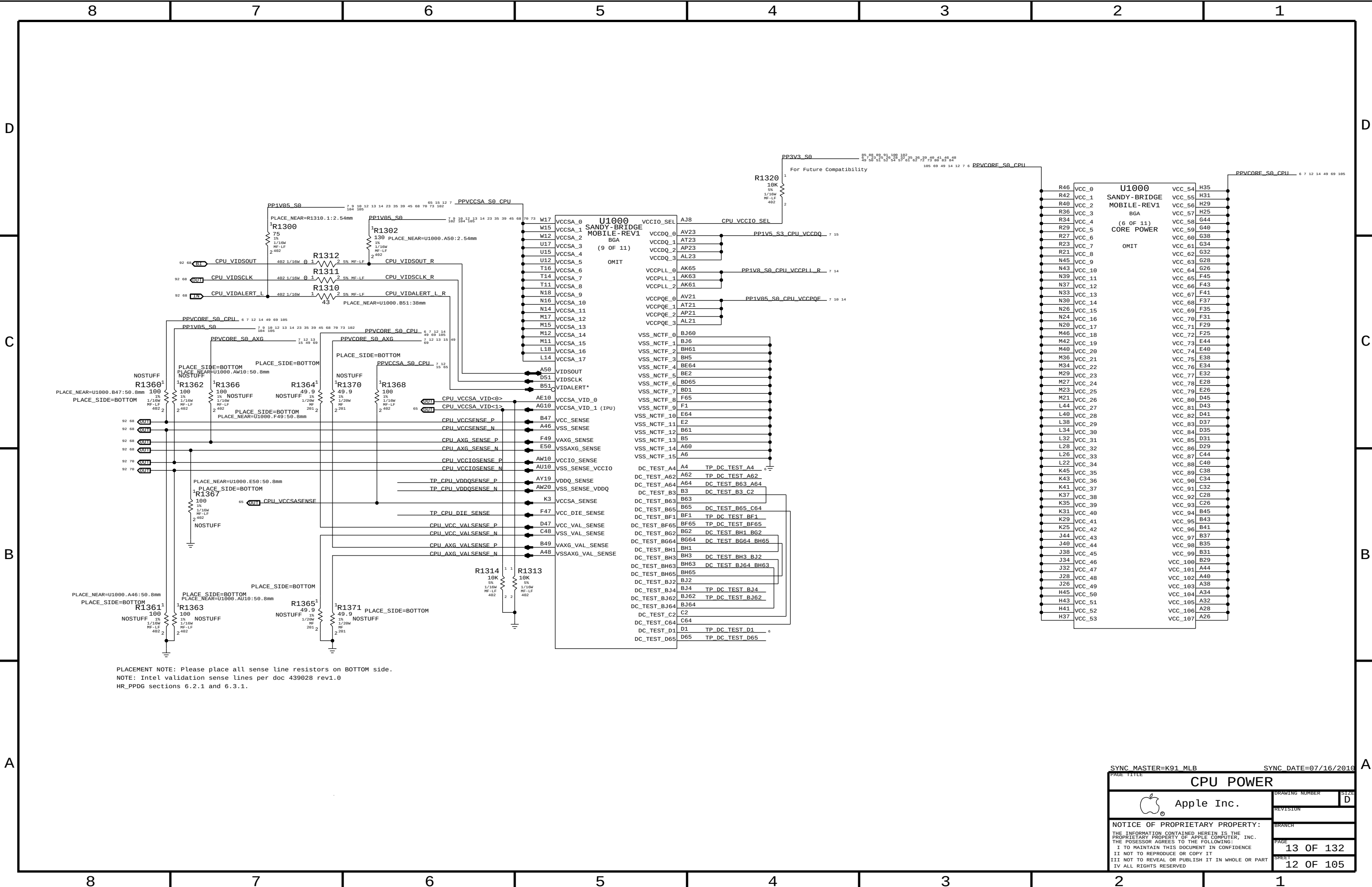


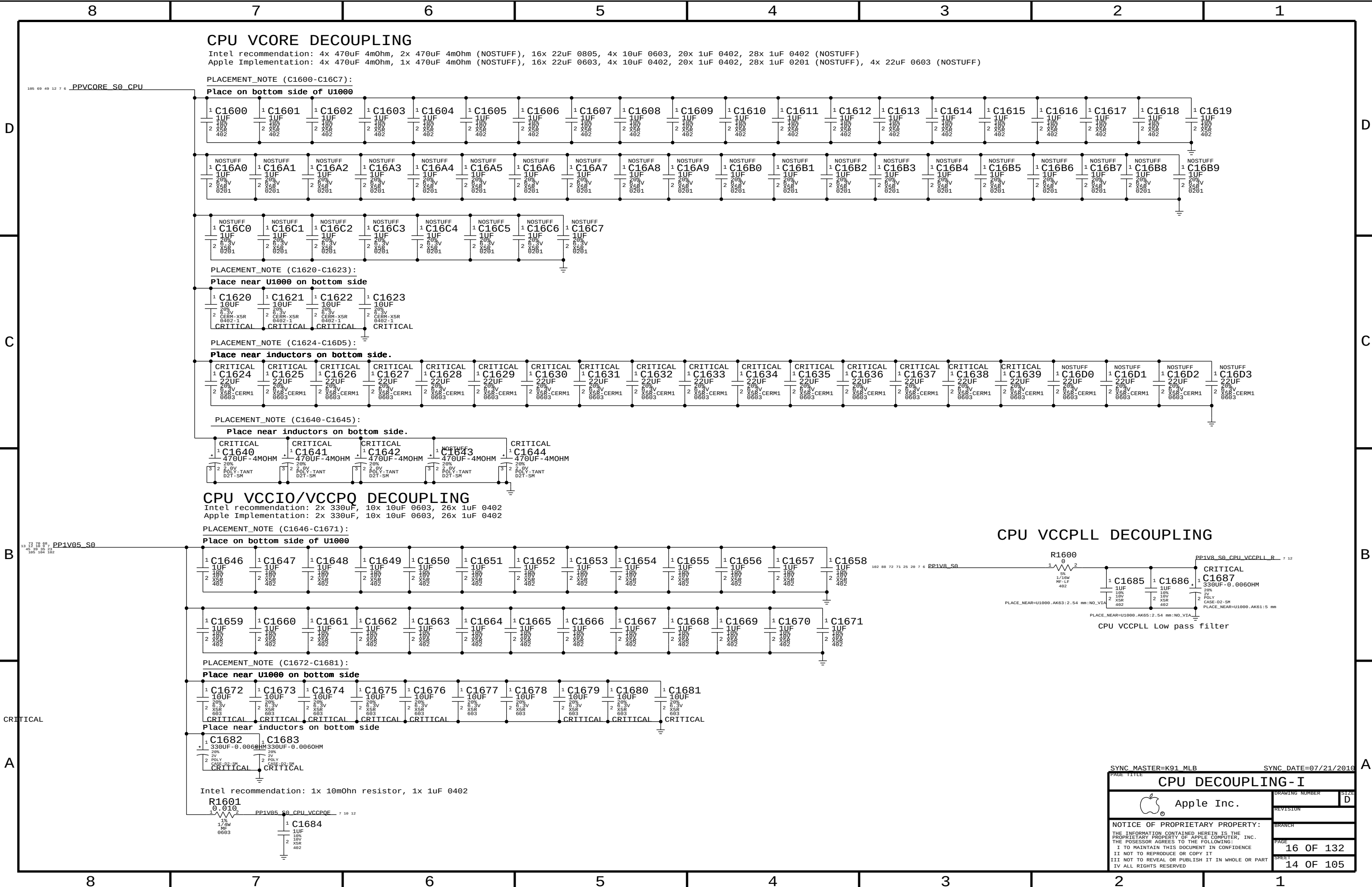
SYNC MASTER=K17_MLB		SYNC DATE=84/26/2816	
PAGE TITLE		DRAWING NUMBER	SIZE
 Apple Inc.		REVISION	D
		BRANCH	
NOTICE OF PROPRIETARY PROPERTY: THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE COMPUTER, INC. THE POSSESSOR AGREES TO THE FOLLOWING: I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE II NOT TO REPRODUCE OR COPY IT III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART IV ALL RIGHTS RESERVED		PAGE	9 OF 132
		SHEET	8 OF 105









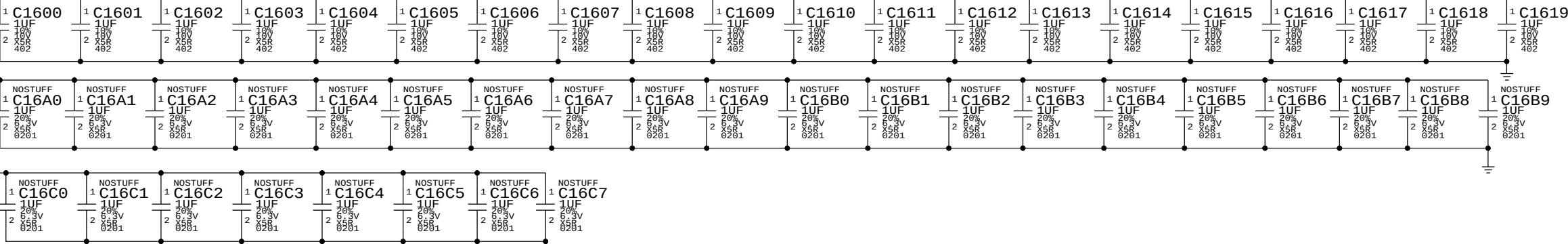


CPU VCORE DECOUPLING

Intel recommendation: 4x 470uF 4mOhm, 2x 470uF 4mOhm (NOSTUFF), 16x 22uF 0805, 4x 10uF 0603, 20x 1uF 0402, 28x 1uF 0402 (NOSTUFF)
Apple Implementation: 4x 470uF 4mOhm, 1x 470uF 4mOhm (NOSTUFF), 16x 22uF 0603, 4x 10uF 0402, 20x 1uF 0402, 28x 1uF 0201 (NOSTUFF), 4x 22uF 0603 (NOSTUFF)

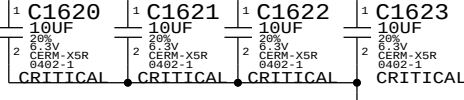
PLACEMENT_NOTE (C1600-C16C7):

Place on bottom side of U1000



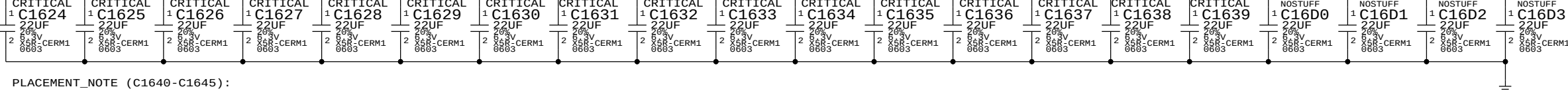
PLACEMENT_NOTE (C1620-C1623):

Place near U1000 on bottom side



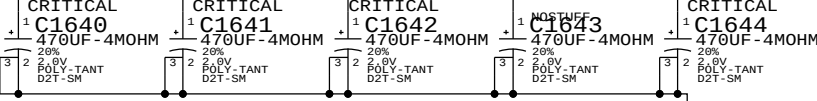
PLACEMENT_NOTE (C1624-C16D5):

Place near inductors on bottom side.



PLACEMENT_NOTE (C1640-C1645):

Place near inductors on bottom side.

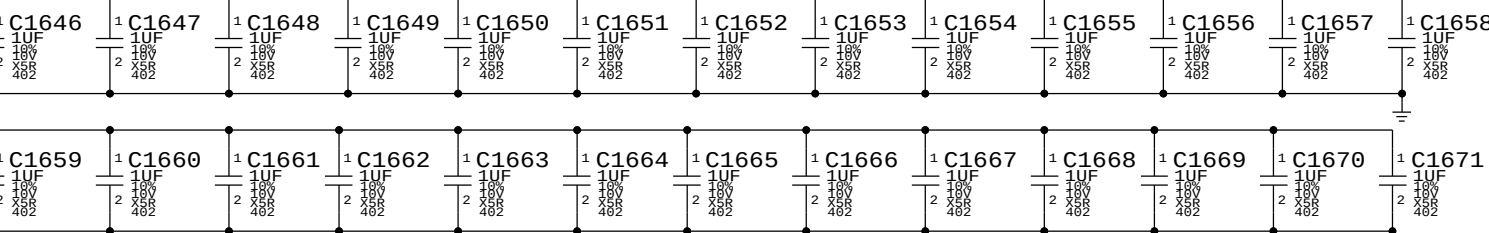


CPU VCCIO/VCCPQ DECOUPLING

Intel recommendation: 2x 330uF, 10x 10uF 0603, 26x 1uF 0402
Apple Implementation: 2x 330uF, 10x 10uF 0603, 26x 1uF 0402

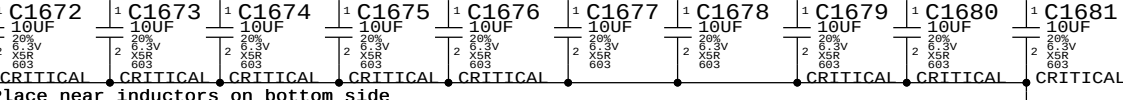
PLACEMENT_NOTE (C1646-C1671):

Place on bottom side of U1000

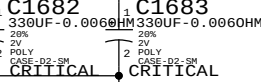


PLACEMENT_NOTE (C1672-C1681):

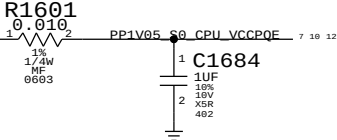
Place near U1000 on bottom side



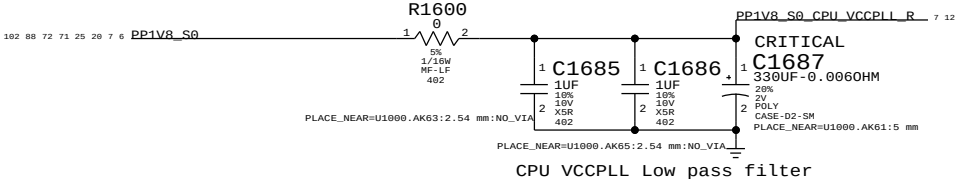
Place near inductors on bottom side



Intel recommendation: 1x 10mOhm resistor, 1x 1uF 0402

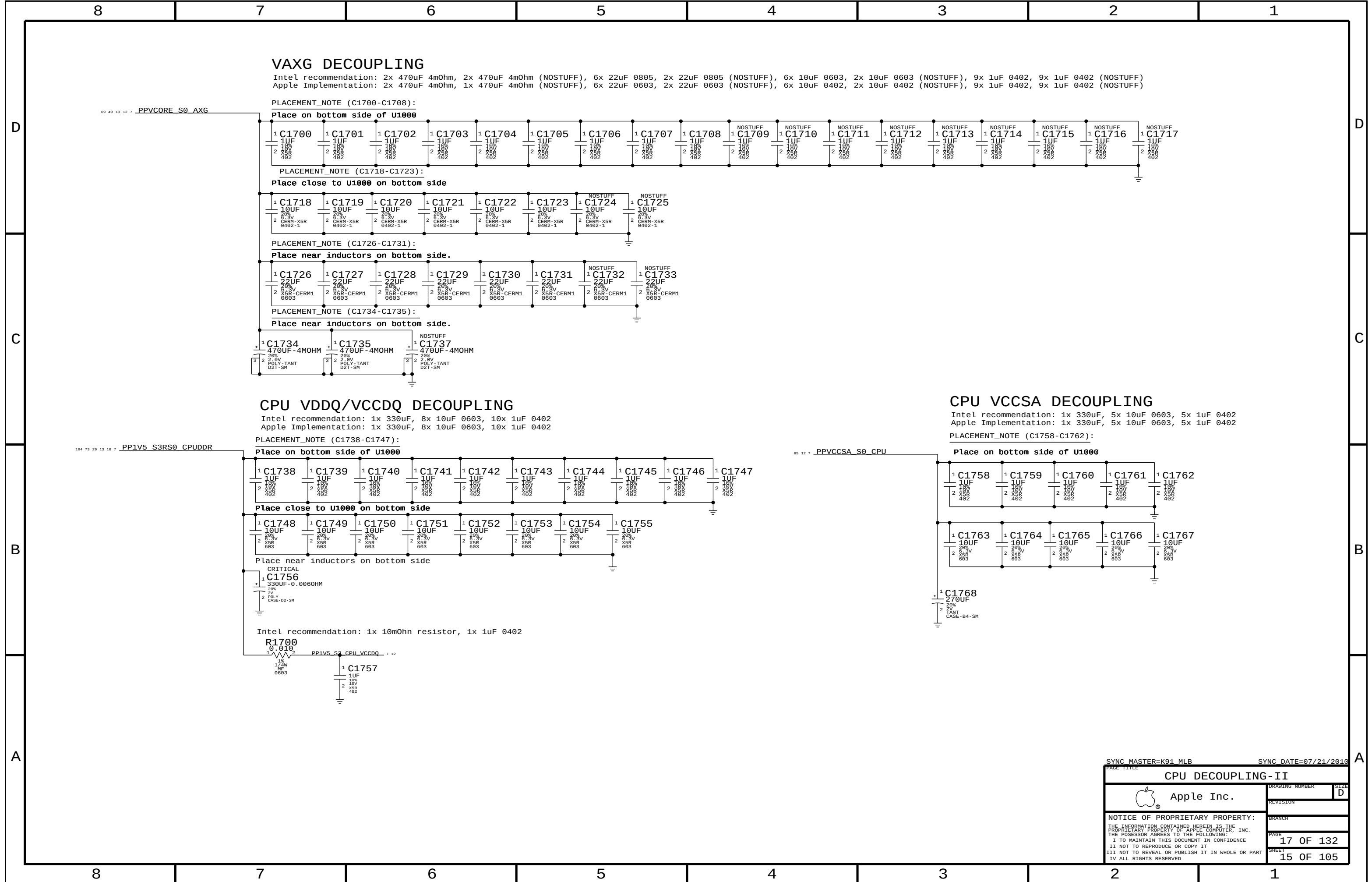


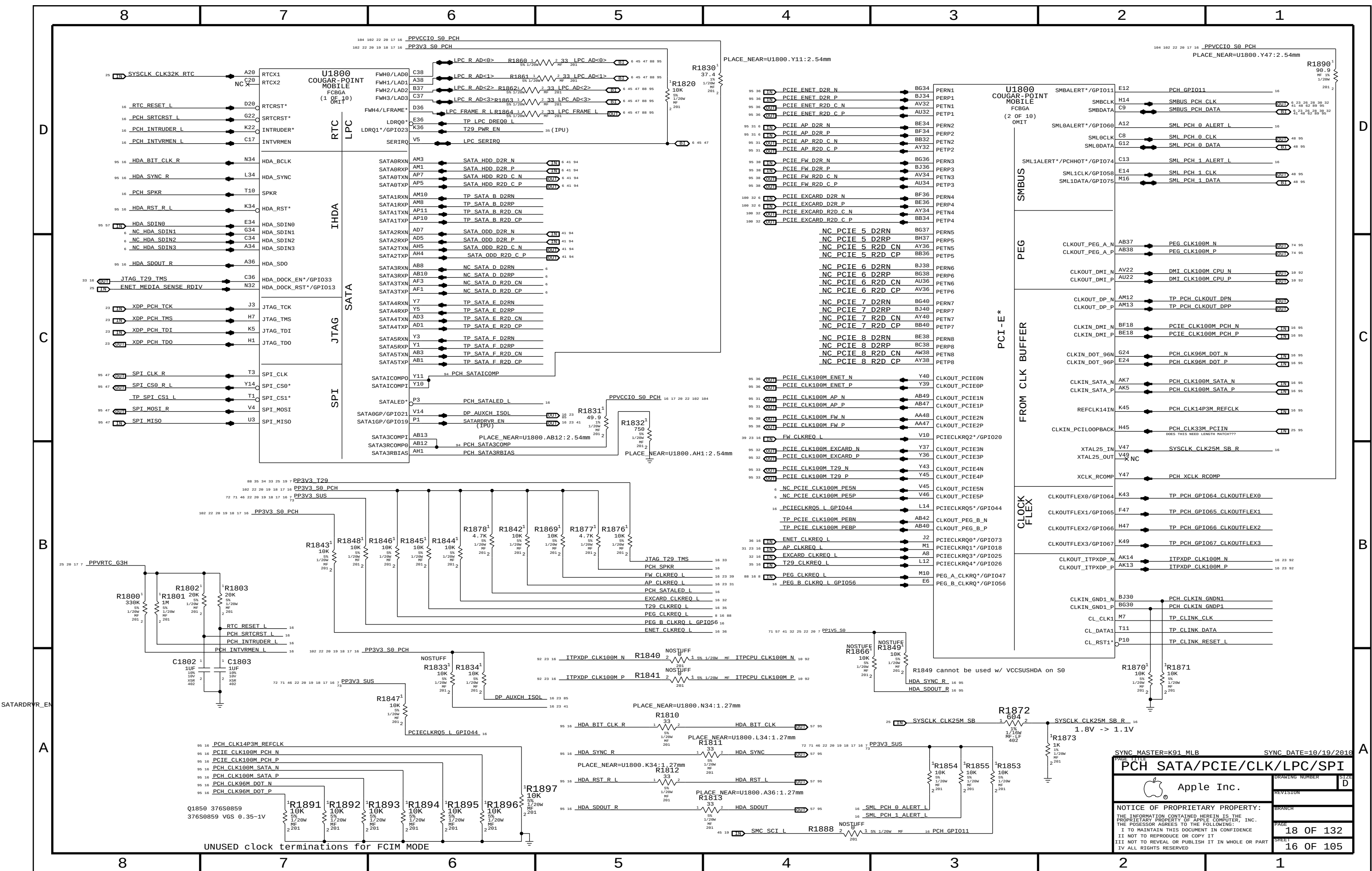
CPU VCCPLL DECOUPLING

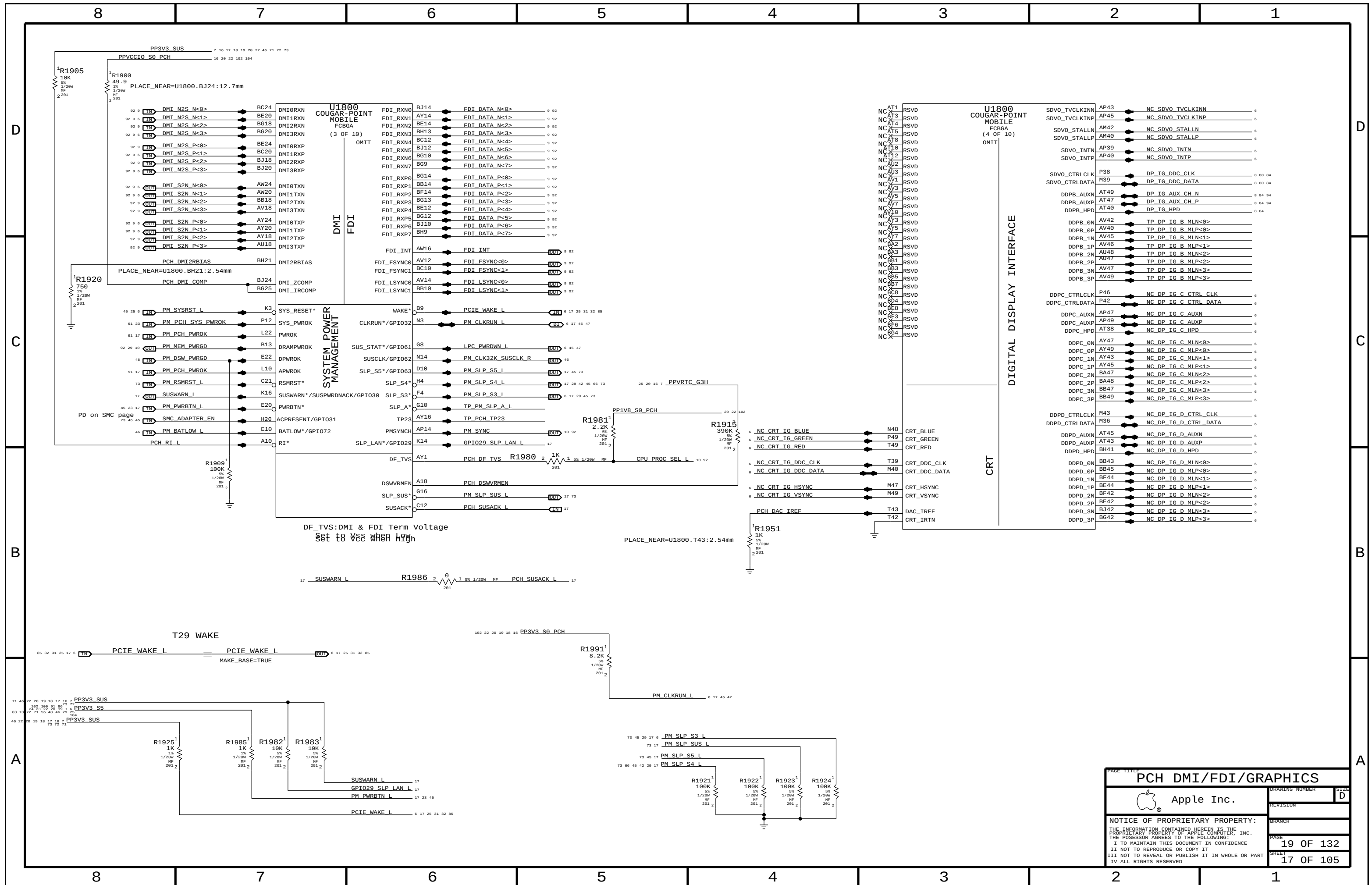


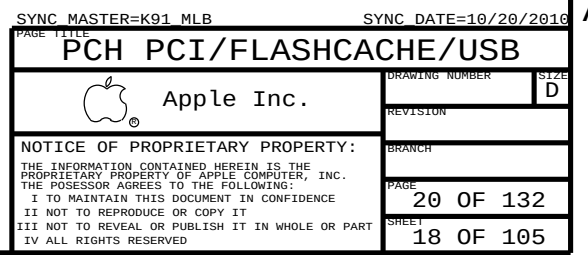
CPU VCCPLL Low pass filter

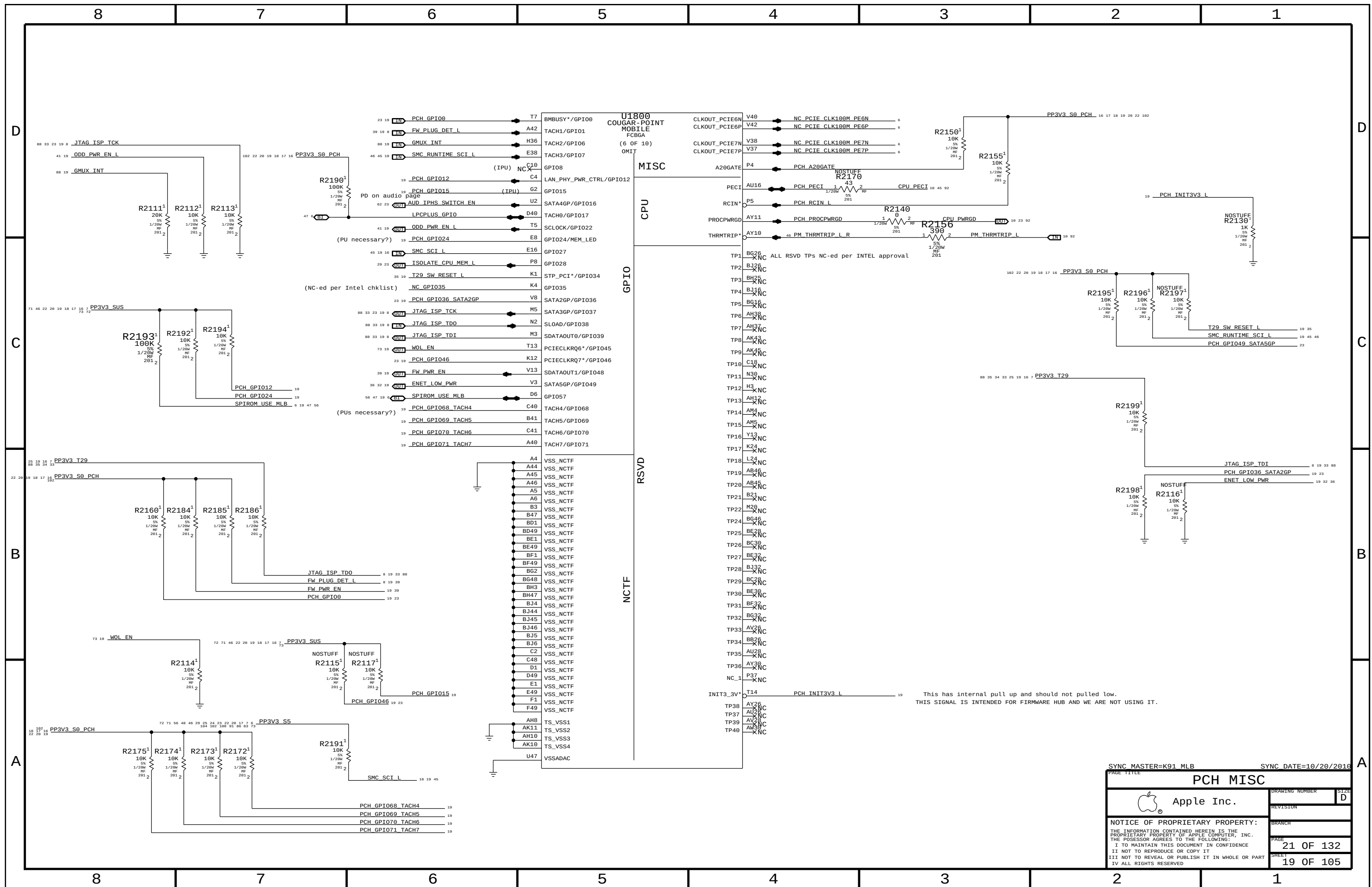
SYNC MASTER=K91 MLB		SYNC DATE=07/21/2016	
PAGE TITLE		CPU DECOUPLING-I	
Apple Inc.		DRAWING NUMBER	SIZE D
NOTICE OF PROPRIETARY PROPERTY: THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE COMPUTER, INC. THE POSSESSOR AGREES TO THE FOLLOWING: I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE II NOT TO REPRODUCE OR COPY IT III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART IV ALL RIGHTS RESERVED		REVISION	BRANCH
		PAGE	16 OF 132
		SHEET	14 OF 105

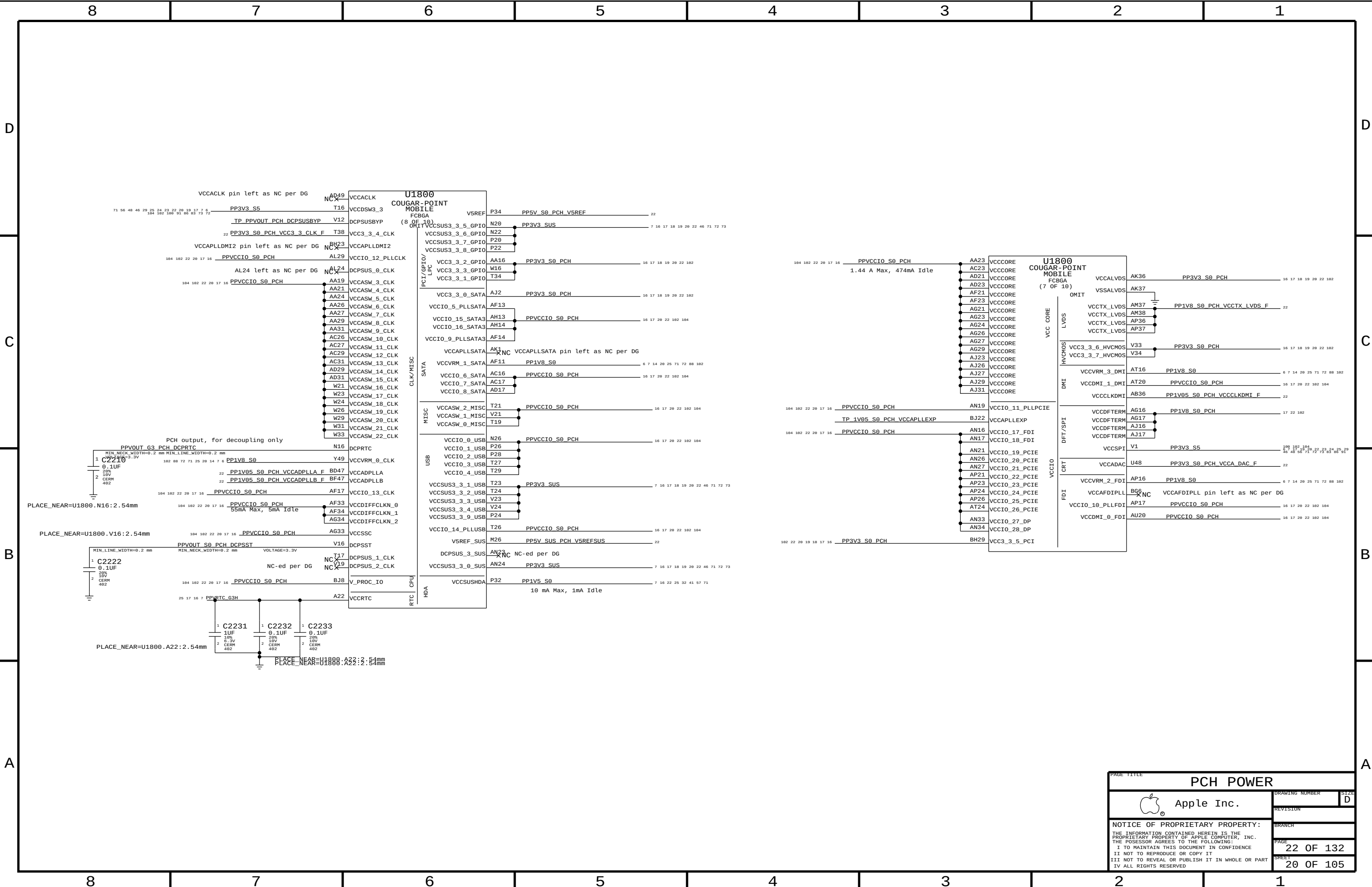


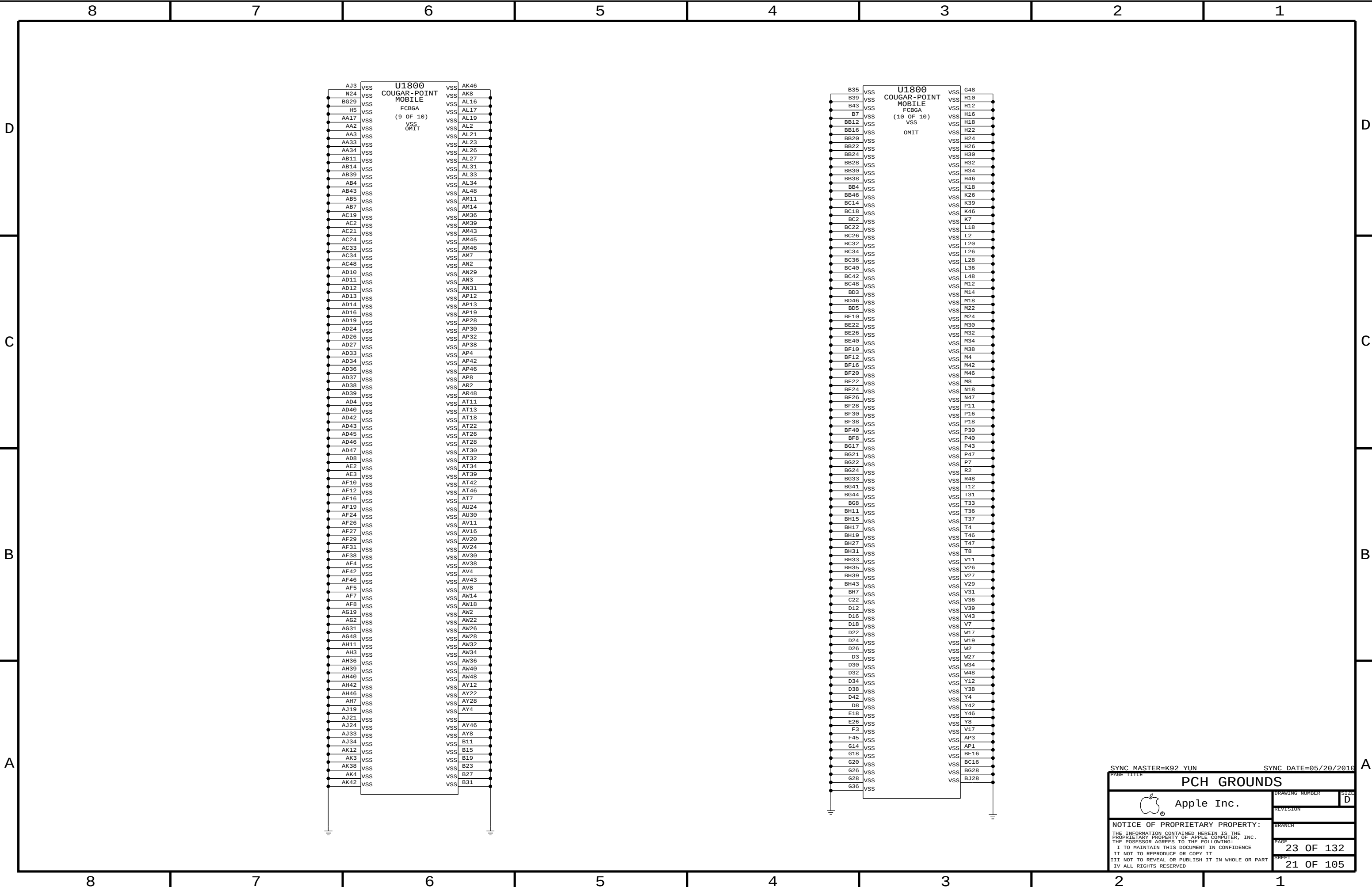








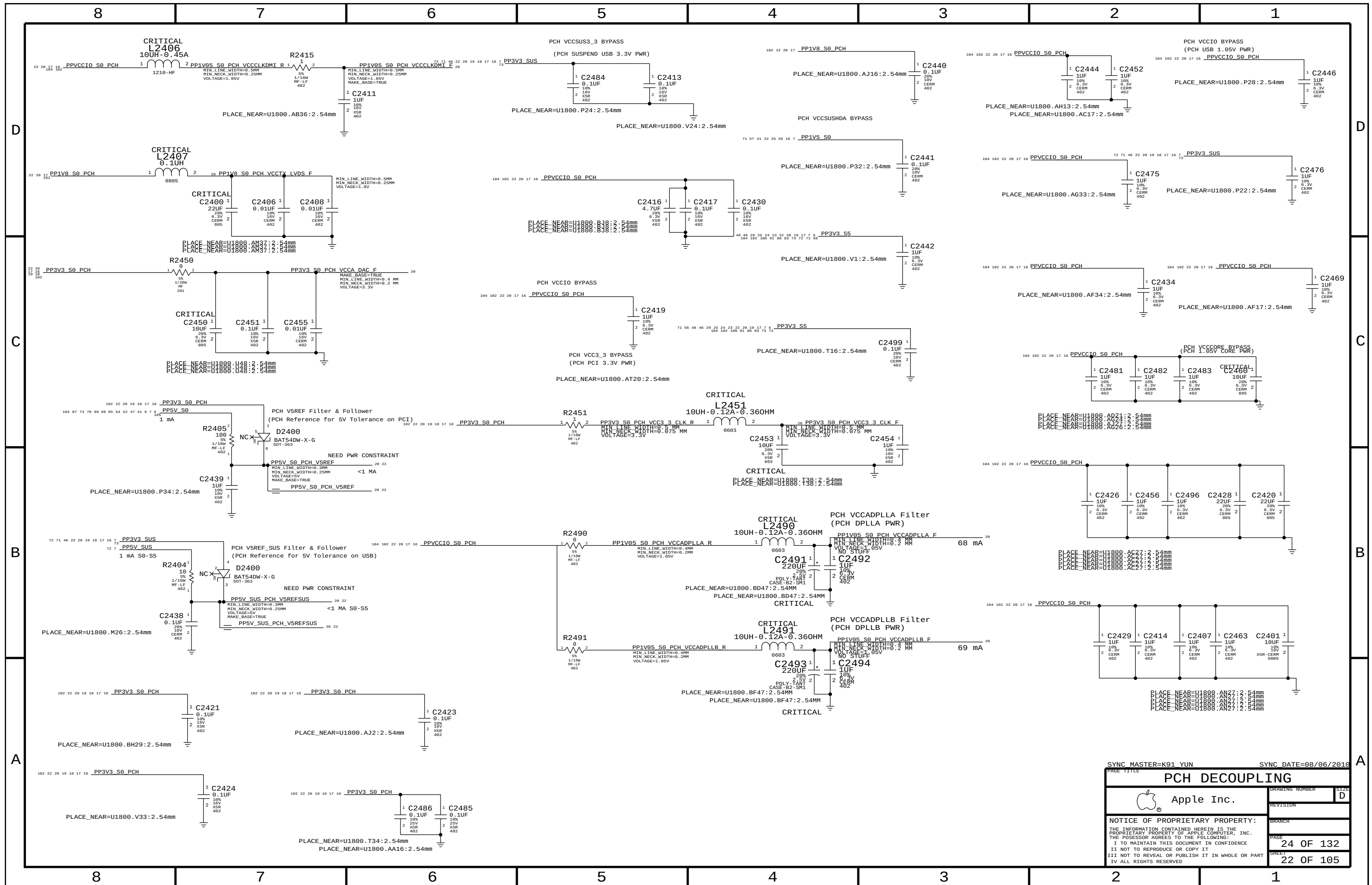


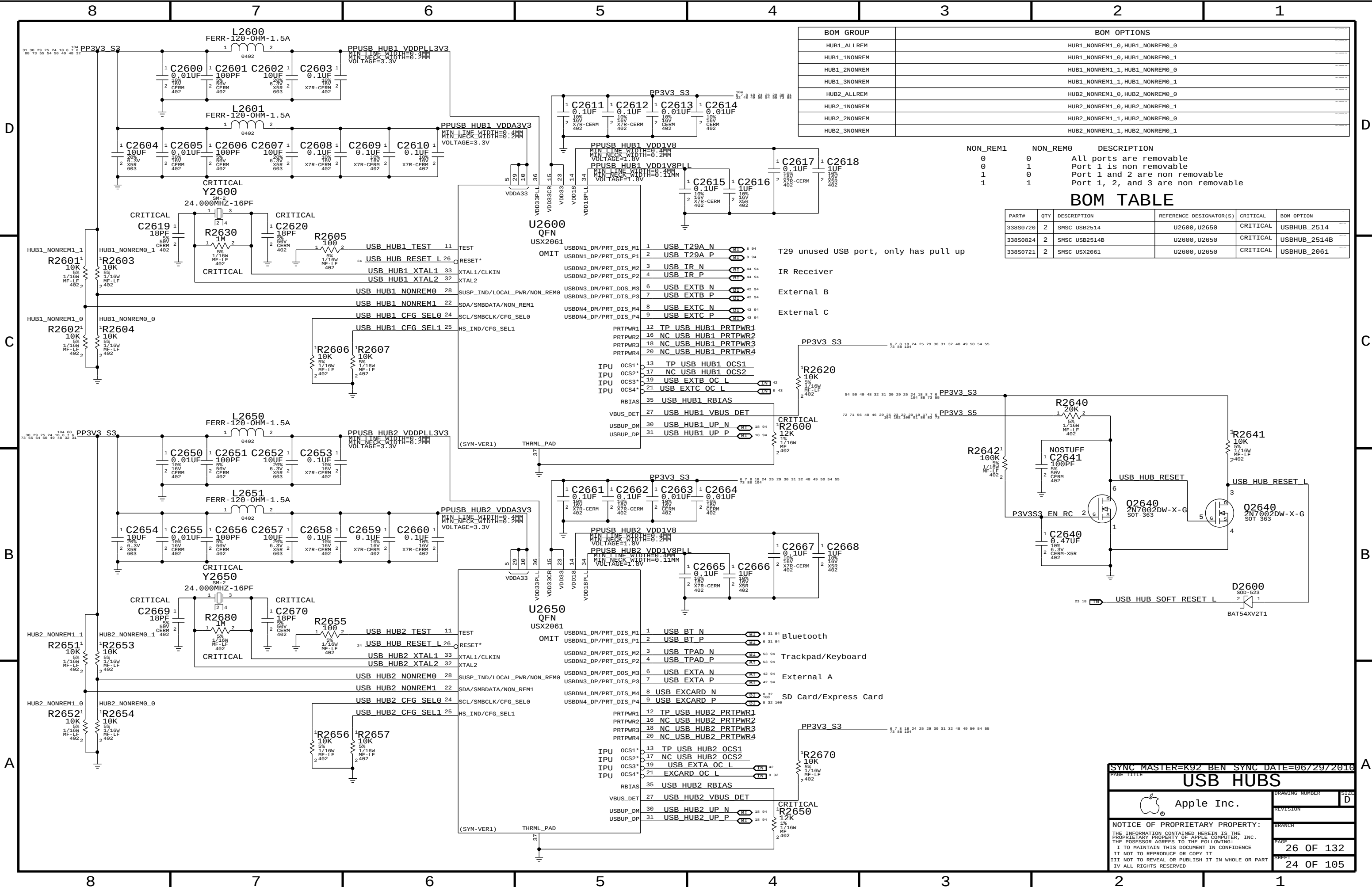


SYNC MASTER=K92_YUN

SYNC DATE=05/20/2016

PAGE TITLE		PCH GROUNDS	
 Apple Inc.		DRAWING NUMBER	SIZE
NOTICE OF PROPRIETARY PROPERTY: THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE COMPUTER, INC. THE POSSESSOR AGREES TO THE FOLLOWING: I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE II NOT TO REPRODUCE OR COPY IT III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART IV ALL RIGHTS RESERVED		REVISION	D
		BRANCH	
		PAGE	23 OF 132
		SHEET	21 OF 105





BOM GROUP		BOM OPTIONS	
HUB1_ALLREM		HUB1_NONREM1_0, HUB1_NONREM0_0	
HUB1_1NONREM		HUB1_NONREM1_0, HUB1_NONREM0_1	
HUB1_2NONREM		HUB1_NONREM1_1, HUB1_NONREM0_0	
HUB1_3NONREM		HUB1_NONREM1_1, HUB1_NONREM0_1	
HUB2_NONREM		HUB2_NONREM1_0, HUB2_NONREM0_0	
HUB2_1NONREM		HUB2_NONREM1_0, HUB2_NONREM0_1	
HUB2_2NONREM		HUB2_NONREM1_1, HUB2_NONREM0_0	
HUB2_3NONREM		HUB2_NONREM1_1, HUB2_NONREM0_1	

NON_REM1	NON_REM0	DESCRIPTION
0	0	All ports are removable
0	1	Port 1 is non removable
1	0	Port 1 and 2 are non removable
1	1	Port 1, 2, and 3 are non removable

BOM TABLE

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
338S0720	2	SMSC USB2514	U2600, U2650	CRITICAL	USBHUB_2514
338S0824	2	SMSC USB2514B	U2600, U2650	CRITICAL	USBHUB_2514B
338S0721	2	SMSC USX2061	U2600, U2650	CRITICAL	USBHUB_2061

T29 unused USB port, only has pull up

IR Receiver

External B

External C

Bluetooth

Trackpad/Keyboard

External A

SD Card/Express Card

SHEET

24 OF 105

PAGE

26 OF 132

NOTICE OF PROPRIETARY PROPERTY:
THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE COMPUTER, INC. THE POSSESSOR AGREES TO THE FOLLOWING:
I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE
II NOT TO REPRODUCE OR COPY IT
III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART
IV ALL RIGHTS RESERVED

Apple Inc.

DRAWING NUMBER

REVISION

BRANCH

SIZE

D

SHEET

24 OF 105

PAGE

26 OF 132

USB HUBS

Apple Inc.

DRAWING NUMBER

REVISION

BRANCH

SIZE

D

SHEET

24 OF 105

PAGE

26 OF 132

SYNC MASTER=K92 BEN SYNC DATE=06/29/2010

PAGE TITLE

USB HUBS

Apple Inc.

DRAWING NUMBER

REVISION

BRANCH

SIZE

D

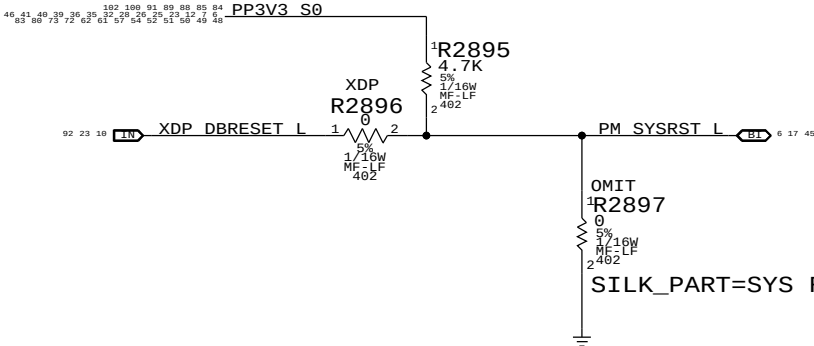
SHEET

24 OF 105

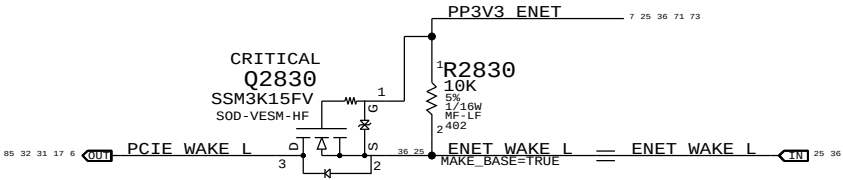
PAGE

26 OF 132

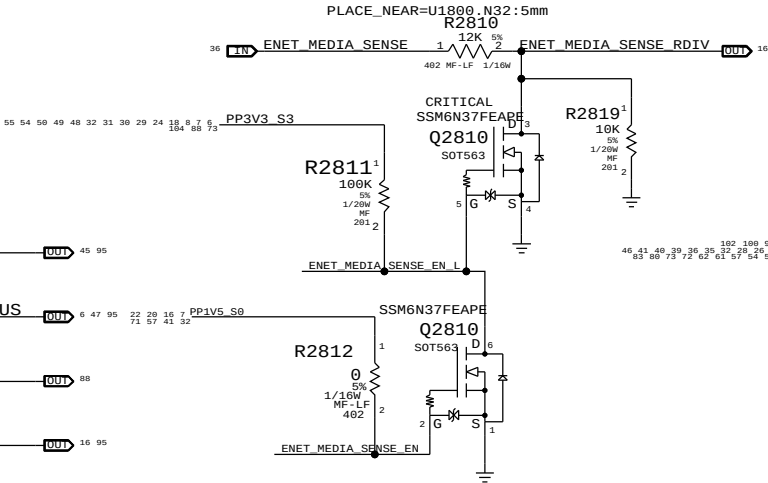
PCH Reset Button



Ethernet WAKE# Isolation

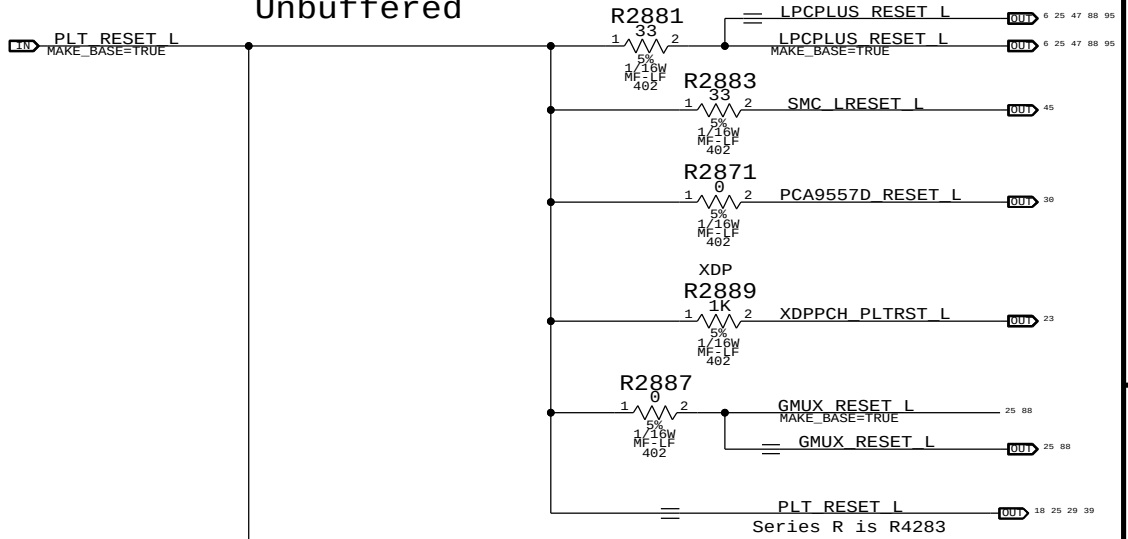


ENET_MEDIA_SENSE ISOLATION CIRCUIT



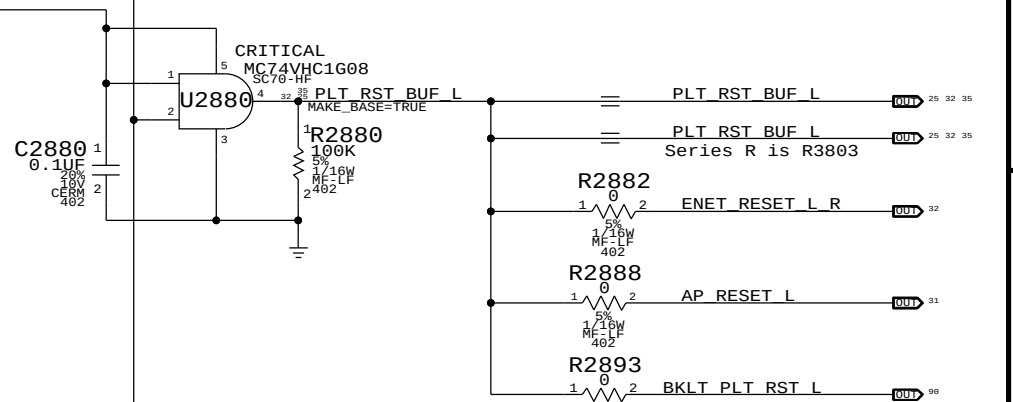
Platform Reset Connections

Unbuffered

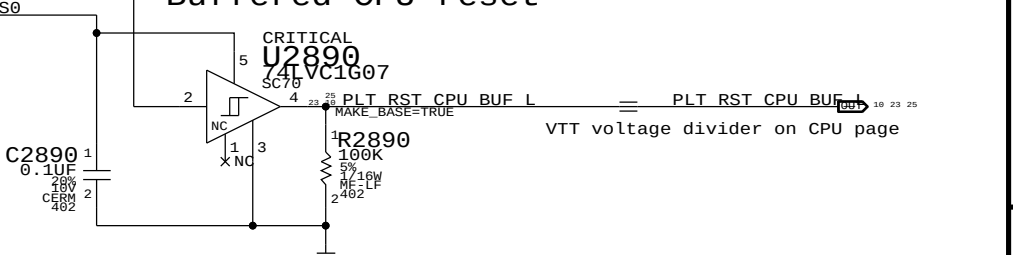


Buffered

Note: Based on K91/K92 layout, ENET,AP and BKLT are moved to Buffered reset.

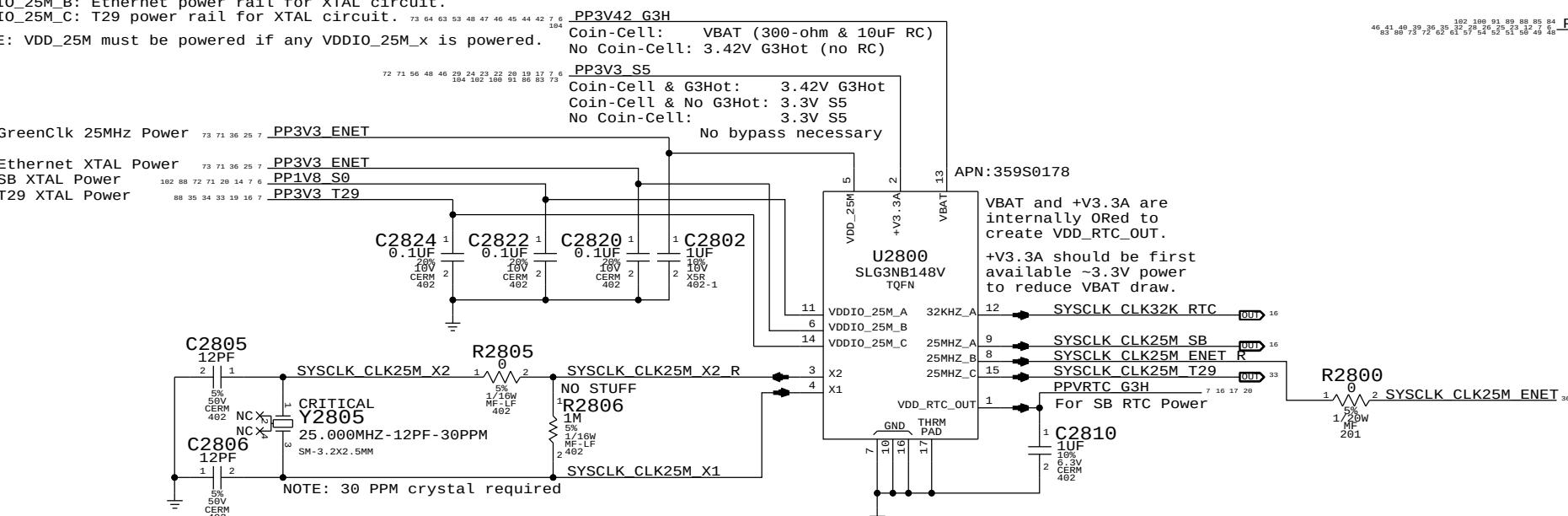


Buffered CPU reset

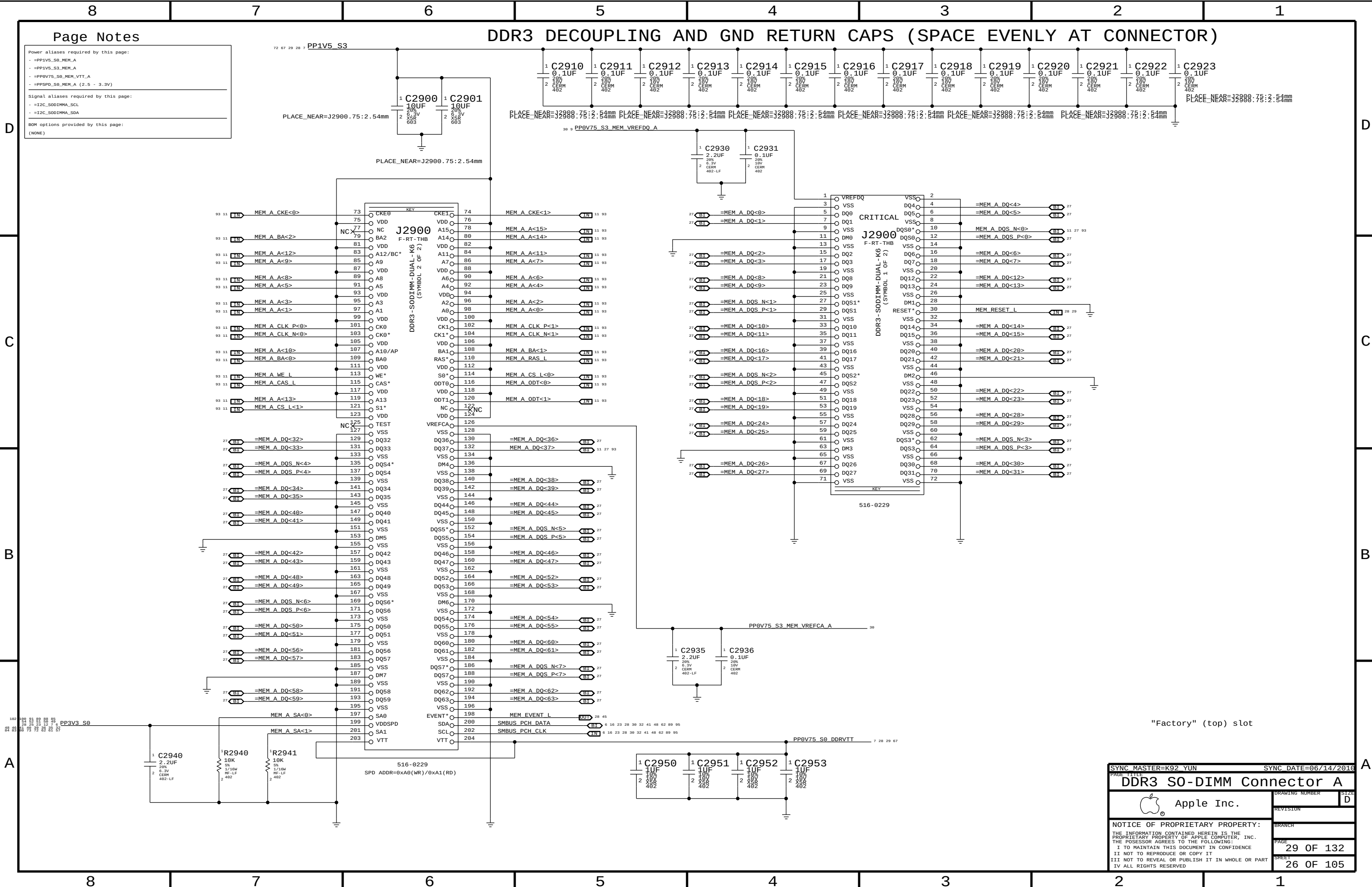


System RTC Power Source & 32kHz / 25MHz Clock Generator

VDDIO_25M_A: SB power rail for XTAL circuit.
VDDIO_25M_B: Ethernet power rail for XTAL circuit.
VDDIO_25M_C: T29 power rail for XTAL circuit.
NOTE: VDD_25M must be powered if any VDDIO_25M_x is powered.



SYNC MASTER=K91 MLB		SYNC DATE=06/29/2016	
PAGE TITLE		Chipset Support	
Apple Inc.		DRAWING NUMBER	SIZE D
NOTICE OF PROPRIETARY PROPERTY: THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE COMPUTER, INC. THE POSSESSOR AGREES TO THE FOLLOWING: I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE II NOT TO REPRODUCE OR COPY IT III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART IV ALL RIGHTS RESERVED		REVISION	BRANCH
		PAGE	28 OF 132
		SHEET	25 OF 105



Page Notes

Power aliases required by this page:

- =PP1V5_S0_MEM_A
- =PP1V5_S3_MEM_A
- =PP0V75_S0_MEM_VTT_A
- =PPSPD_S0_MEM_A (2.5 - 3.3V)

Signal aliases required by this page:

- =I2C_SODIMM_SCL
- =I2C_SODIMM_SDA

BOM options provided by this page:

(NONE)

DDR3 DECOUPLING AND GND RETURN CAPS (SPACE EVENLY AT CONNECTOR)

SYNC MASTER=K92_YUN		SYNC DATE=06/14/2016	
PAGE TITLE			
DDR3 S0-DIMM Connector A			
Apple Inc.		DRAWING NUMBER	SIZE
		REVISION	D
NOTICE OF PROPRIETARY PROPERTY:		BRANCH	
THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE COMPUTER, INC. THE POSSESSOR AGREES TO THE FOLLOWING:		PAGE	
I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE		29 OF 132	
II NOT TO REPRODUCE OR COPY IT		SHEET	
III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART		26 OF 105	
IV ALL RIGHTS RESERVED			

[illegible]

Page Notes

Power aliases required by this page:

- =PP1V5_S0_MEM_B
- =PP1V5_S3_MEM_B
- =PP0V75_S0_MEM_VTT_B
- =PPSPD_S0_MEM_B (2.5 - 3.3V)

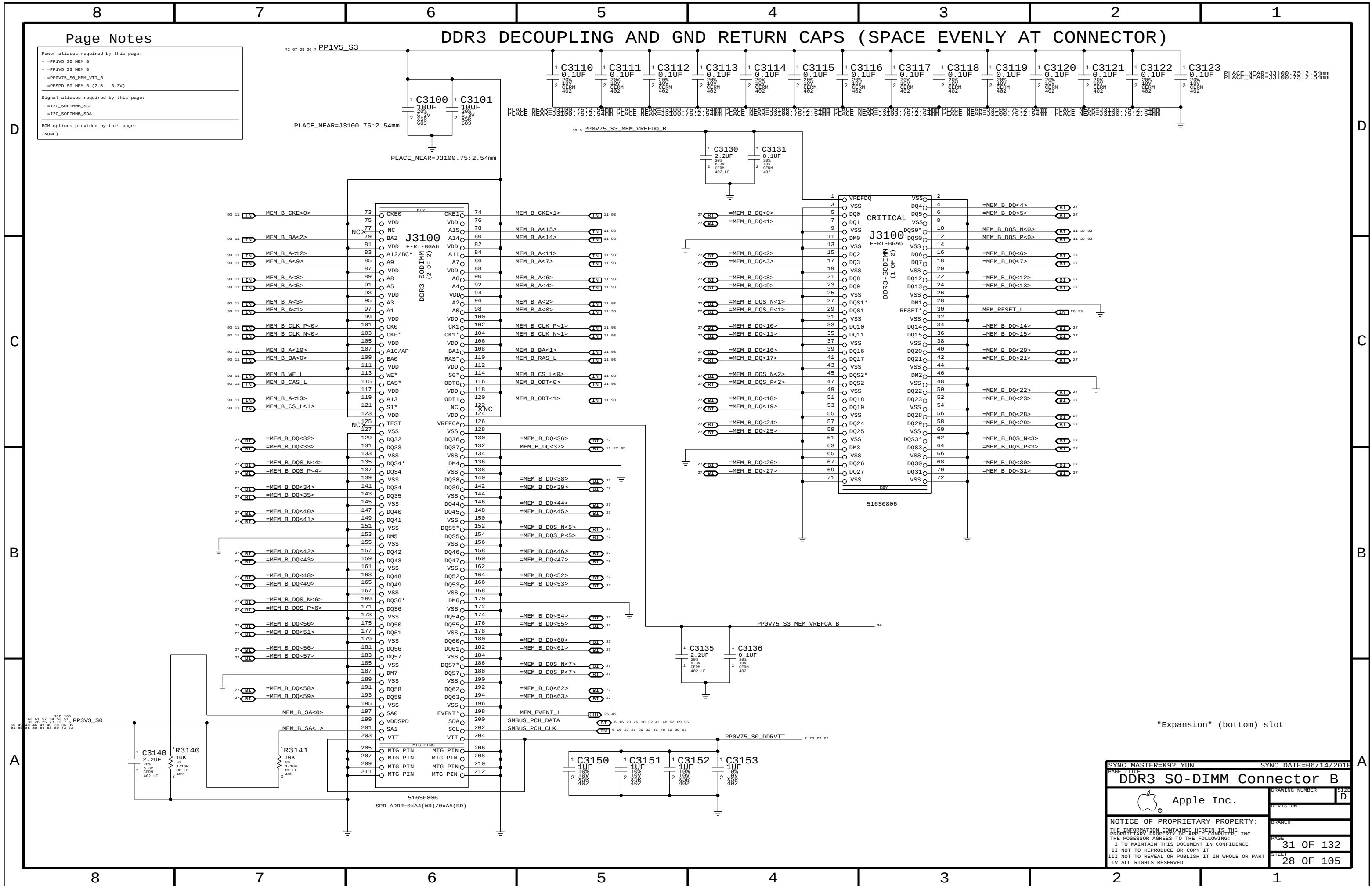
Signal aliases required by this page:

- =I2C_S0DIMM_SCL
- =I2C_S0DIMM_SDA

BOM options provided by this page:

(NONE)

DDR3 DECOUPLING AND GND RETURN CAPS (SPACE EVENLY AT CONNECTOR)



"Expansion" (bottom) slot

SYNC MASTER=K92_YUN		SYNC DATE=06/14/2016	
PAGE TITLE			
DDR3 S0-DIMM Connector B			
Apple Inc.		DRAWING NUMBER	SIZE
NOTICE OF PROPRIETARY PROPERTY:		REVISION	
THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE COMPUTER, INC. THE POSSESSOR AGREES TO THE FOLLOWING:		BRANCH	
I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE		PAGE	31 OF 132
II NOT TO REPRODUCE OR COPY IT		SHEET	28 OF 105
III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART			
IV ALL RIGHTS RESERVED			

The circuit below handles CPU and VTT power during S0->S3->S0 transitions, as well as isolating the CPU's SM_DRAMRST# output from the S0-DIMMs when necessary.

ISOLATE_CPU_MEM_L GPIO state during S3<->S0 transitions determines behavior of signals.

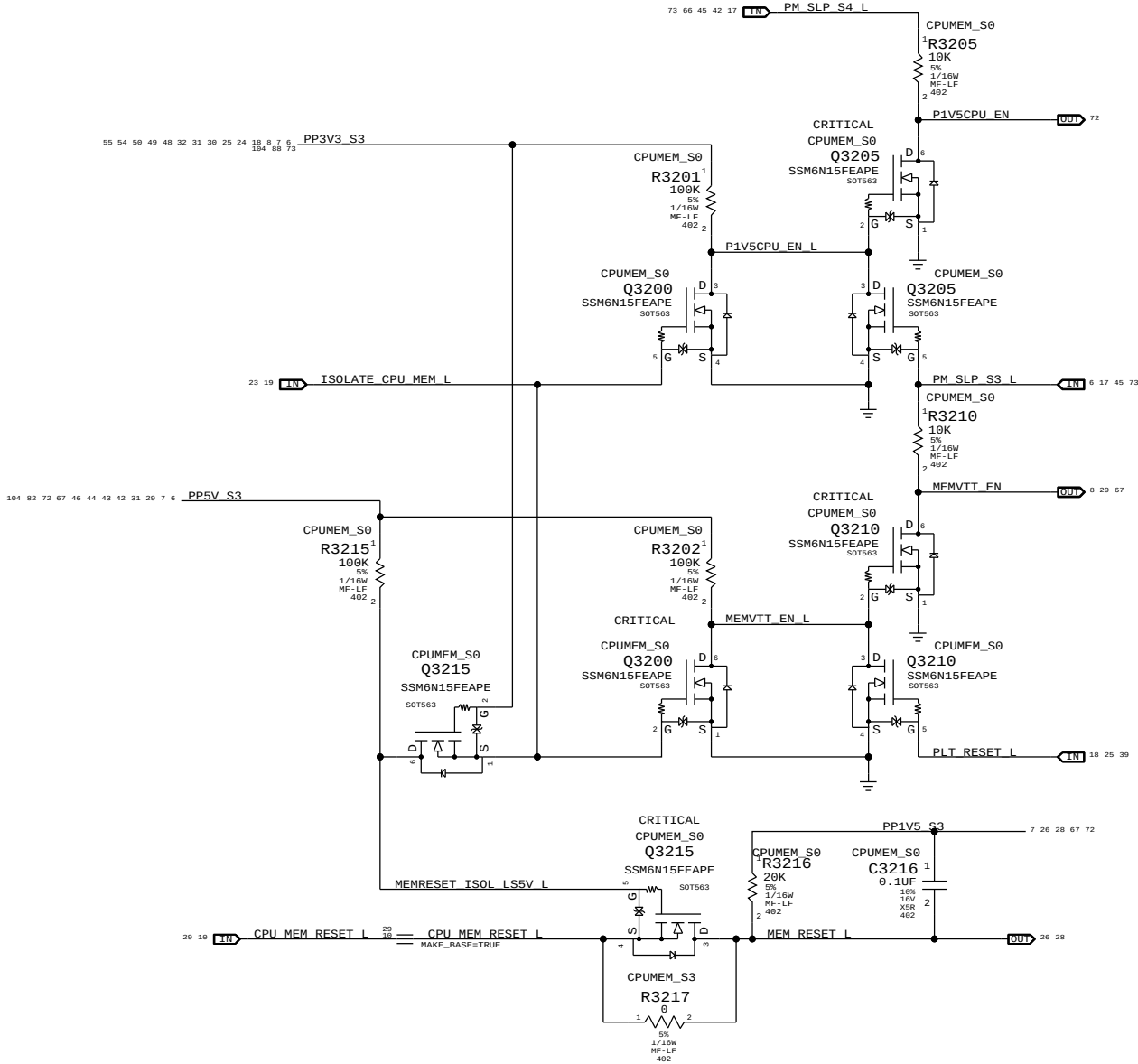
WHEN HIGH: CPU 1.5V remains powered in S3, VTT follows S0 rails, MEM_RESET_L not isolated.

WHEN LOW: CPU 1.5V follows S0 rails, VTT ensures clean CKE transition, MEM_RESET_L isolated.

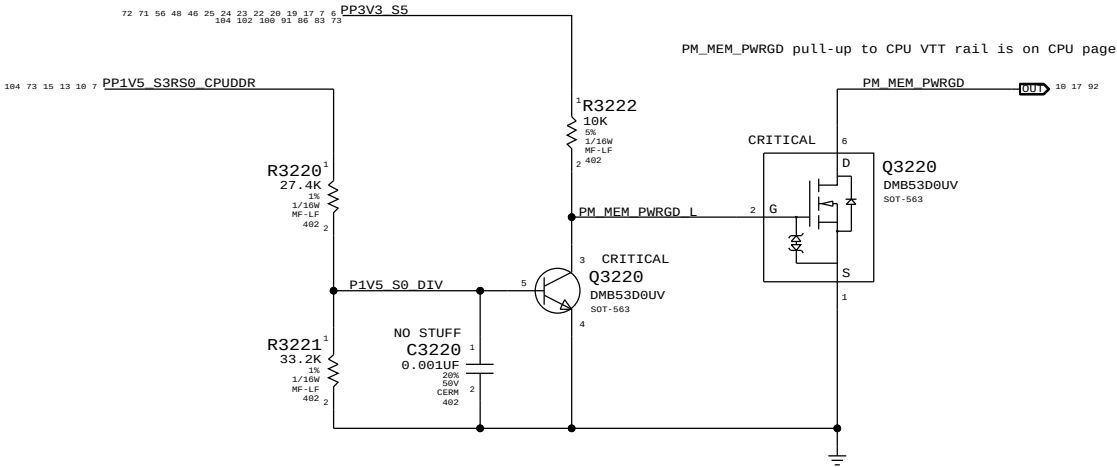
P1V5CPU_EN = (ISOLATE_CPU_MEM_L + PM_SLP_S3_L) * PM_SLP_S4_L

MEMVTT_EN = (ISOLATE_CPU_MEM_L + PLT_RST_L) * PM_SLP_S3_L

MEM_RESET_L = !ISOLATE_CPU_MEM_L + CPU_MEM_RESET_L

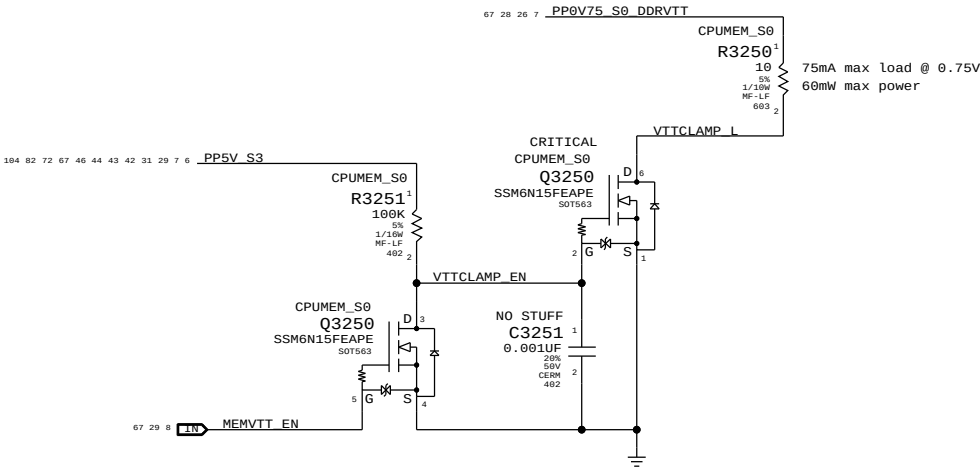


1V5 S0 "PGOOD" for CPU



MEMVTT Clamp

Ensures CKE signals are held low in S3

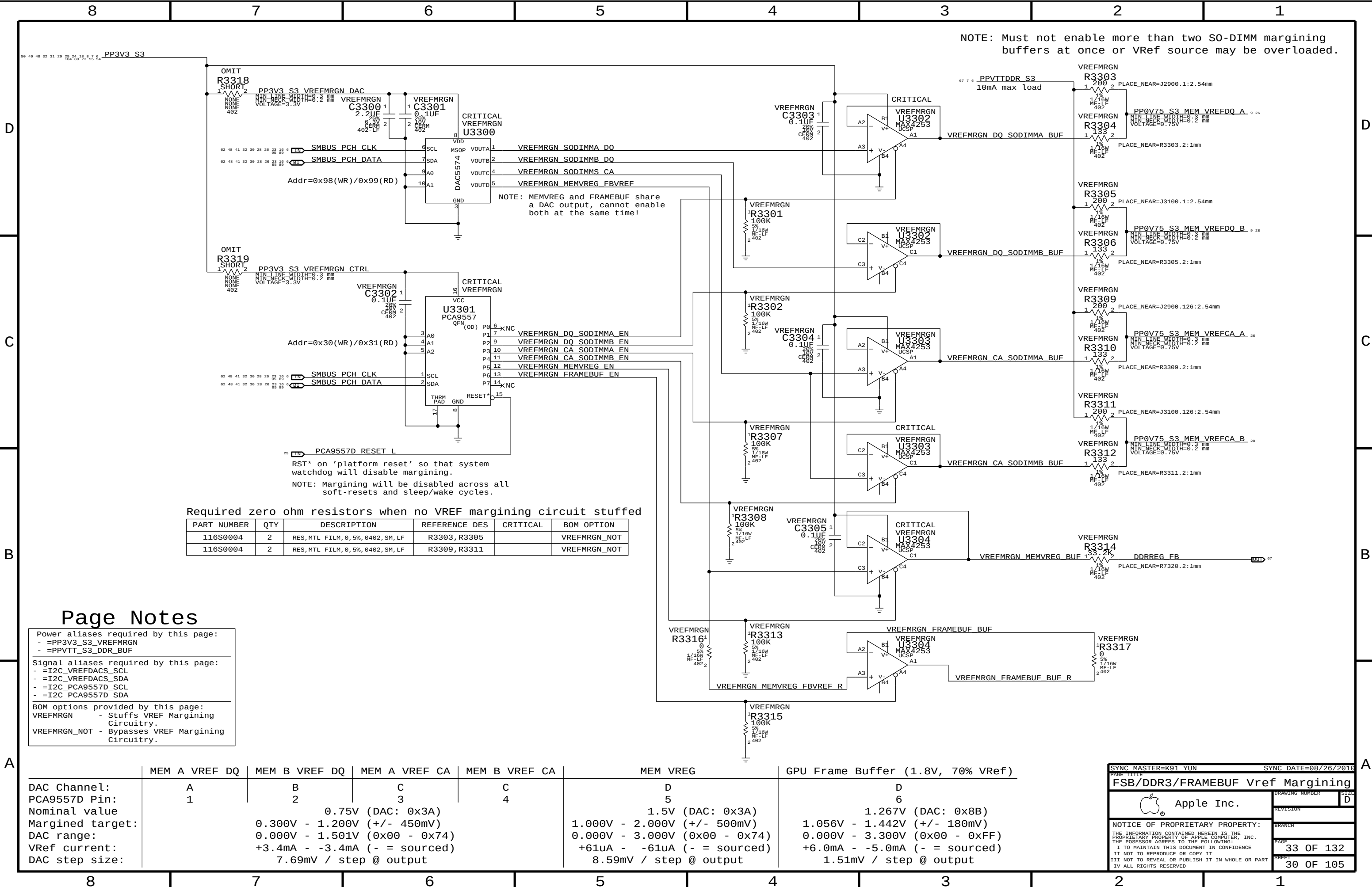


Step	SOLATE_CPU_MEM_L	PLT_RESET_L	PM_SLP_S3_L	PM_SLP_S4_L	CPU_MEM_RESET_L	MEM_RESET_L	MEMVTT_EN	P1V5CPU_EN
S0	0	1	1	1	1	CPU_MEM_RESET_L	1	1
1	1	0	1	1	1	1	1	1
2	0	0	1	1	1	1	0	1
3	0	0	0	1	X	1	0	0
4	0	0	1	1	X	1	0	1
5	0	1	1	1	0 (*)	1	1	1
6	0	1	1	1	1	1	1	1
S0	7	1	1	1	1	CPU_MEM_RESET_L	1	1

(*) CPU_MEM_RESET_L asserts due to loss of PM_MEM_PWRGD, must wait for software to clear before deasserting ISOLATE_CPU_MEM_L GPIO.

NOTE: In the event of a S3->S5 transition ISOLATE_CPU_MEM_L will still be asserted on next S5->S0 transition. Rails will power-up as if from S3, but MEM_RESET_L will not properly assert. Software must deassert ISOLATE_CPU_MEM_L and then generate a valid reset cycle on CPU_MEM_RESET_L.

SYNC MASTER=K17_MLB		SYNC DATE=04/26/2010	
PAGE TITLE		CPU Memory S3 Support	
Apple Inc.		DRAWING NUMBER	SIZE D
NOTICE OF PROPRIETARY PROPERTY:		REVISION	
THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE COMPUTER, INC. THE POSSESSOR AGREES TO THE FOLLOWING:		BRANCH	
I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE		PAGE 32 OF 132	
II NOT TO REPRODUCE OR COPY IT		SHEET 29 OF 105	
III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART			
IV ALL RIGHTS RESERVED			



Page Notes

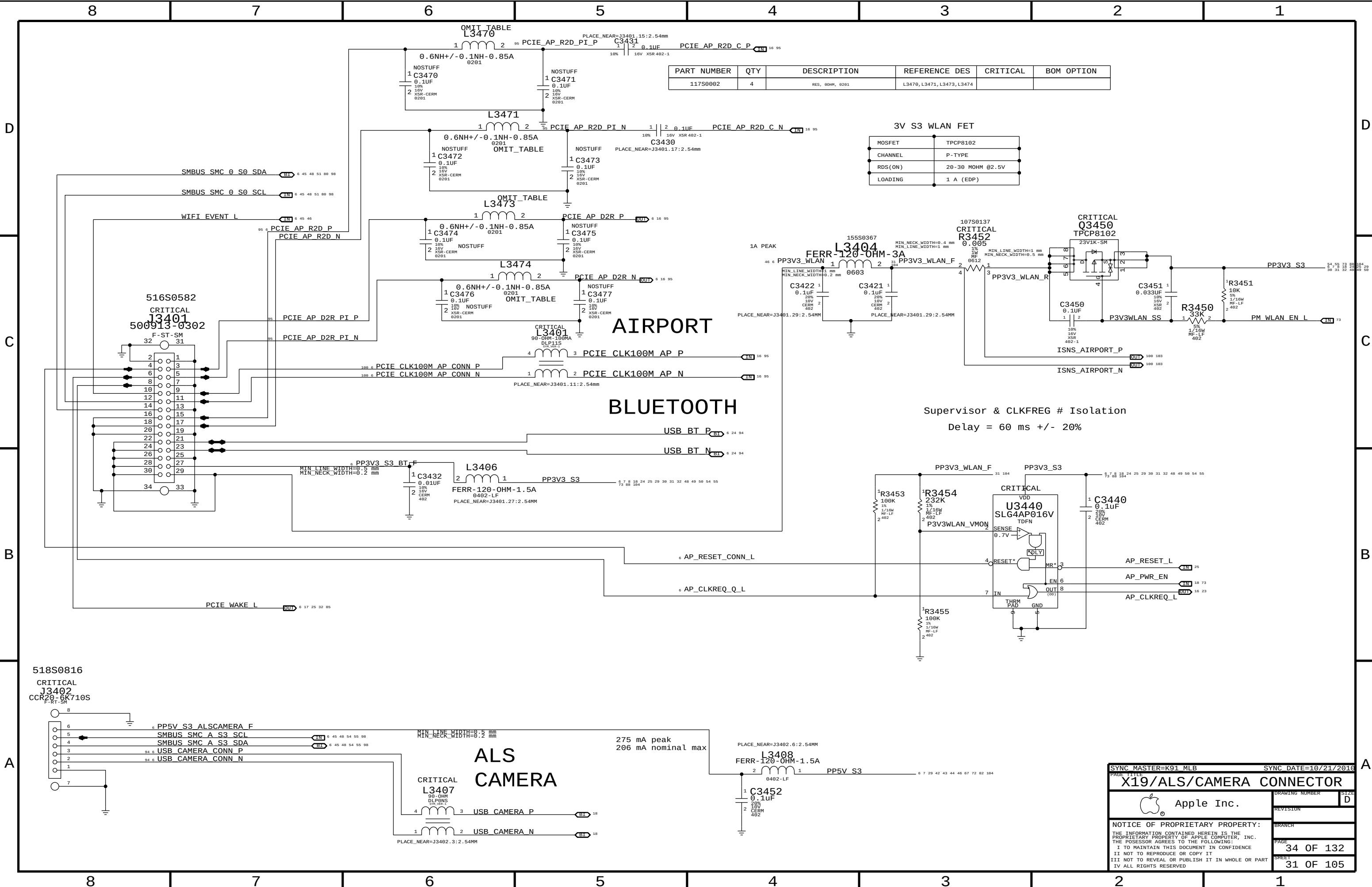
Power aliases required by this page:
- =PP3V3_S3_VREFMRGN
- =PPVTT_S3_DDR_BUF

Signal aliases required by this page:
- =I2C_VREFDAC_SCL
- =I2C_VREFDAC_SDA
- =I2C_PCA9557D_SCL
- =I2C_PCA9557D_SDA

BOM options provided by this page:
VREFMRGN - Stuffs VREF Margining Circuitry.
VREFMRGN_NOT - Bypasses VREF Margining Circuitry.

	MEM A VREF DQ	MEM B VREF DQ	MEM A VREF CA	MEM B VREF CA	MEM VREG	GPU Frame Buffer (1.8V, 70% Vref)
DAC Channel:	A	B	C	C	D	D
PCA9557D Pin:	1	2	3	4	5	6
Nominal value		0.75V (DAC: 0x3A)			1.5V (DAC: 0x3A)	1.267V (DAC: 0x8B)
Margined target:		0.300V - 1.200V (+/- 450mV)			1.000V - 2.000V (+/- 500mV)	1.056V - 1.442V (+/- 180mV)
DAC range:		0.000V - 1.501V (0x00 - 0x74)			0.000V - 3.000V (0x00 - 0x74)	0.000V - 3.300V (0x00 - 0xFF)
Vref current:		+3.4mA - -3.4mA (- = sourced)			+61uA - -61uA (- = sourced)	+6.0mA - -5.0mA (- = sourced)
DAC step size:		7.69mV / step @ output			8.59mV / step @ output	1.51mV / step @ output

SYNC MASTER=K91 YUN		SYNC DATE=08/26/2016	
PAGE TITLE		FSB/DDR3/FRAMEBUF Vref Margining	
 Apple Inc.		DRAWING NUMBER	SIZE
		REVISION	D
NOTICE OF PROPRIETARY PROPERTY:		BRANCH	
THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE COMPUTER, INC. THE POSSESSOR AGREES TO THE FOLLOWING:		PAGE	33 OF 132
I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE		SHEET	30 OF 105
II NOT TO REPRODUCE OR COPY IT			
III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART			
IV ALL RIGHTS RESERVED			



PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
117S0002	4	RES, 600M, 0201	L3470, L3471, L3473, L3474		

3V S3 WLAN FET	
MOSFET	TPCP8102
CHANNEL	P-TYPE
RDS(ON)	20-30 MOHM @2.5V
LOADING	1 A (EDP)

AIRPORT

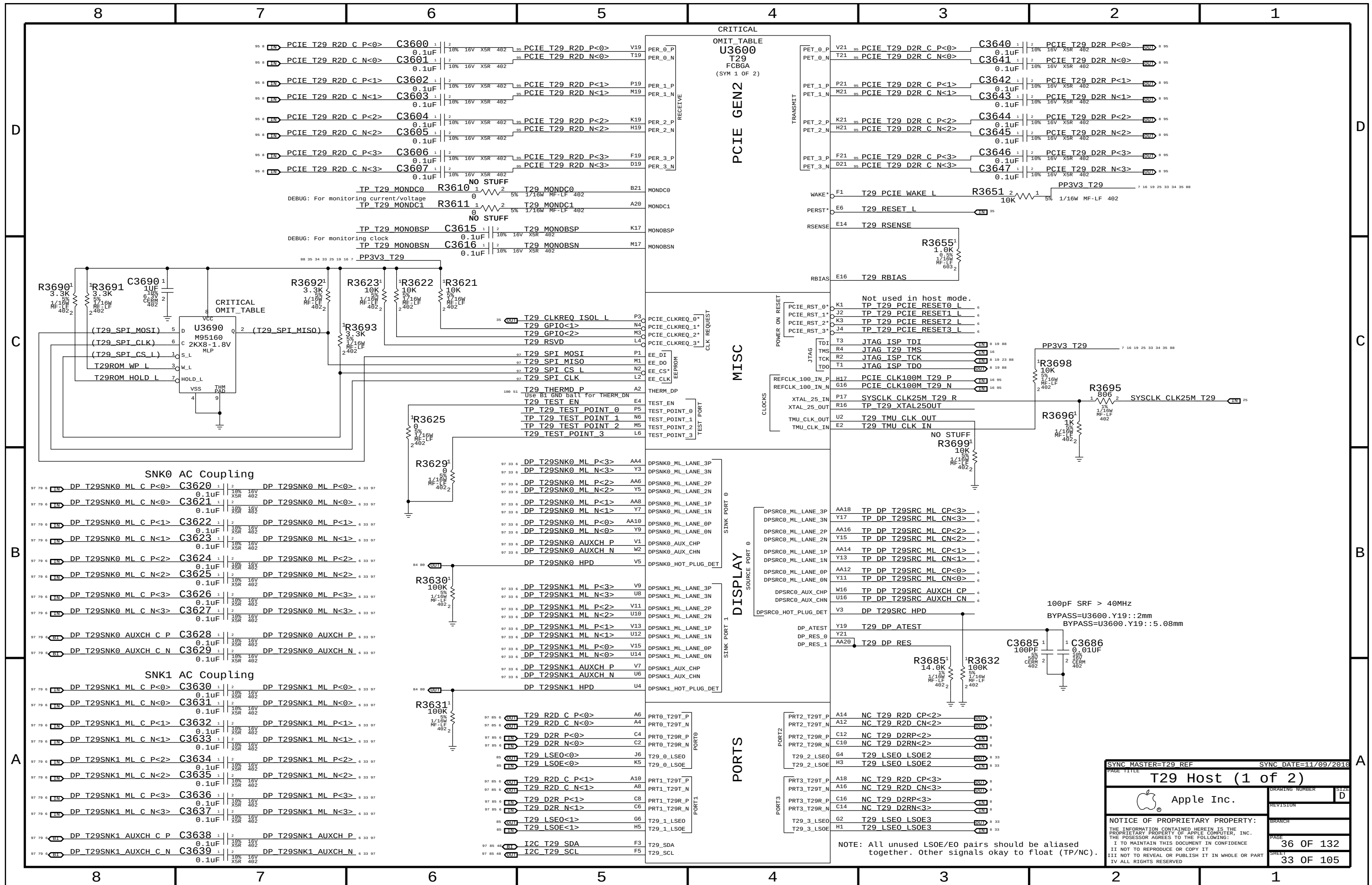
BLUETOOTH

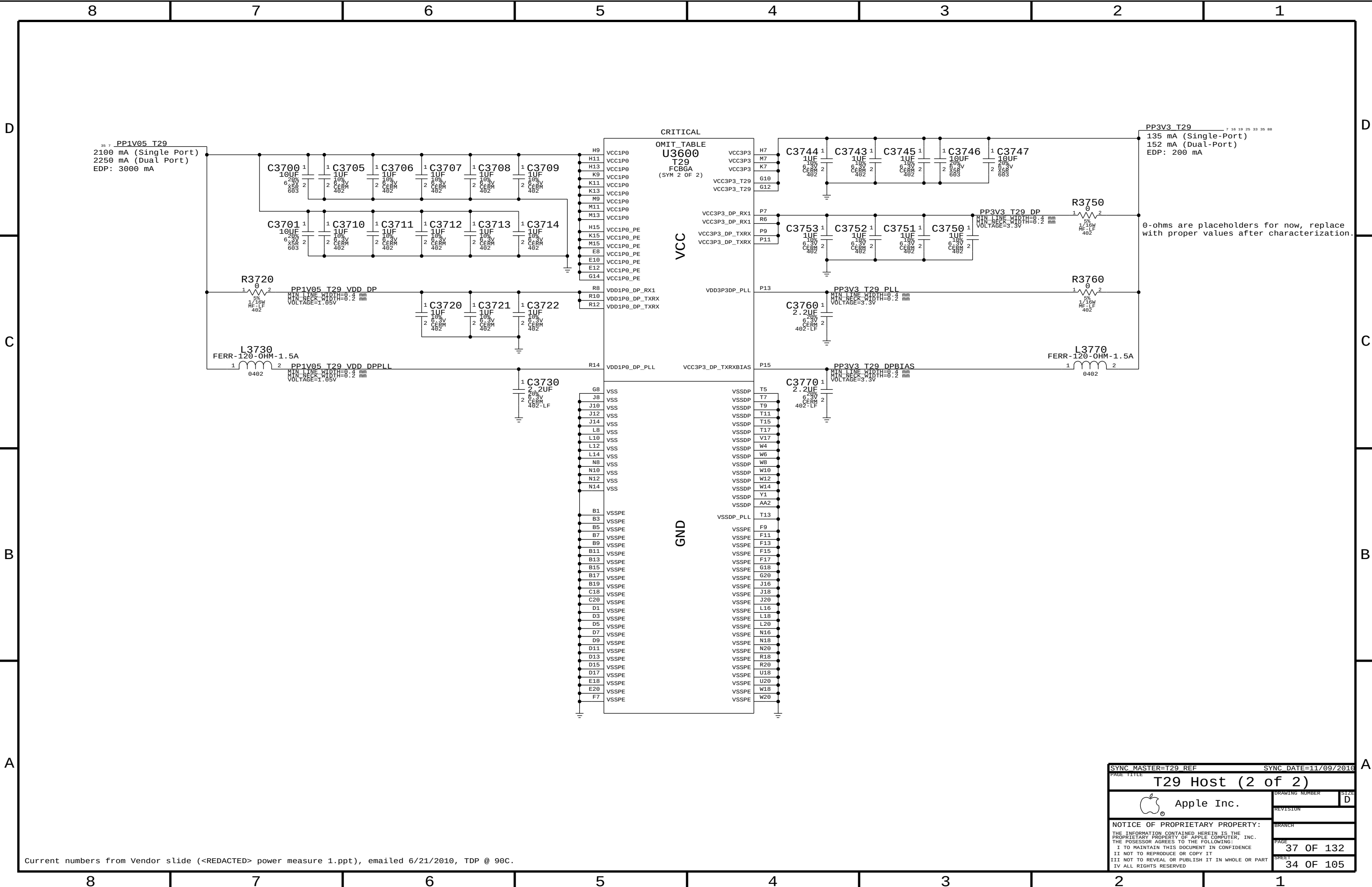
Supervisor & CLKFREG # Isolation
Delay = 60 ms +/- 20%

518S0816
CRITICAL
J3402
CCR20-6K710S

ALS
CAMERA

SYNC MASTER=K91 MLB		SYNC DATE=10/21/2016	
PAGE TITLE			
X19/ALS/CAMERA		CONNECTOR	
 Apple Inc.		DRAWING NUMBER	SIZE
			D
		REVISION	
		BRANCH	
		PAGE	34 OF 132
		SHEET	31 OF 105
NOTICE OF PROPRIETARY PROPERTY: THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE COMPUTER, INC. THE POSSESSOR AGREES TO THE FOLLOWING: I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE II NOT TO REPRODUCE OR COPY IT III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART IV ALL RIGHTS RESERVED			





Current numbers from Vendor slide (<REDACTED> power measure 1.ppt), emailed 6/21/2010, TDP @ 90C.

SYNC MASTER=T29 REF

SYNC DATE=11/09/2010

T29 Host (2 of 2)

Apple Inc.

NOTICE OF PROPRIETARY PROPERTY:

THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE COMPUTER, INC. THE POSSESSOR AGREES TO THE FOLLOWING:
I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE
II NOT TO REPRODUCE OR COPY IT
III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART
IV ALL RIGHTS RESERVED

DRAWING NUMBER

REVISION

BRANCH

PAGE

SHEET

SIZE

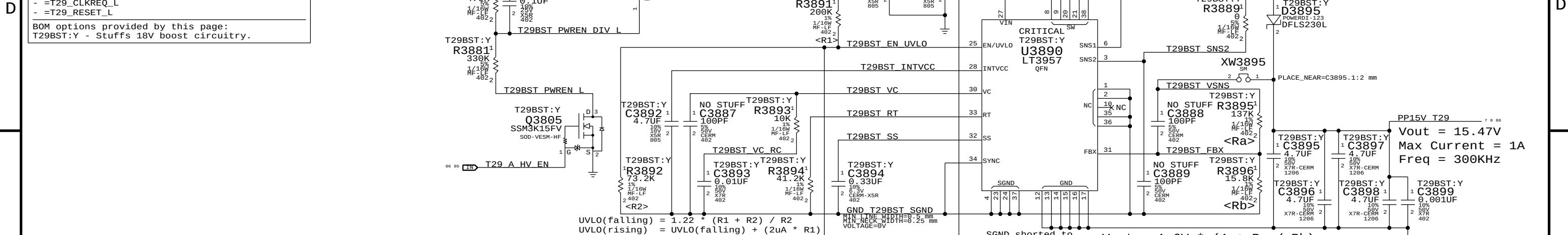
D

37 OF 132

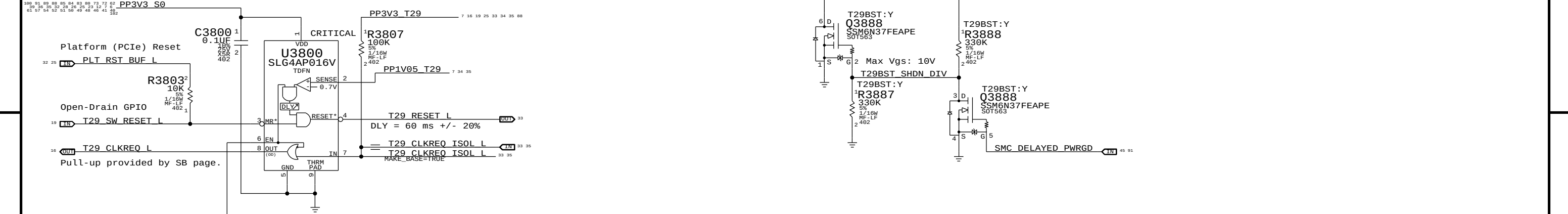
34 OF 105

8	7	6	5	4	3	2	1
---	---	---	---	---	---	---	---

T29BST:Y - Stuffs 18V boost circuitry.



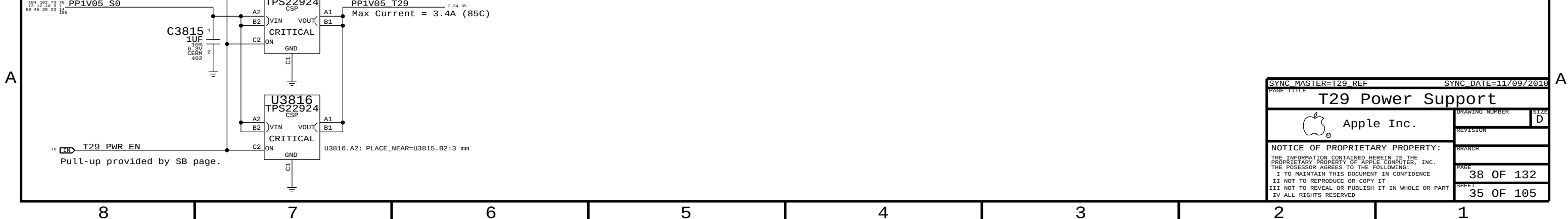
C	Supervisor & CLKREQ# Isolation	UVLO = 4.55V (falling), 4.95 (rising)	Some shipped to GND inside package, no XW necessary. $V_{out} = 1.6V * (1 + R_a / R_b)$		C
---	--------------------------------	---------------------------------------	--	--	---



B	3.3V T29 Switch	B
---	-----------------	---



	1.05V T29 Switch	
	U3815	

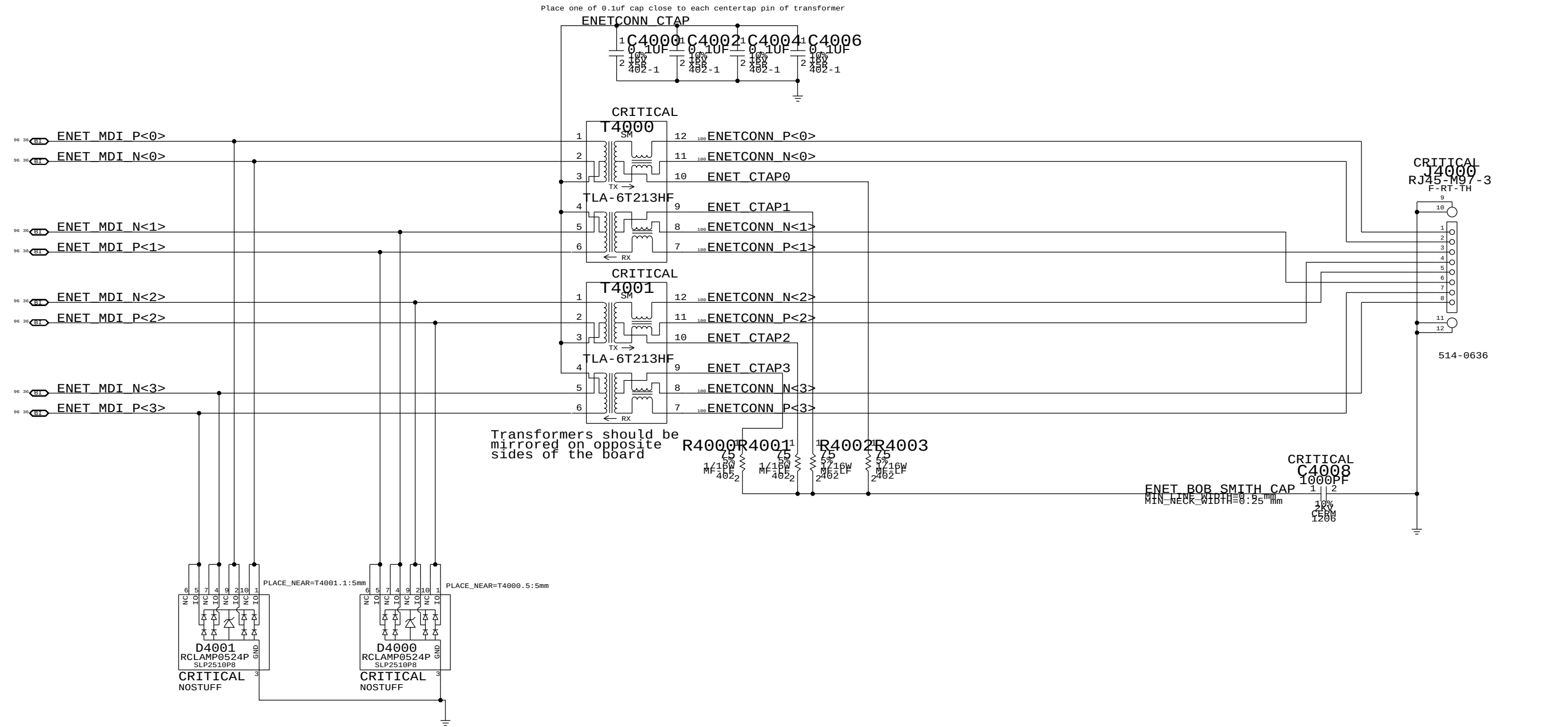


Page Notes

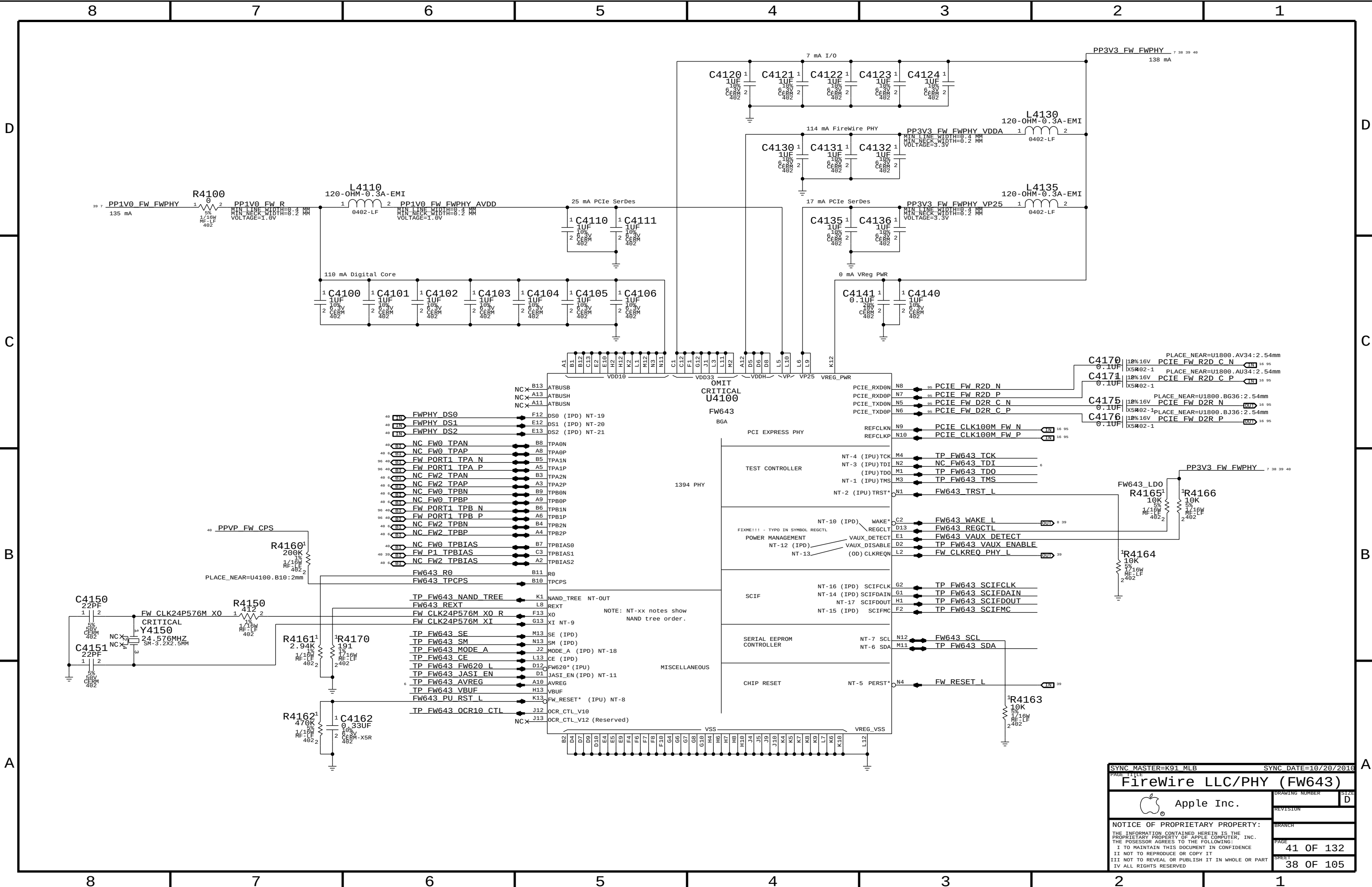
Power aliases required by this page:
(NONE)

Signal aliases required by this page:
(NONE)

BOM options provided by this page:
(NONE)



SYNC_MASTER=K92_ERIC		SYNC_DATE=08/24/2016	
PAGE TITLE			
Ethernet Connector			
 Apple Inc.		DRAWING NUMBER	SIZE
NOTICE OF PROPRIETARY PROPERTY: THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE COMPUTER, INC. THE POSSESSOR AGREES TO THE FOLLOWING: I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE II NOT TO REPRODUCE OR COPY IT III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART IV ALL RIGHTS RESERVED		D	D
		REVISION	
		BRANCH	
		PAGE	
		40 OF 132	
		SHEET	
		37 OF 105	



Page Notes

Power aliases required by this page:

- =PPBUS_S5_FWPWRSW (FW VP FET Input)
- =PPBUS_FW_FET (FW VP FET Output)
- =PP3V3_FW_P3V3FWFET (3.3V FET Input)
- =PP3V3_FW_FET (3.3V FET Output)
- =PP3V3_FW_FWPHY (PHY 3.3V Power)
- =PP3V3_S0_FWLATEVG
- =PP3V3_S0_FWPWRCTL
- =PP1V05_S0_FWPWRCTL (5KPD Bias Rail)
- =PP1V05_FW_P1V0FWFET (1.0V FET Input)
- =PP1V0_FW_FET_R (1.0V FET Output)
- =PP1V0_FW_FWPHY (PHY 1.0V)

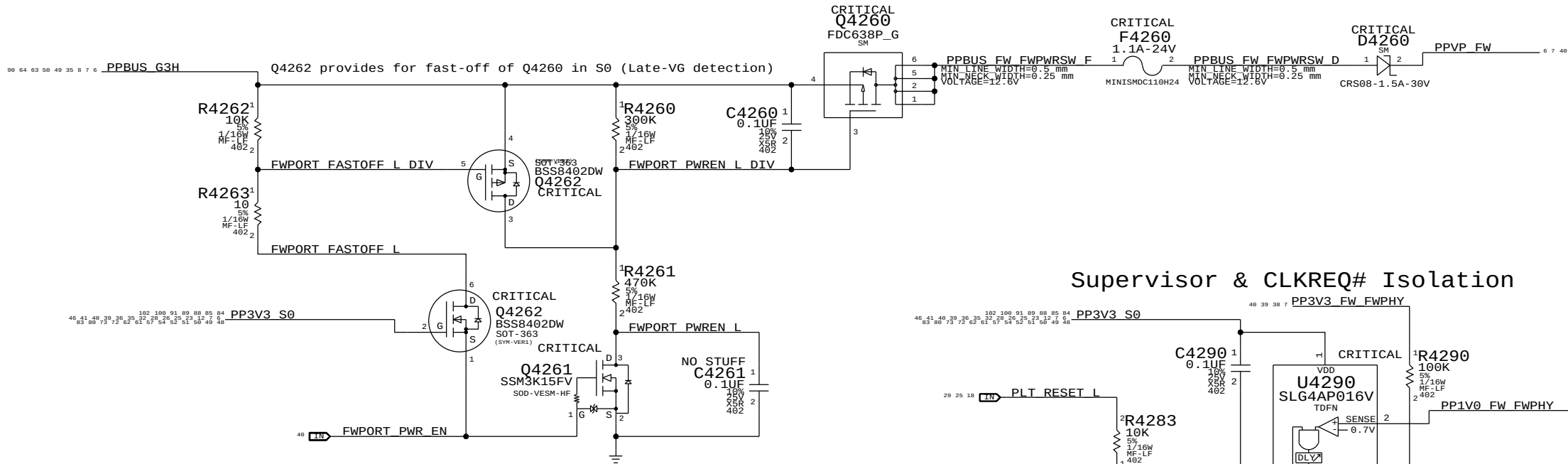
Signal aliases required by this page:

- =FW_CLKREQ_L
- =FW_PME_L

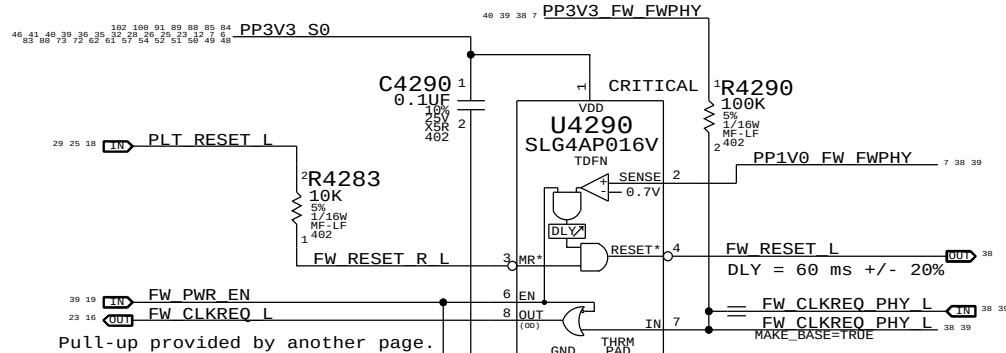
BOM options provided by this page:

(NONE)

FireWire Port Power Switch

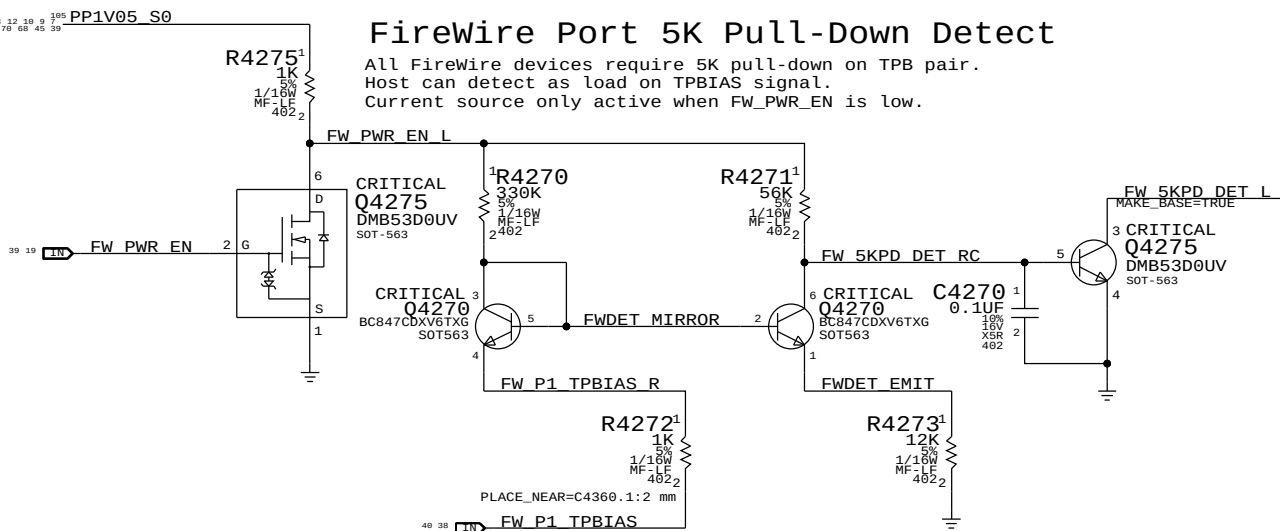


Supervisor & CLKREQ# Isolation



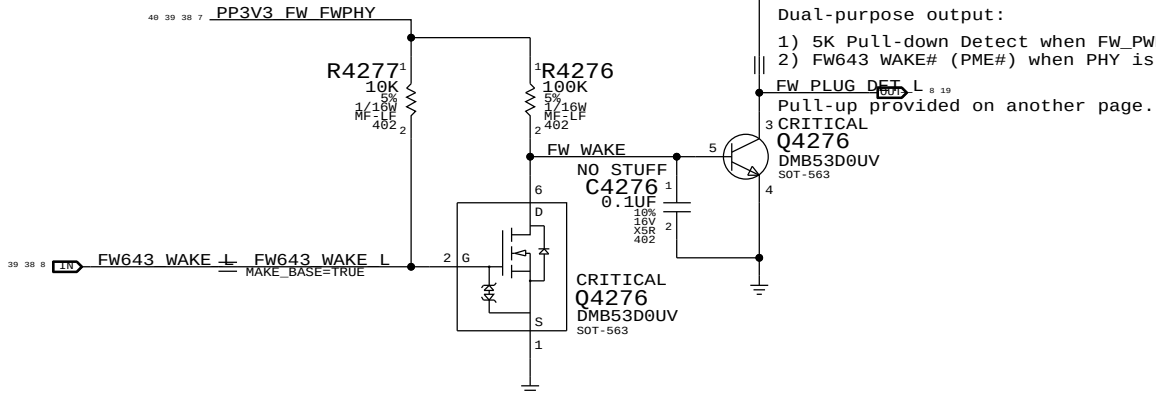
FireWire Port 5K Pull-Down Detect

All FireWire devices require 5K pull-down on TPB pair.
Host can detect as load on TPBIAS signal.
Current source only active when FW_PWR_EN is low.



FireWire PHY WAKE# Support

When PHY is powered, FW_5KPD_DET_L acts as legacy PME# signal.

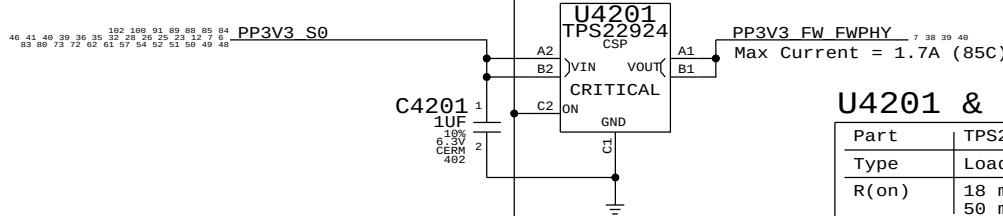


Dual-purpose output:

- 1) 5K Pull-down Detect when FW_PWR_EN is low.
- 2) FW643 WAKE# (PME#) when PHY is powered.

Pull-up provided on another page.

3.3V FW Switch

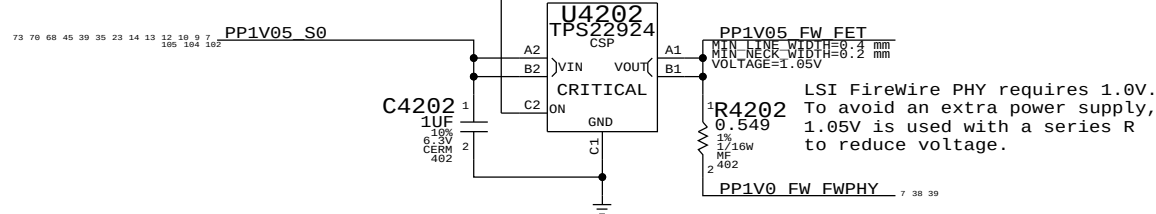


U4201 & U4202

Part	TPS22924C
Type	Load Switch
R(on)	18 mOhm Typ 50 mOhm Max

Max Output: 2A

1.0V FW Switch



SYNC MASTER=K91 MLB		SYNC DATE=10/20/2016	
PAGE 17171		FireWire Port & PHY Power	
Apple Inc.		DRAWING NUMBER	SIZE D
NOTICE OF PROPRIETARY PROPERTY: THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE COMPUTER, INC. THE POSSESSOR AGREES TO THE FOLLOWING: I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE II NOT TO REPRODUCE OR COPY IT III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART IV ALL RIGHTS RESERVED		REVISION	BRANCH
		PAGE	42 OF 132
		SHEET	39 OF 105

Page Notes

Power aliases required by this page:

- PPVP_FW_PORT1
- PPVP_FW_PHY_CPS_FET (From Port)
- PPVP_FW_PHY_CPS (To PHY)
- PP3V3_FW_FWPHY
- PP3V3_S0_FWLATEVG

Signal aliases required by this page:

- FW_PHY_DS0
- FW_PHY_DS1
- FW_PHY_DS2

NOTE: This page is expected to contain the necessary aliases to map the FireWire TPA/TPB pairs to their appropriate connectors and/or to properly terminate unused signals.

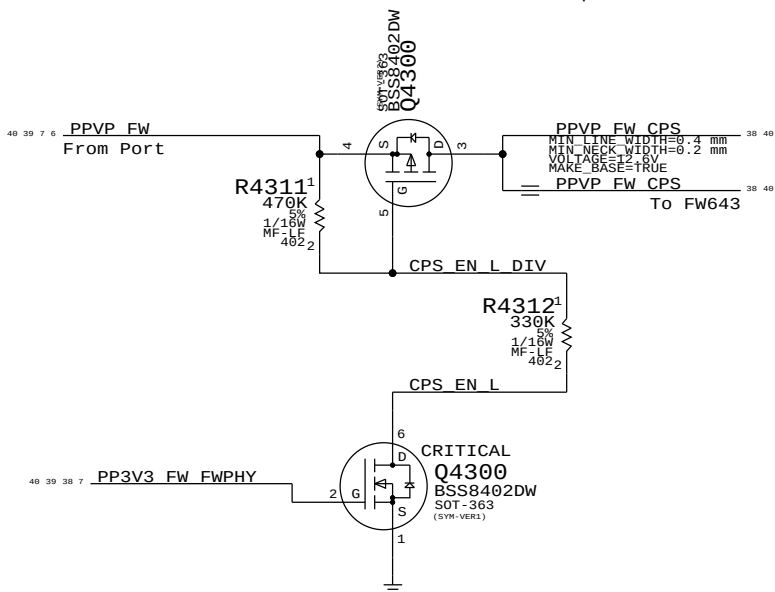
BOM options provided by this page:

(NONE)

1394b implementation based on Apple FireWire Design Guide (FWDG 0.6, 5/14/03)

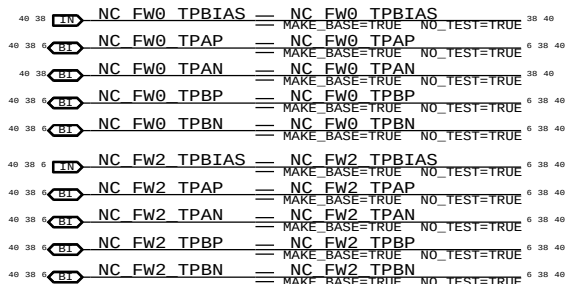
FW643 TPCPS Leakage Protection

FW643 has internal leakage path from TPCPS pin to VDD33.
FET blocks current to TPCPS until VDD33 is powered.



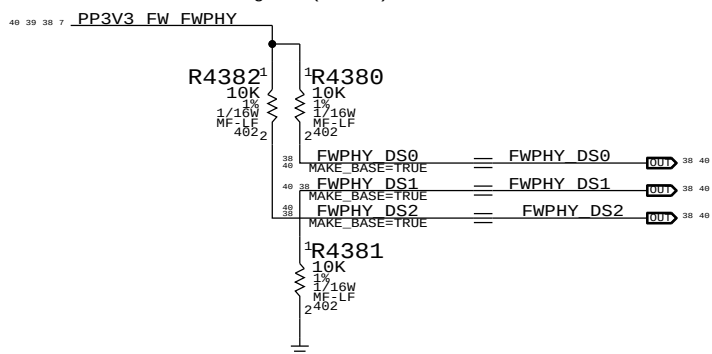
Unused FireWire Ports

Disabled per LSI instructions
(All unused port signals TP/NC)



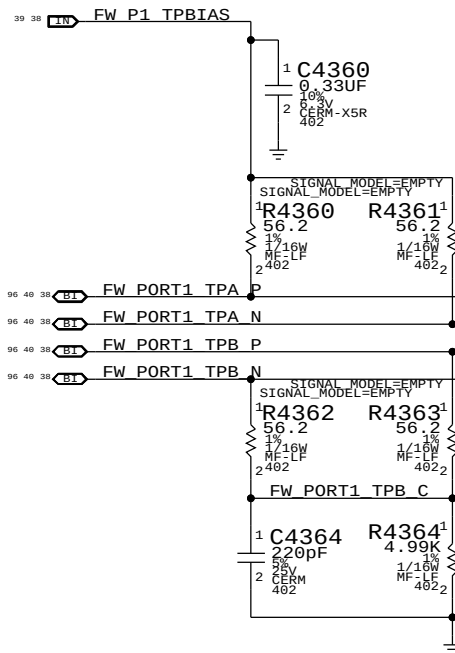
FireWire PHY Config Straps

Configures PHY for:
- Port "1" Bilingual (1394B)

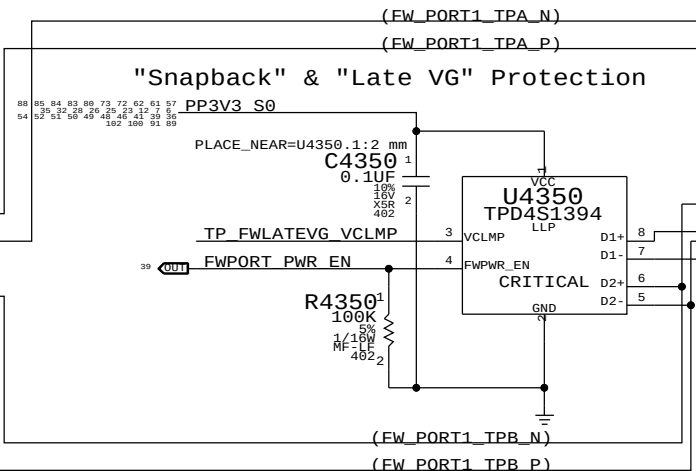


Termination

Place close to FireWire PHY



"Snapback" & "Late VG" Protection

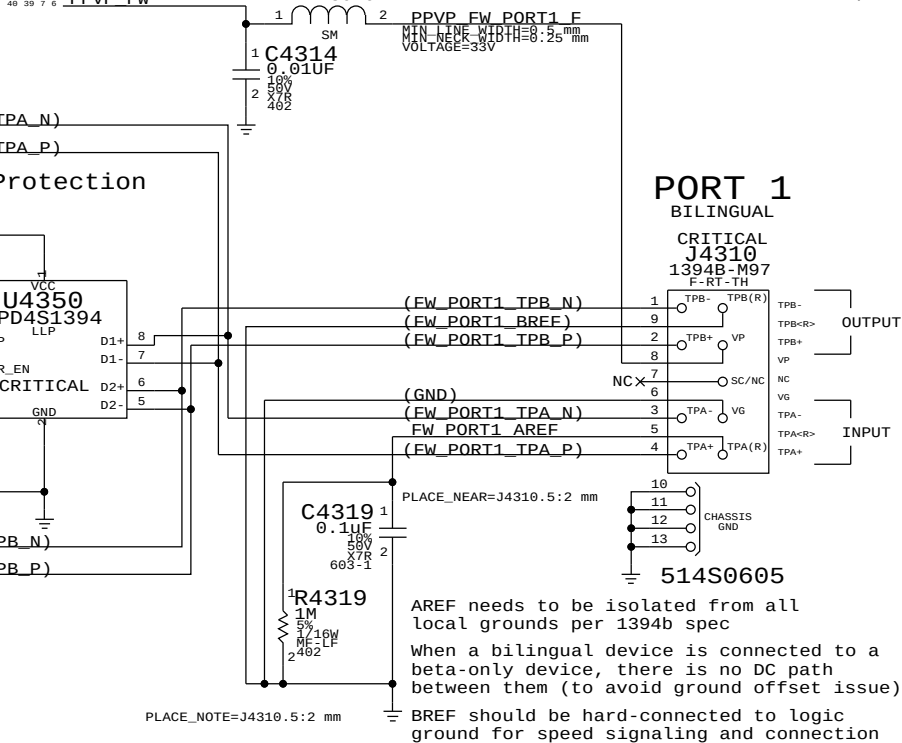


Cable Power

CRITICAL L4310

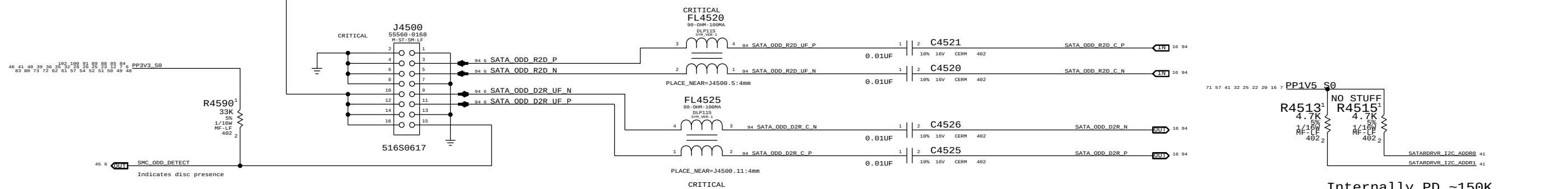
FERR-250-OHM

Note: Trace PPVP_FW_PORT1 must handle up to 5A



SYNC MASTER=K91 MLB		SYNC DATE=07/22/2016	
PAGE TITLE		PAGE	
FireWire Connector		DRAWING NUMBER	
Apple Inc.		SIZE	
NOTICE OF PROPRIETARY PROPERTY:		REVISION	
THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE COMPUTER, INC. THE POSSESSOR AGREES TO THE FOLLOWING:		BRANCH	
I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE		PAGE	
II NOT TO REPRODUCE OR COPY IT		43 OF 132	
III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART		SHEET	
IV ALL RIGHTS RESERVED		40 OF 105	

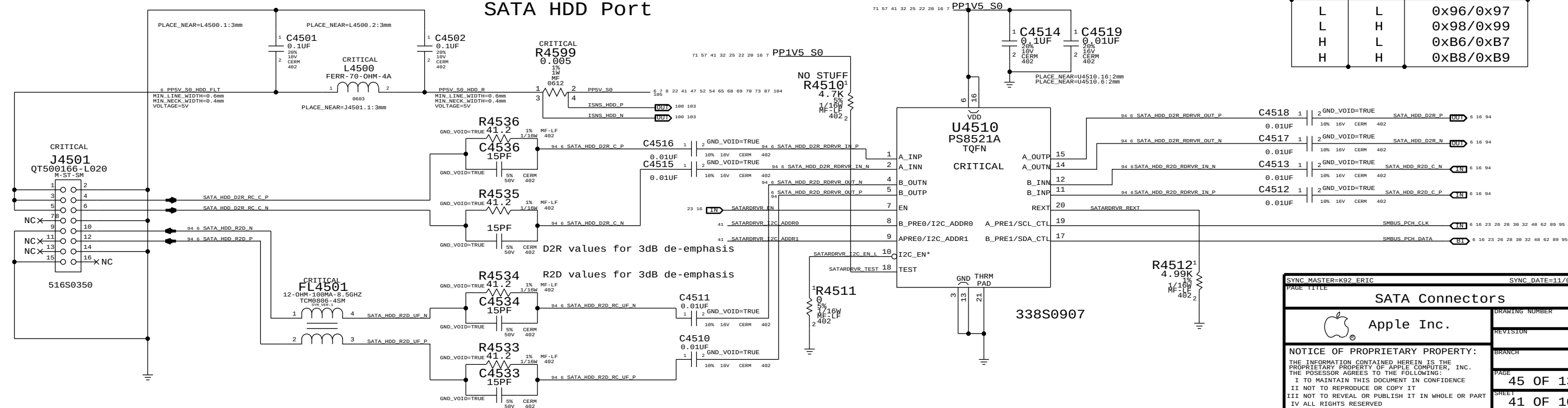
NOTE: 3.3V must be S0 if 5V is S3 or S5 to ensure the drive is unpowered in S3/S5.



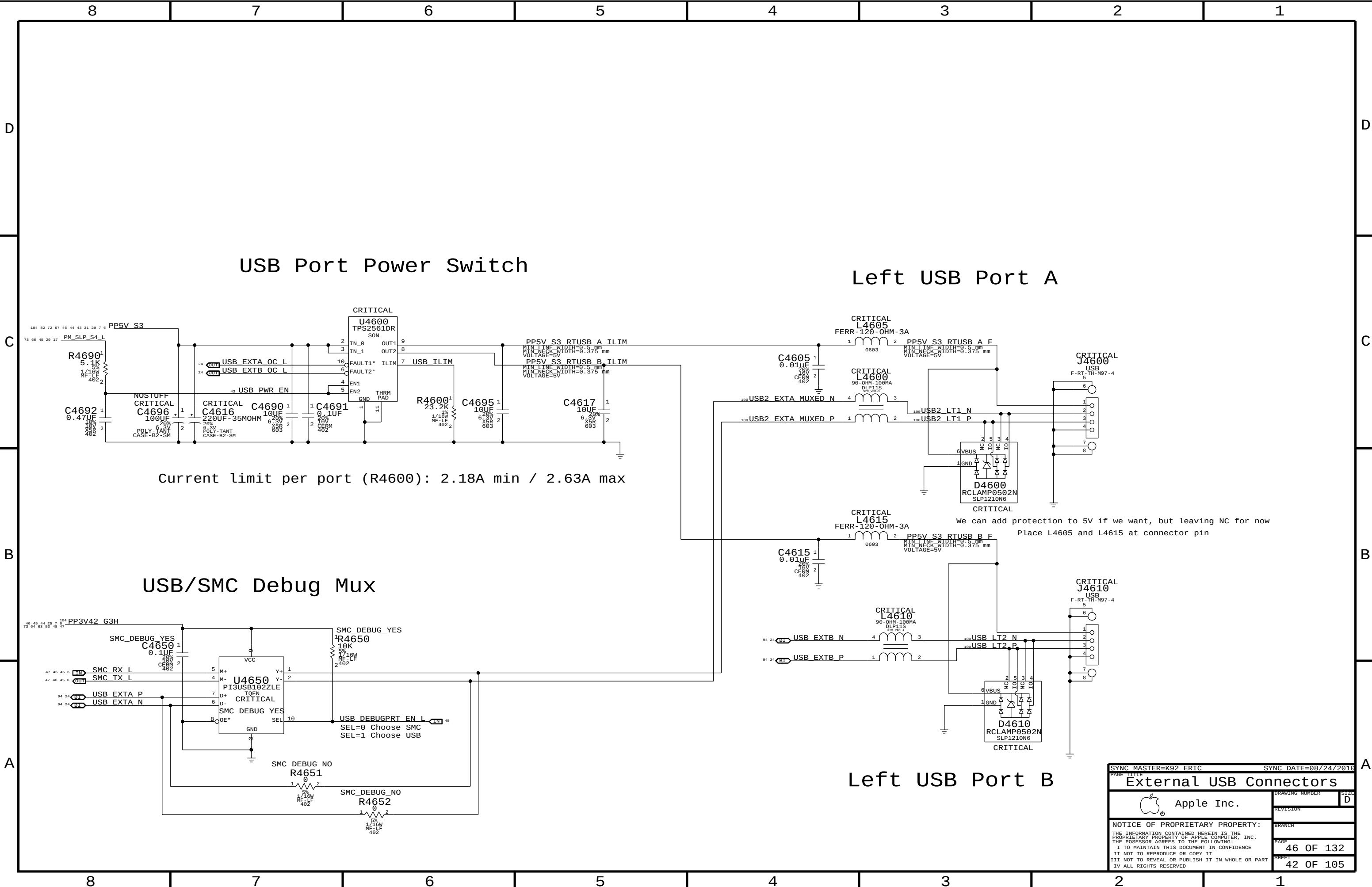
Internally PD ~150K
Write:0xB6 Read:0xB7

ADDR1	ADDR0	Address (R/W)
L	L	0x96/0x97
L	H	0x98/0x99
H	L	0xB6/0xB7
H	H	0xB8/0xB9

41 D2R values for 3dB de-emphasis



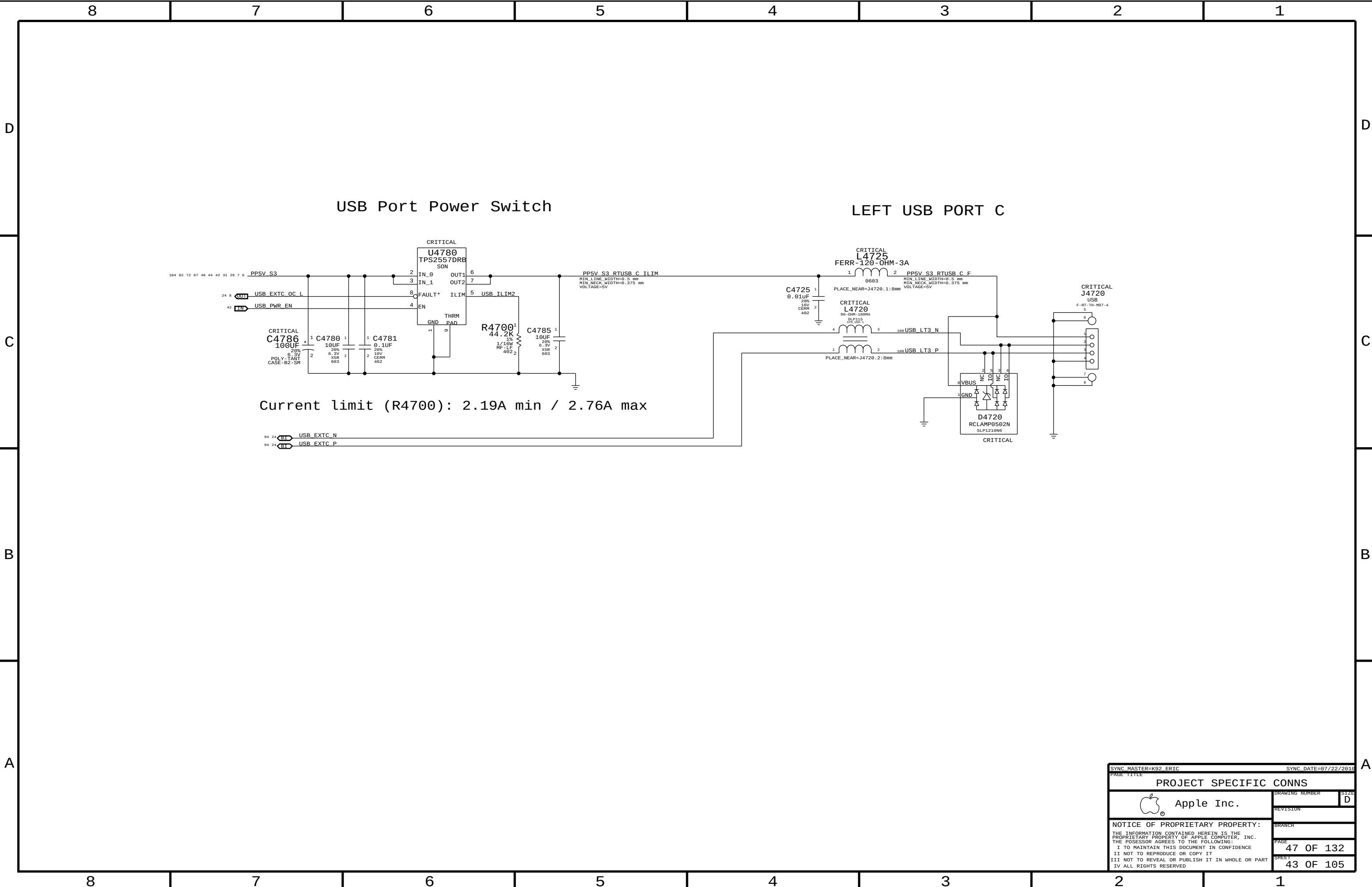
SYNC MASTER-K92 ERIC		SYNC DATE=11/08/2016	
PAGE TITLE			
SATA Connectors			
 Apple Inc.		DRAWING NUMBER	
		SIZE	
		D	
		REVISION	
NOTICE OF PROPRIETARY PROPERTY:		BRANCH	
THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE COMPUTER, INC. THE POSSESSOR AGREES TO THE FOLLOWING: I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE II NOT TO REPRODUCE OR COPY IT III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART IV ALL RIGHTS RESERVED		PAGE	
		45 OF 132	
		SHEET	
		41 OF 105	

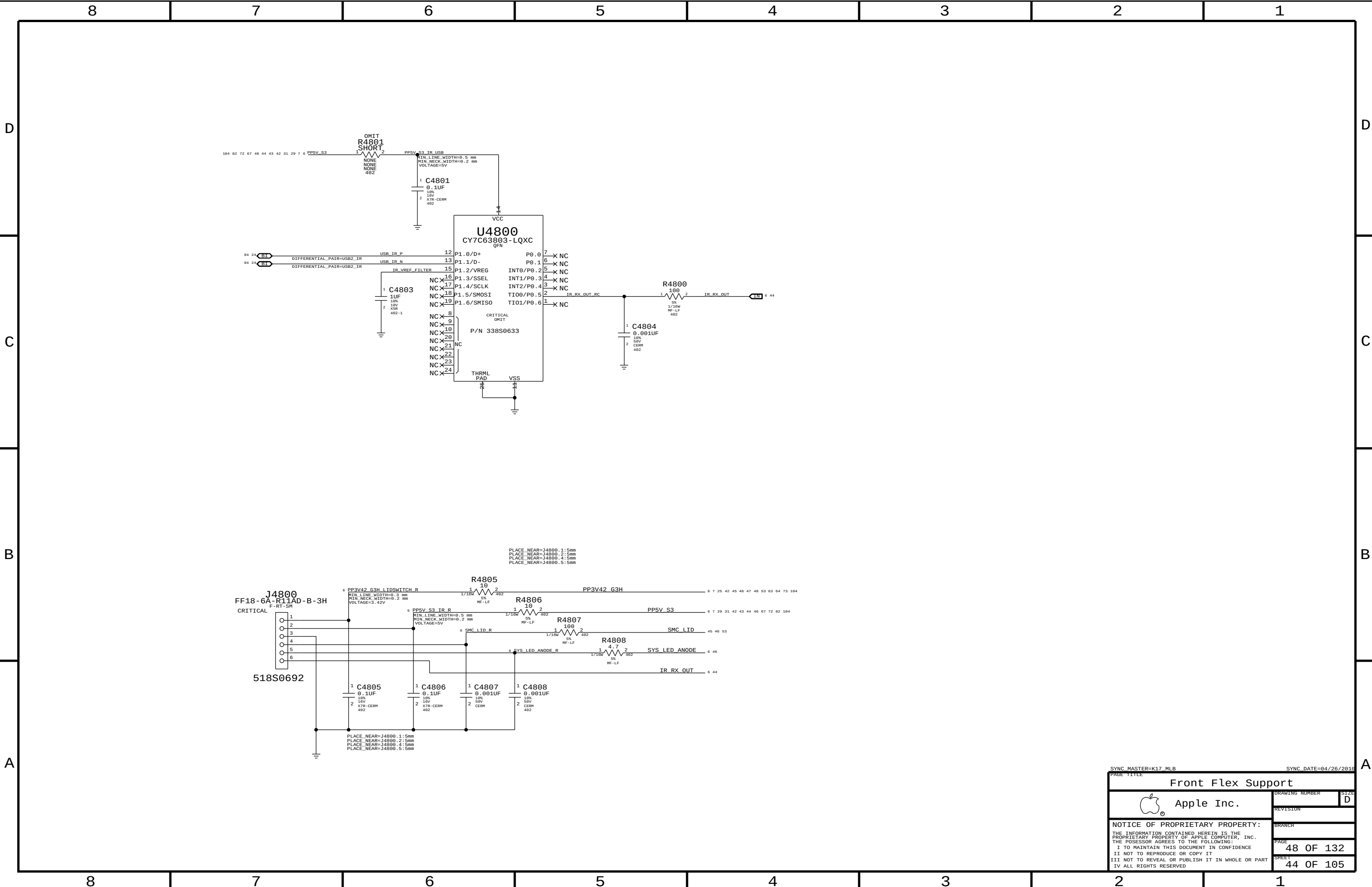


Current limit per port (R4600): 2.18A min / 2.63A max

We can add protection to 5V if we want, but leaving NC for now
Place L4605 and L4615 at connector pin

SYNC MASTER=K92.ERIC		SYNC DATE=08/24/2016	
PAGE TITLE		External USB Connectors	
Apple Inc.		DRAWING NUMBER	SIZE
NOTICE OF PROPRIETARY PROPERTY: THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE COMPUTER, INC. THE POSSESSOR AGREES TO THE FOLLOWING: I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE II NOT TO REPRODUCE OR COPY IT III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART IV ALL RIGHTS RESERVED		REVISION	PAGE
		BRANCH	46 OF 132
		SHEET	42 OF 105





A



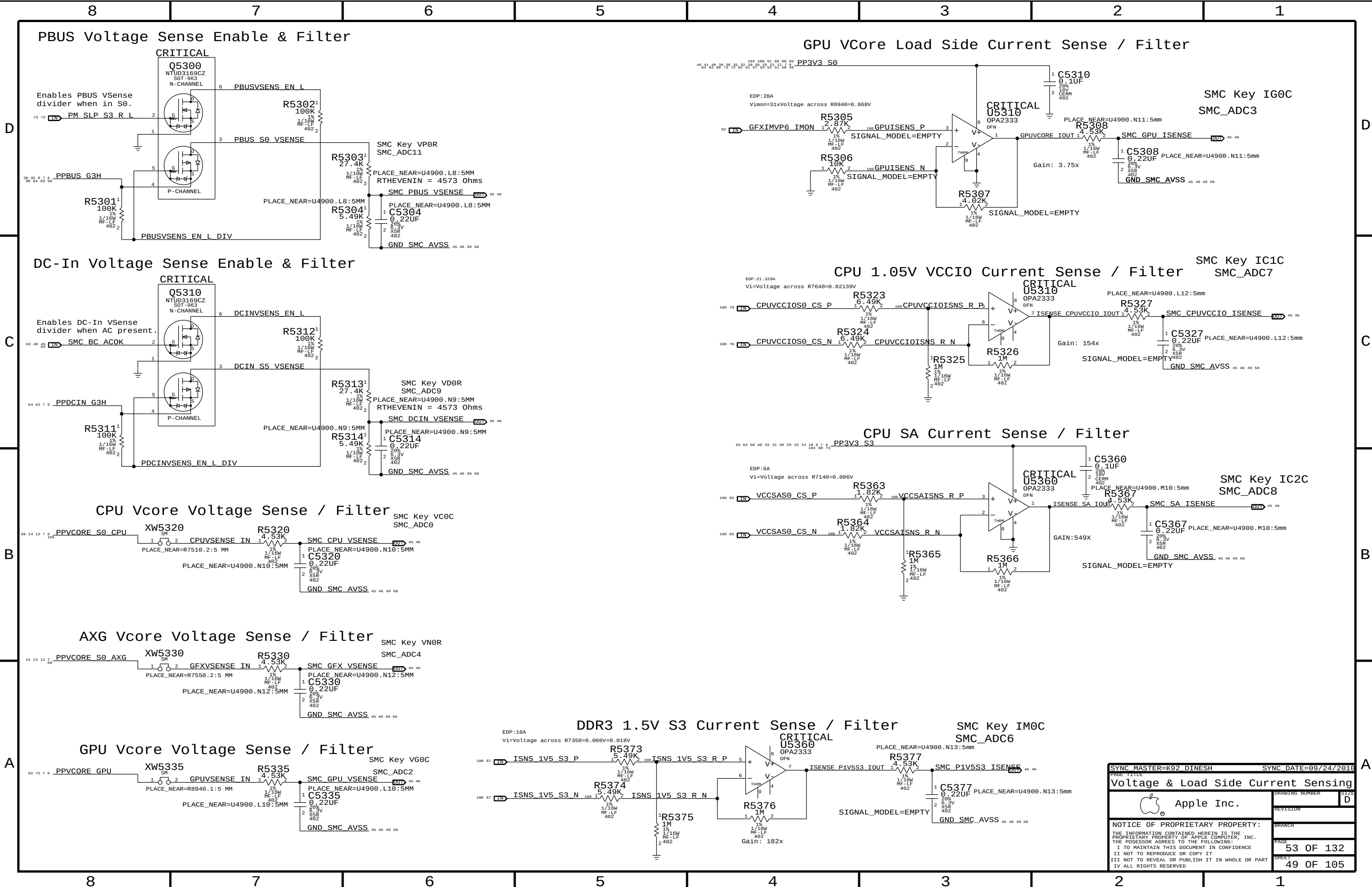
SYNCH MASTER=K91 BEN		SYNCH DATE=07/12/2016	
PAGE TITLE			
SMC			
 Apple Inc.		DRAWING NUMBER	
		SIZE D	
		REVISION	
NOTICE OF PROPRIETARY PROPERTY:		BRANCH	
THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE COMPUTER, INC. THE POSSESSOR AGREES TO THE FOLLOWING: I I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE II NOT TO REPRODUCE OR COPY IT III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART IV ALL RIGHTS RESERVED		PAGE 49 OF 132	
		SHEET 45 OF 105	

D

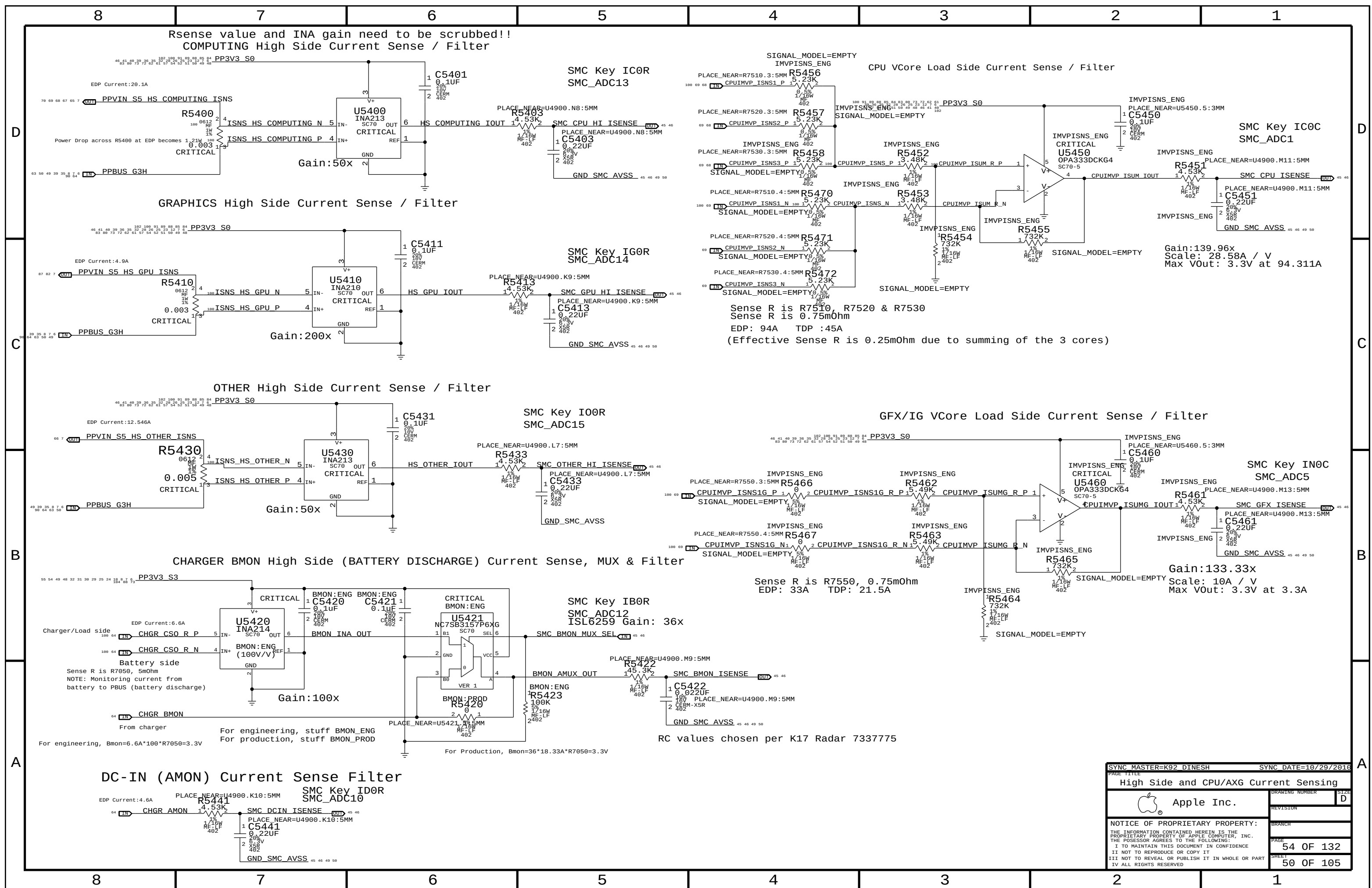


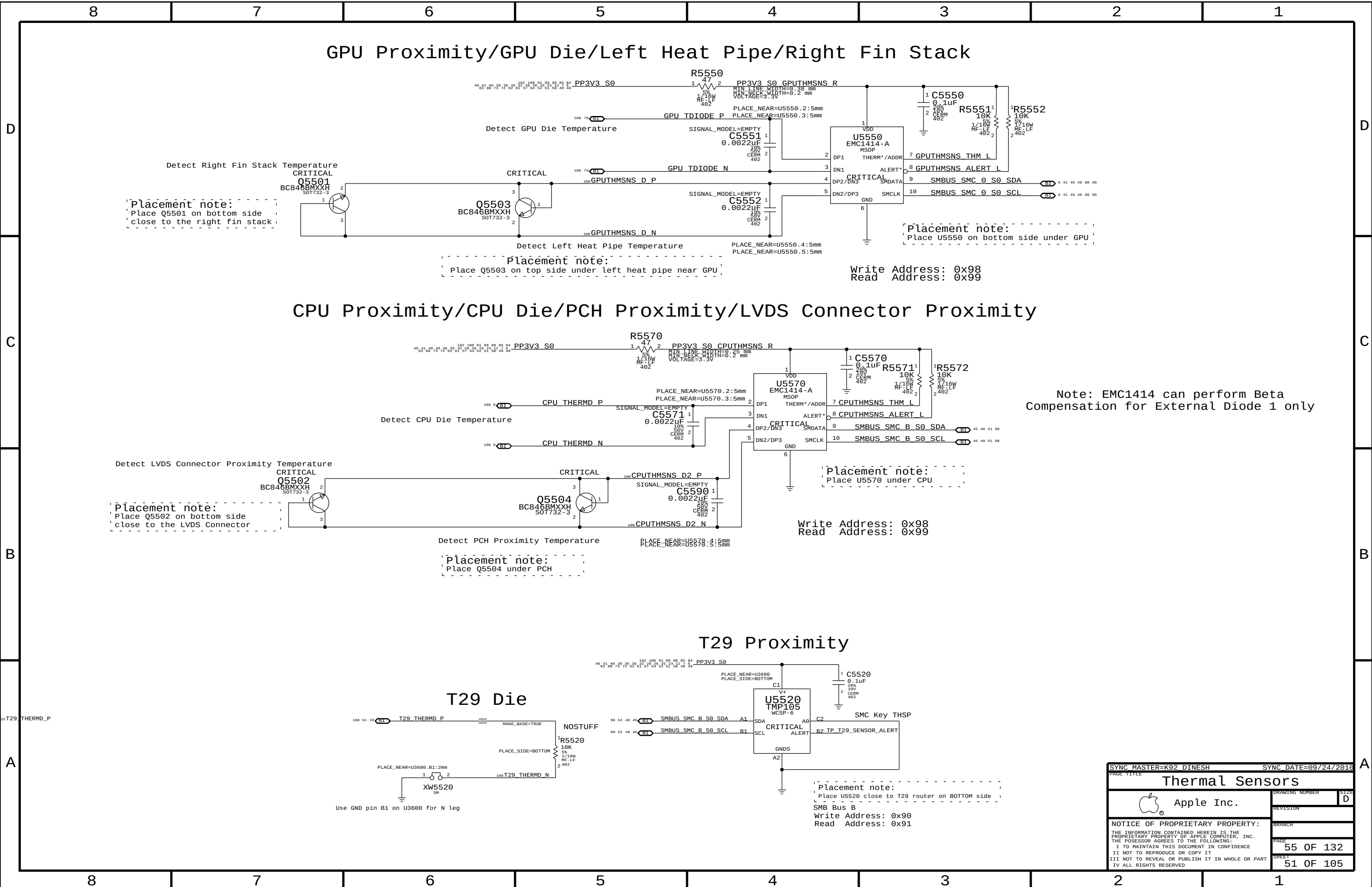
B

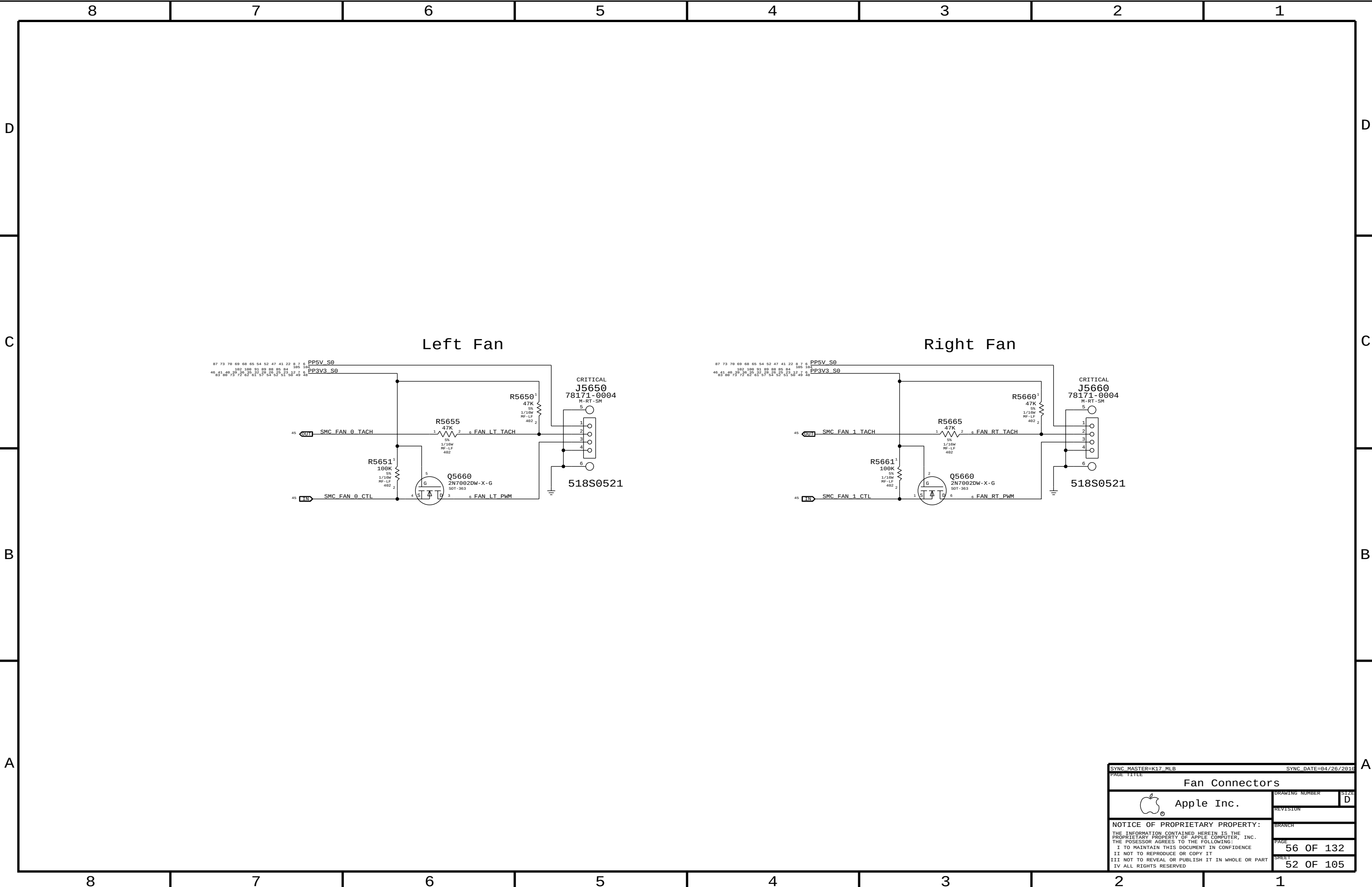
D



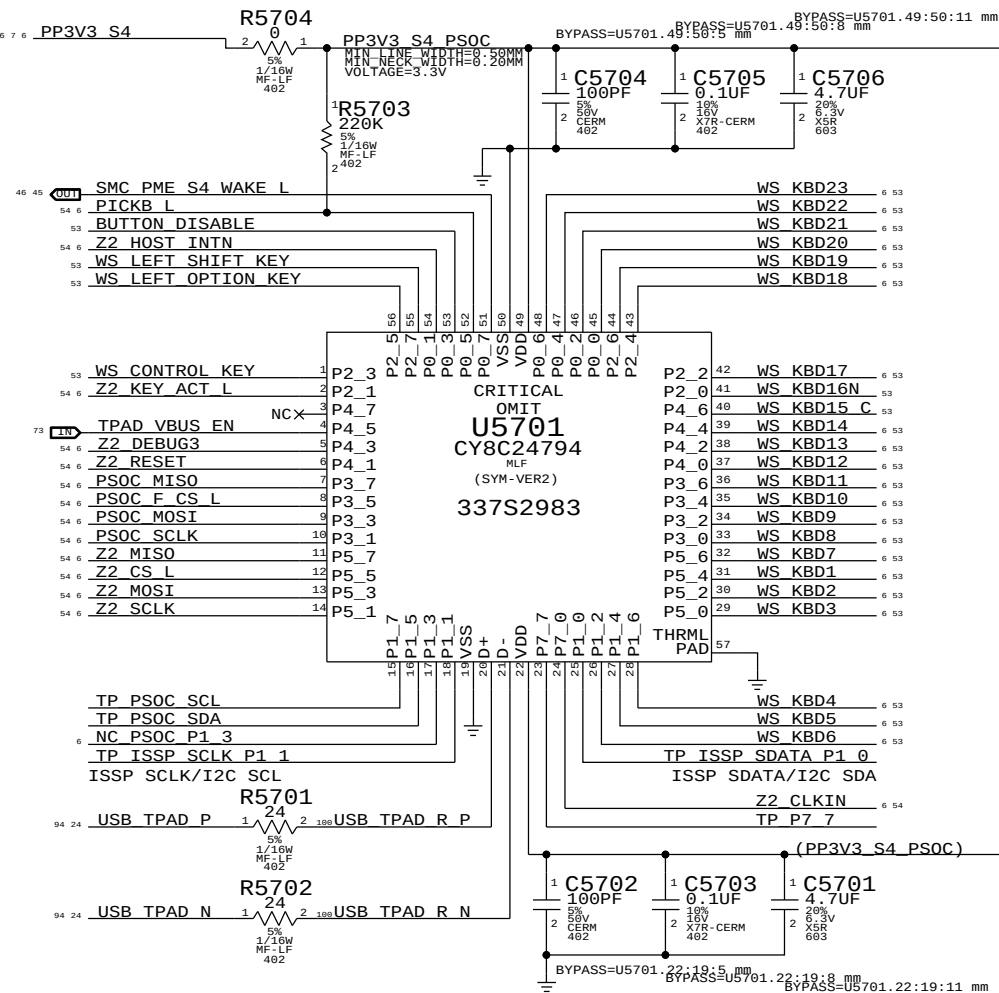
SYNC MASTER=K92 DINESH		SYNC DATE=09/24/2016	
PAGE TITLE		Voltage & Load Side Current Sensing	
 Apple Inc.		DRAWING NUMBER	SIZE
		REVISION	D
NOTICE OF PROPRIETARY PROPERTY: THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE COMPUTER, INC. THE POSSESSOR AGREES TO THE FOLLOWING: I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE II NOT TO REPRODUCE OR COPY IT III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART IV ALL RIGHTS RESERVED		BRANCH	
		PAGE	53 OF 132
		SHEET	49 OF 105



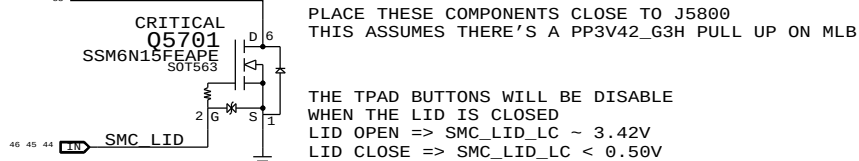




- USB INTERFACES TO MLB
- SPI HOST TO Z2
- TRACKPAD PICK BUTTONS
- KEYBOARD SCANNER

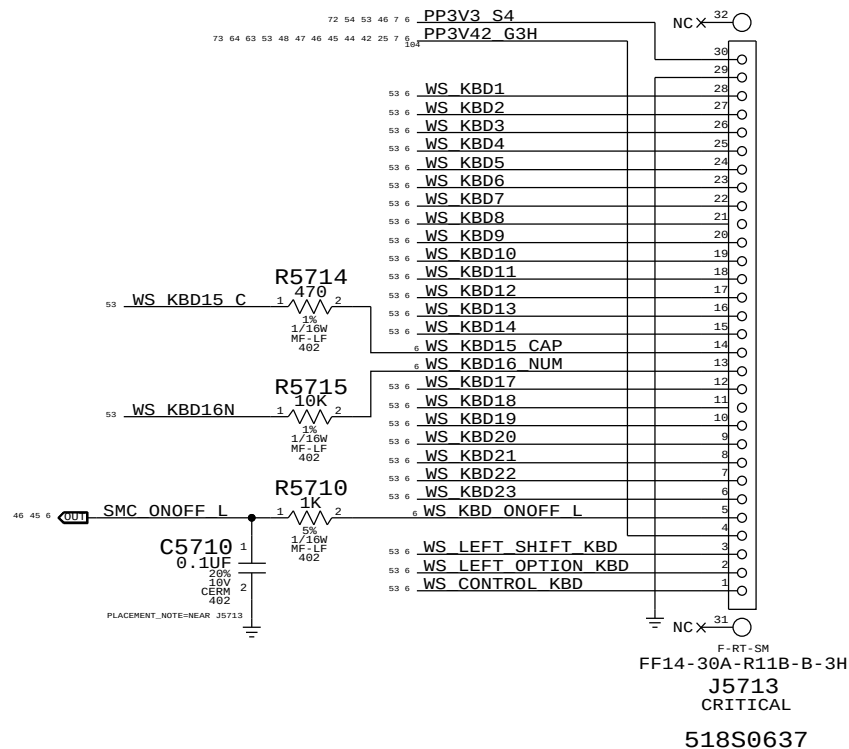


53 BUTTON DISABLE

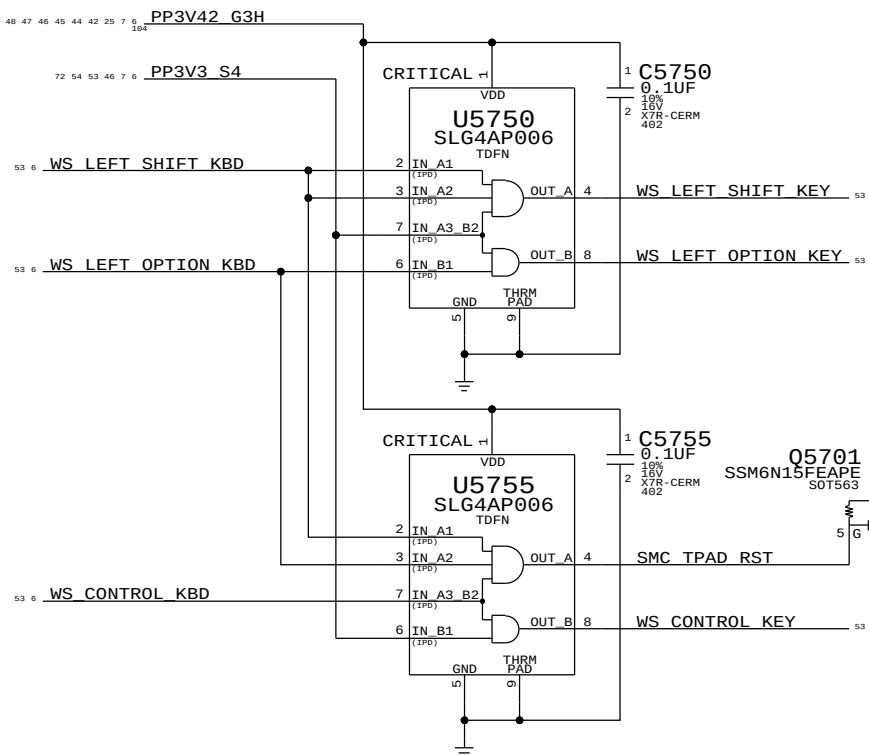


IC	PIN NAME	CURRENT	R_SNS	V_SNS	POWER
TMP102	V+	10UA 80UA	2.55 KOHM	0.0255 V 0.204 V	0.255E-6 W 16.32E-6 W
3V3 LDO	VDD VOUT	60MA (MAX) 60MA (MAX)	10 OHM 0.2 OHM	0.6 V 0.012 V	36E-3 W 0.72E-3 W
PSOC	VDD	8MA (TYP) 14MA (MAX)	1.5 OHM	0.012 V 0.021 V	96E-6 W 294E-6 W
18V BOOSTER	VIN	4MA (MAX)	4.7 OHM	0.0188 V	75.2E-6 W

72 54 53 46 7 6 PP3V3 S4
73 64 63 53 48 47 46 45 44 42 25 7 6 PP3V42 G3H NC X³²



Left shift, option & control keys combined with power button cause SMC RESET# assertion. Keys ANDed with PSOC power to isolate when PSOC is not powered.



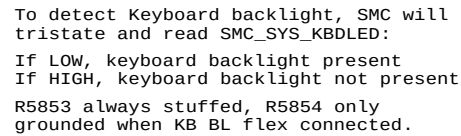
SYNC MASTER=K92 ERIC		SYNC DATE=10/11/2016	
PAGE TITLE			
WELLSPRING 1			
 Apple Inc.		DRAWING NUMBER	SIZE
		REVISION	D
NOTICE OF PROPRIETARY PROPERTY:			
THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE COMPUTER, INC. THE POSSESSOR AGREES TO THE FOLLOWING:		BRANCH	
I. TO MAINTAIN THIS DOCUMENT IN CONFIDENCE		PAGE	
II. NOT TO REPRODUCE OR COPY IT		57 OF 132	
III. NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART		SHEET	
IV. ALL RIGHTS RESERVED		53 OF 105	

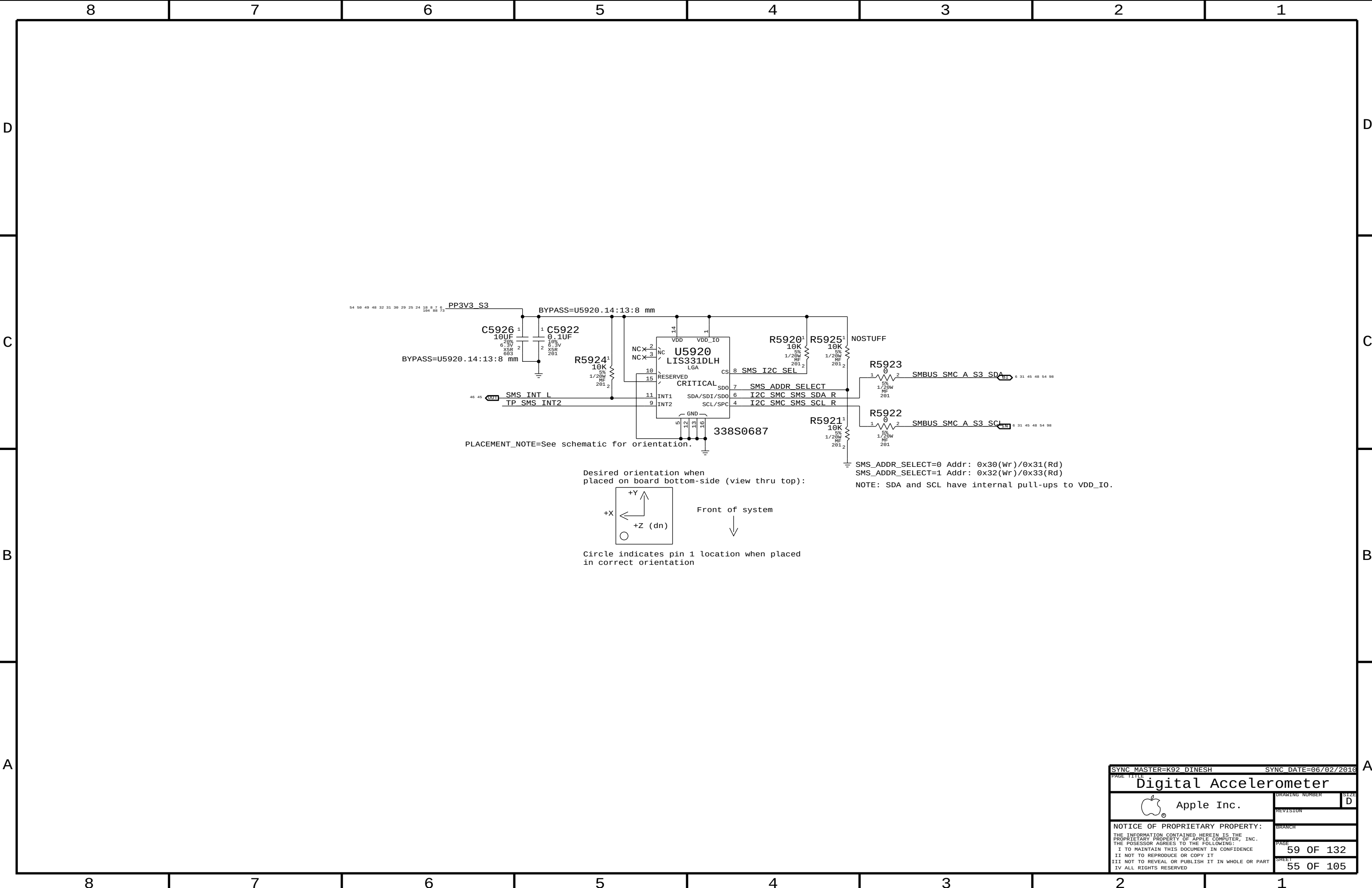
BOOSTER DESIGN CONSIDERATION:

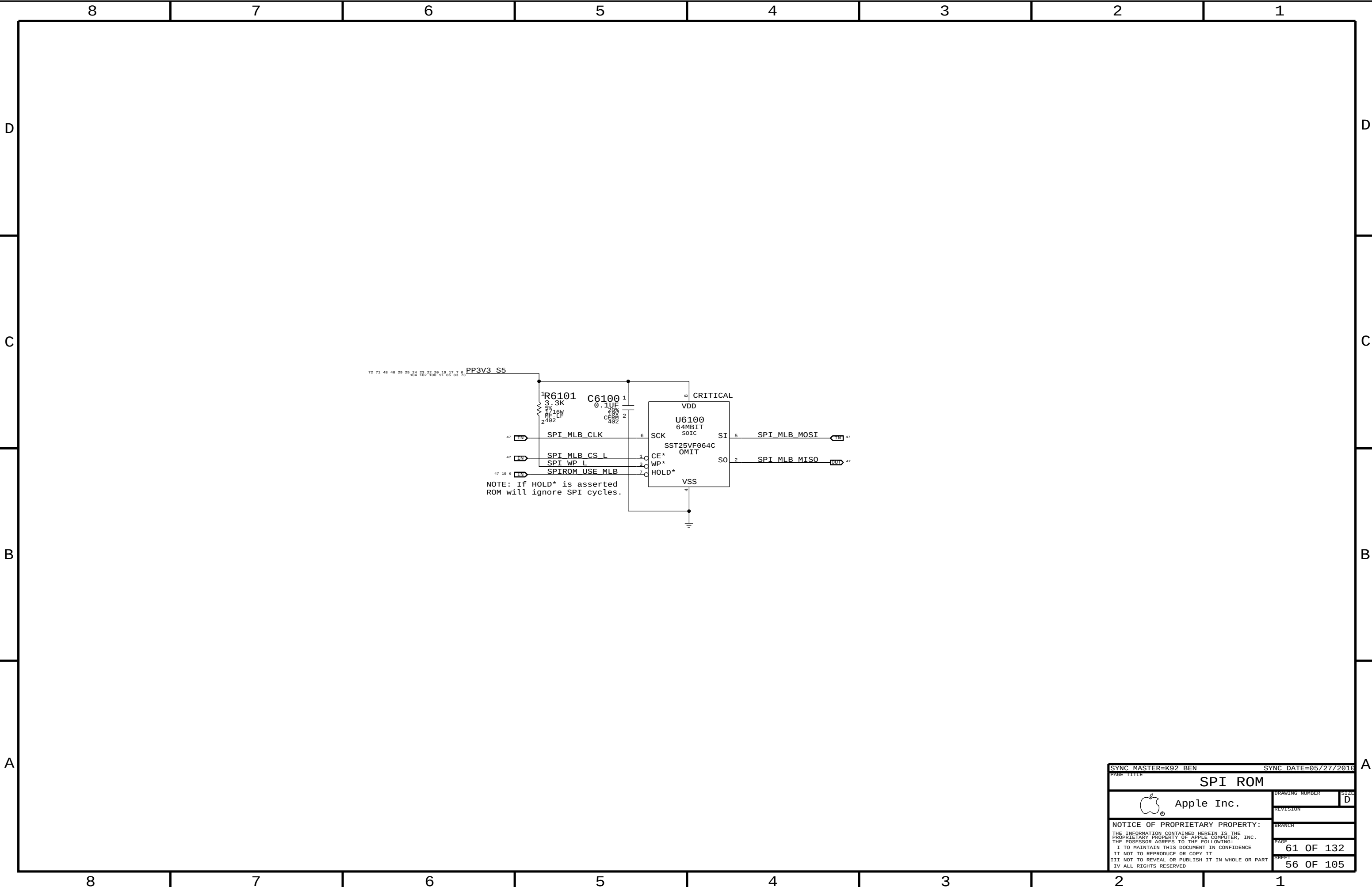
- POWER CONSUMPTION
- DROOP LINE REGULATION
- RIPPLE TO MEET ERS
- 100-300 KHZ CLEAN SPECTRUM
- STARTUP TIME LESS THAN 2MS
- R5812,R5813,C5818 MODIFIED

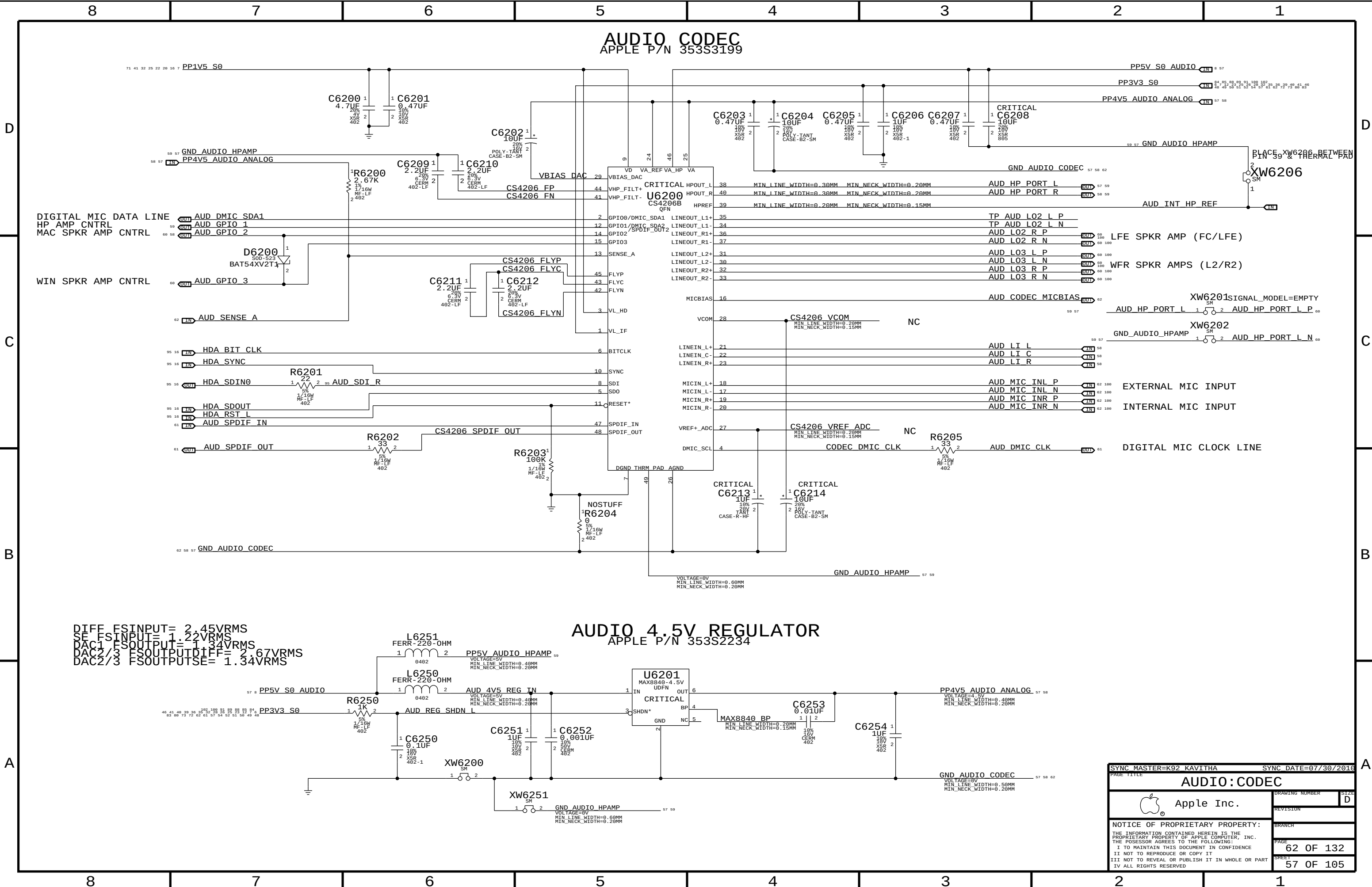


Keyboard Backlight Connector



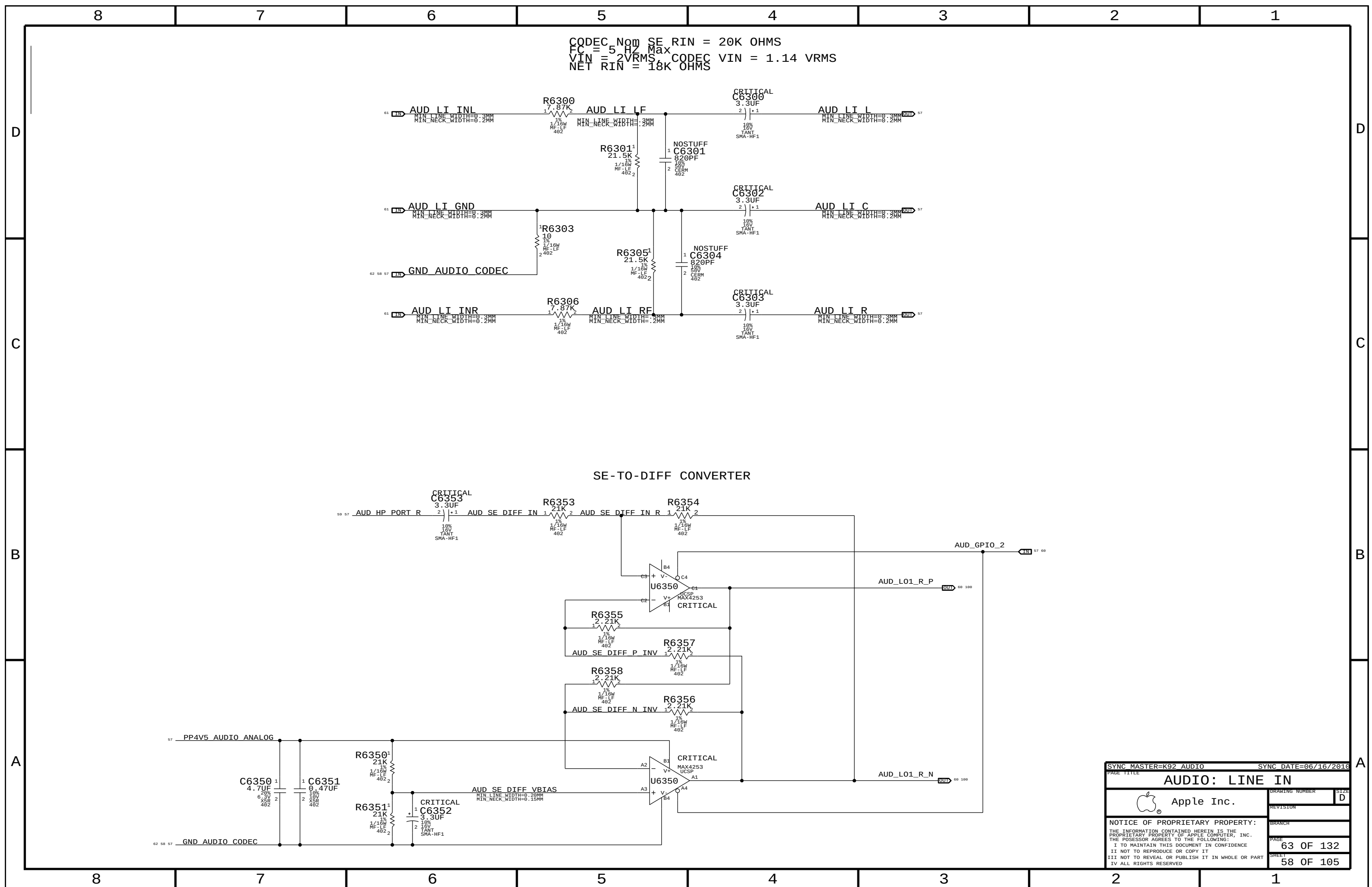


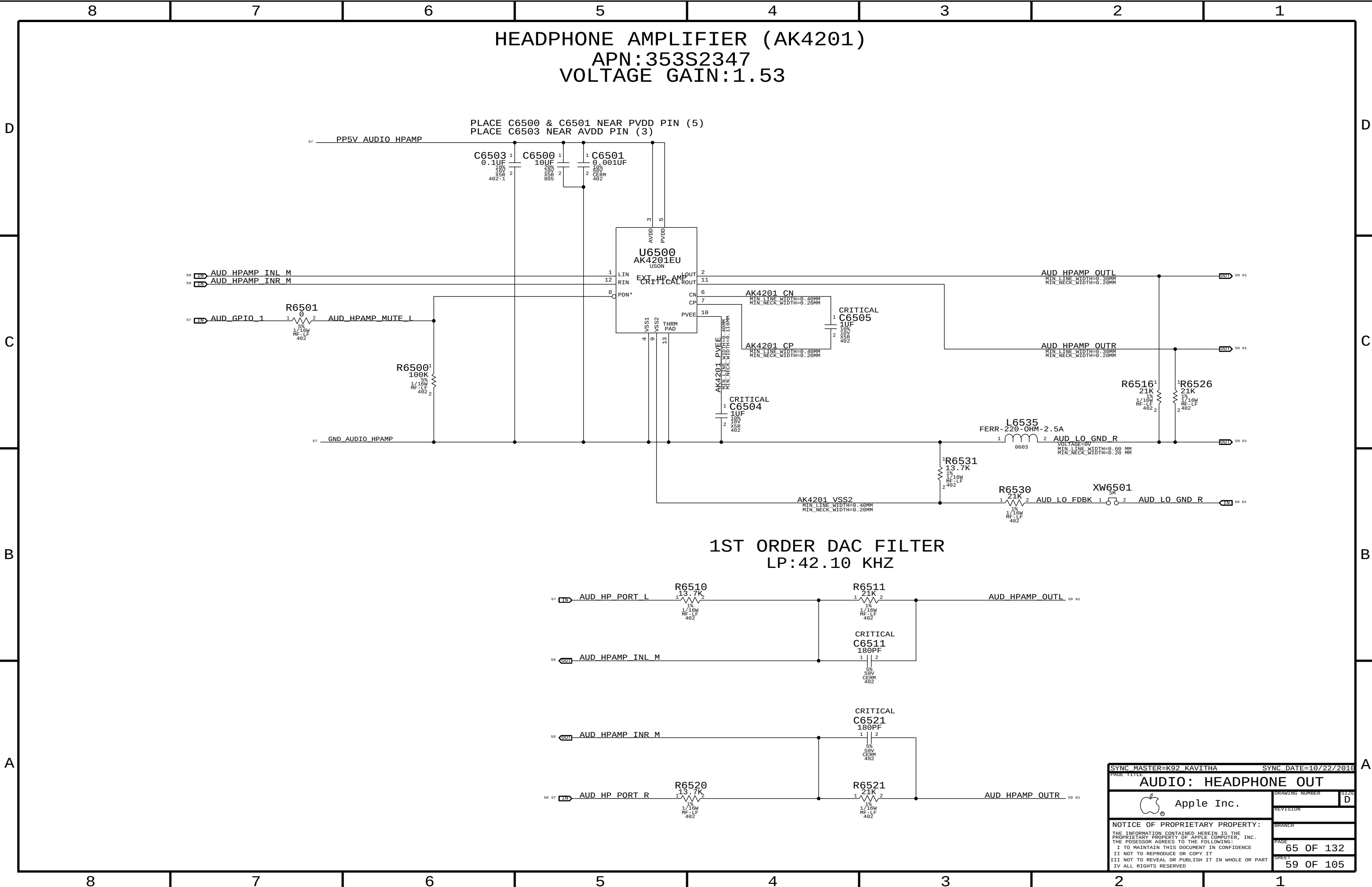


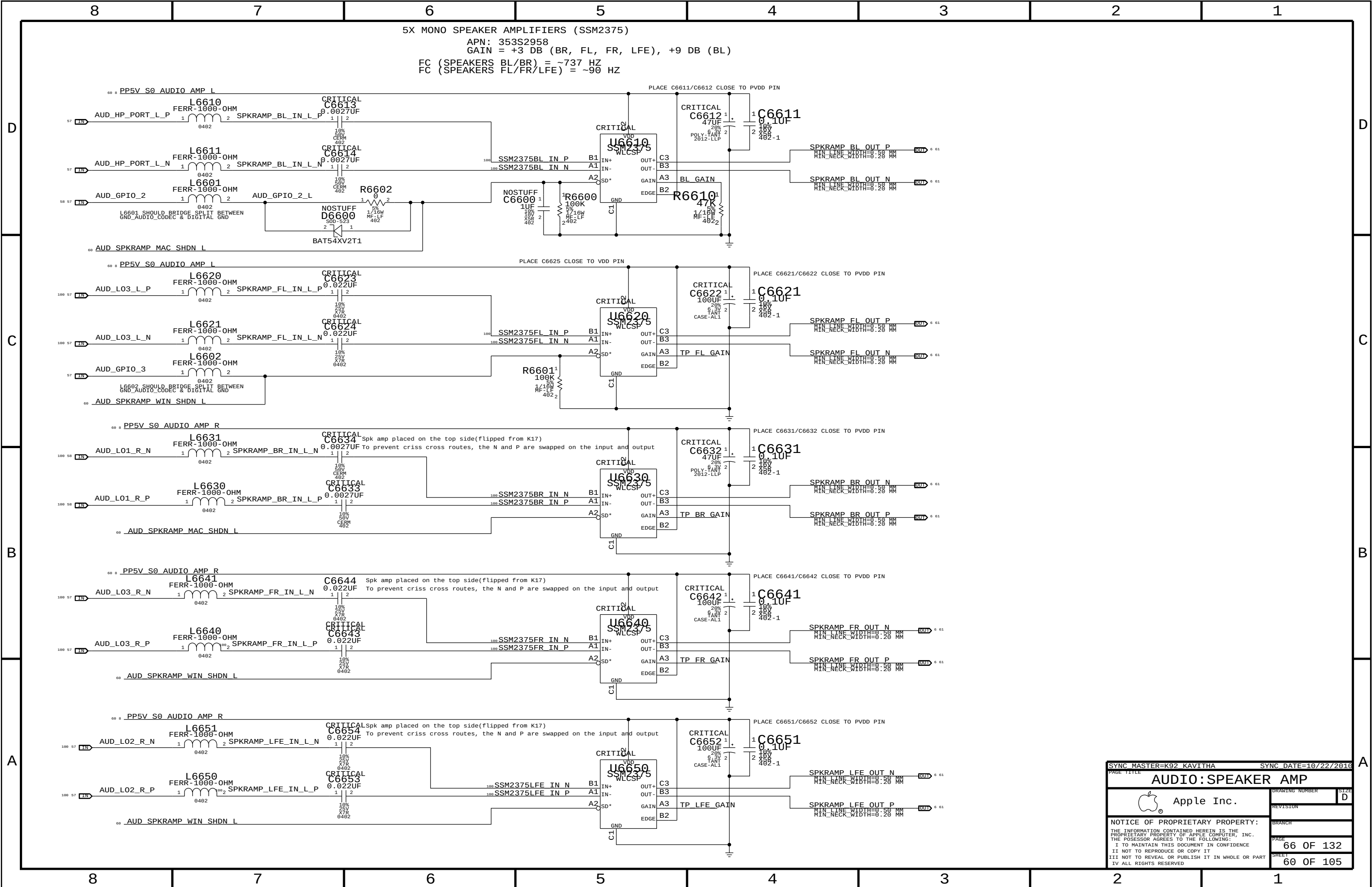


DIFF FSINPUT= 2.45VRMS
SE FSINPUT= 1.22VRMS
DAC1 FSOUTPUT= 1.34VRMS
DAC2/3 FSOUTPUTDIFF= 2.67VRMS
DAC2/3 FSOUTPUTSE= 1.34VRMS

SYNC MASTER=K92_KAVITHA		SYNC DATE=07/30/2016	
PAGE TITLE		AUDIO: CODEC	
		DRAWING NUMBER	SIZE
		62	D
NOTICE OF PROPRIETARY PROPERTY: THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE COMPUTER, INC. THE POSSESSOR AGREES TO THE FOLLOWING: I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE II NOT TO REPRODUCE OR COPY IT III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART IV ALL RIGHTS RESERVED		REVISION	BRANCH
		PAGE	SHEET
		62 OF 132	57 OF 105







AUDIO JACK 1 LO/HP JACK, SPDIF TX

MIC CONNECTORS: single anlg mic + 1 dig mic

D

C

B

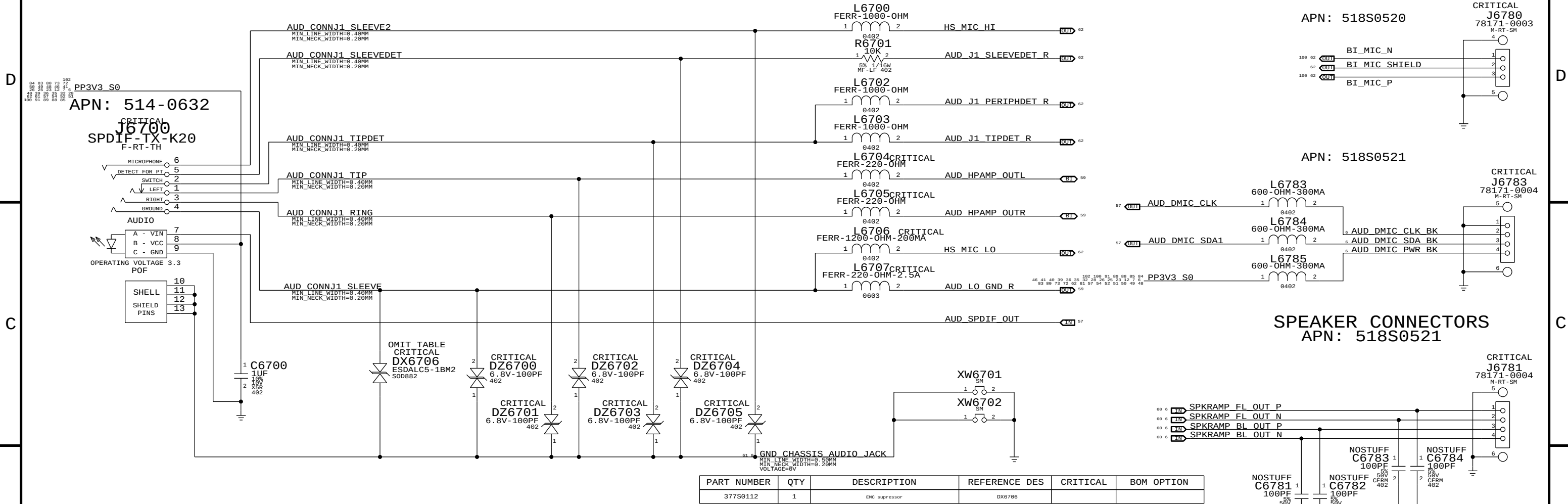
A

D

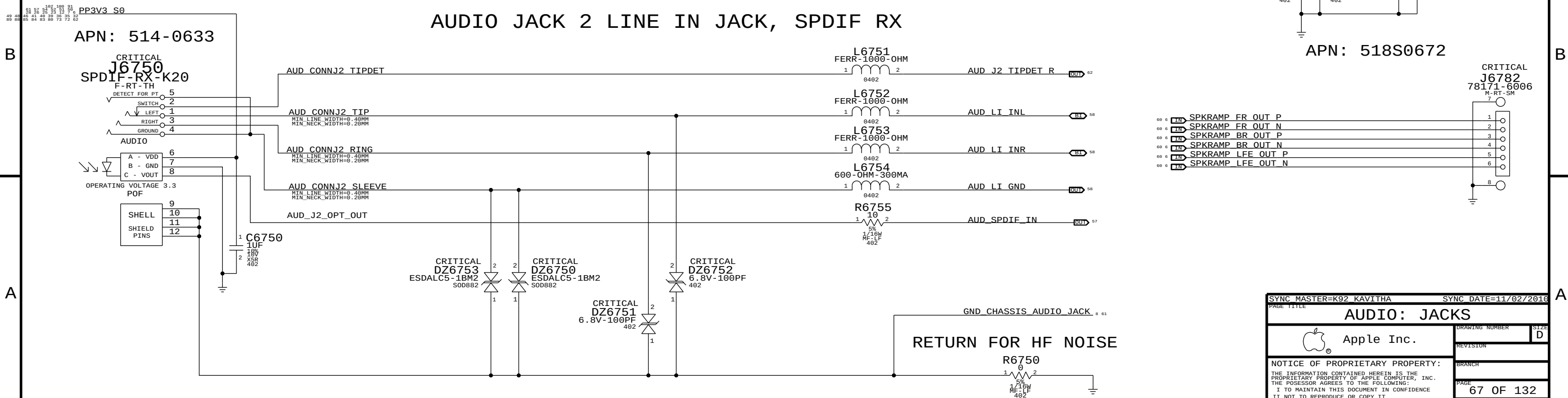
C

B

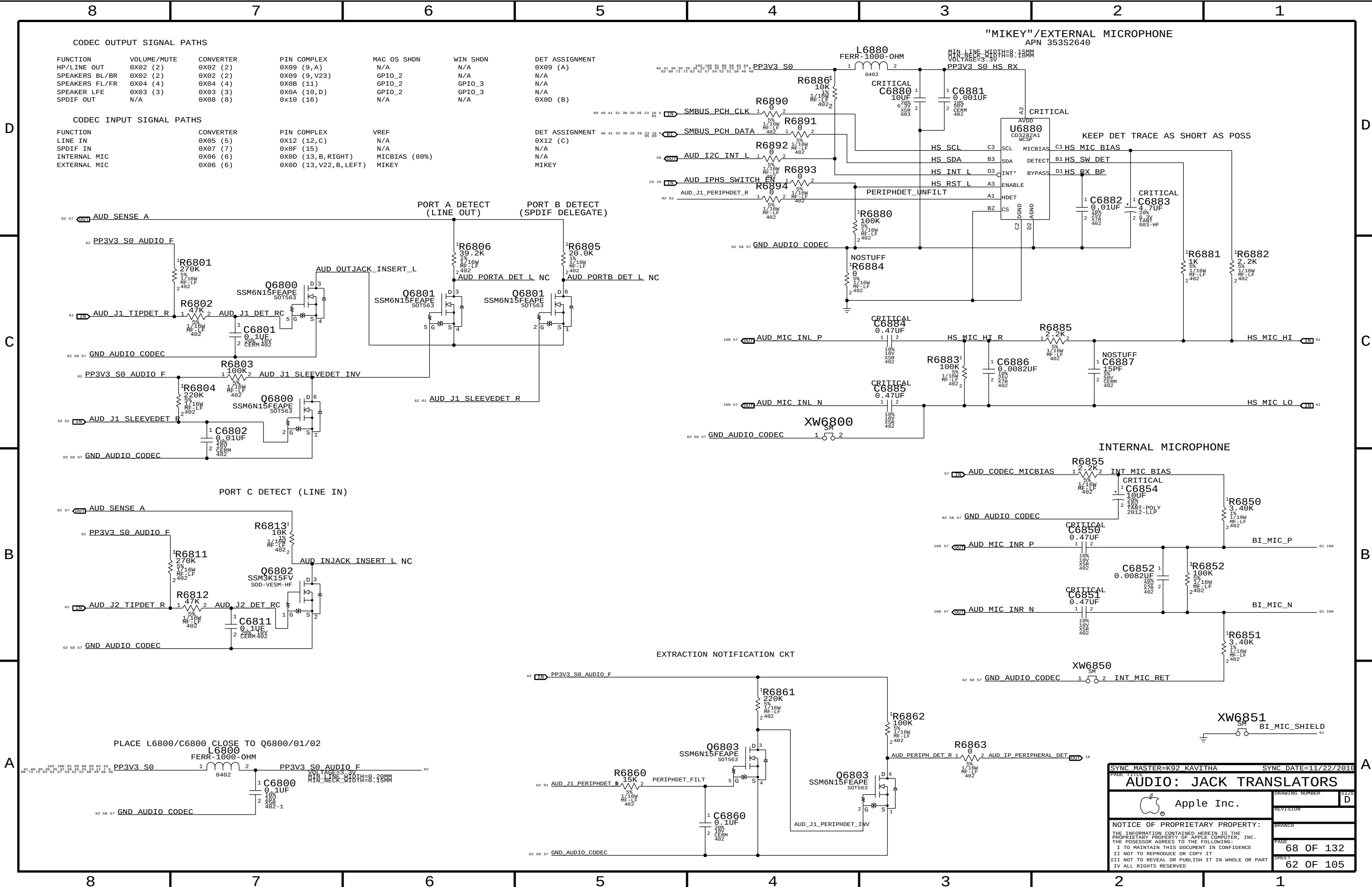
A



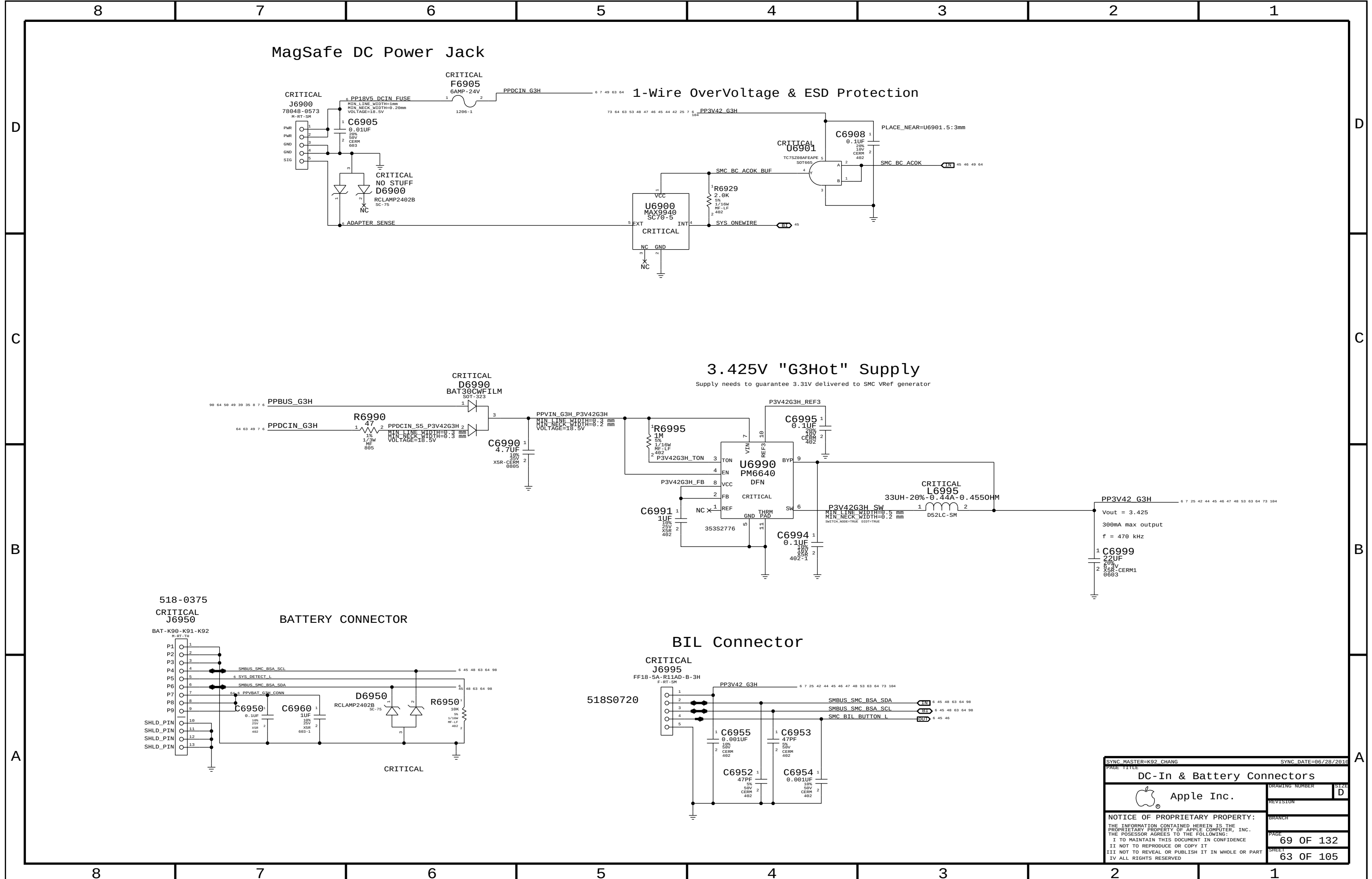
AUDIO JACK 2 LINE IN JACK, SPDIF RX

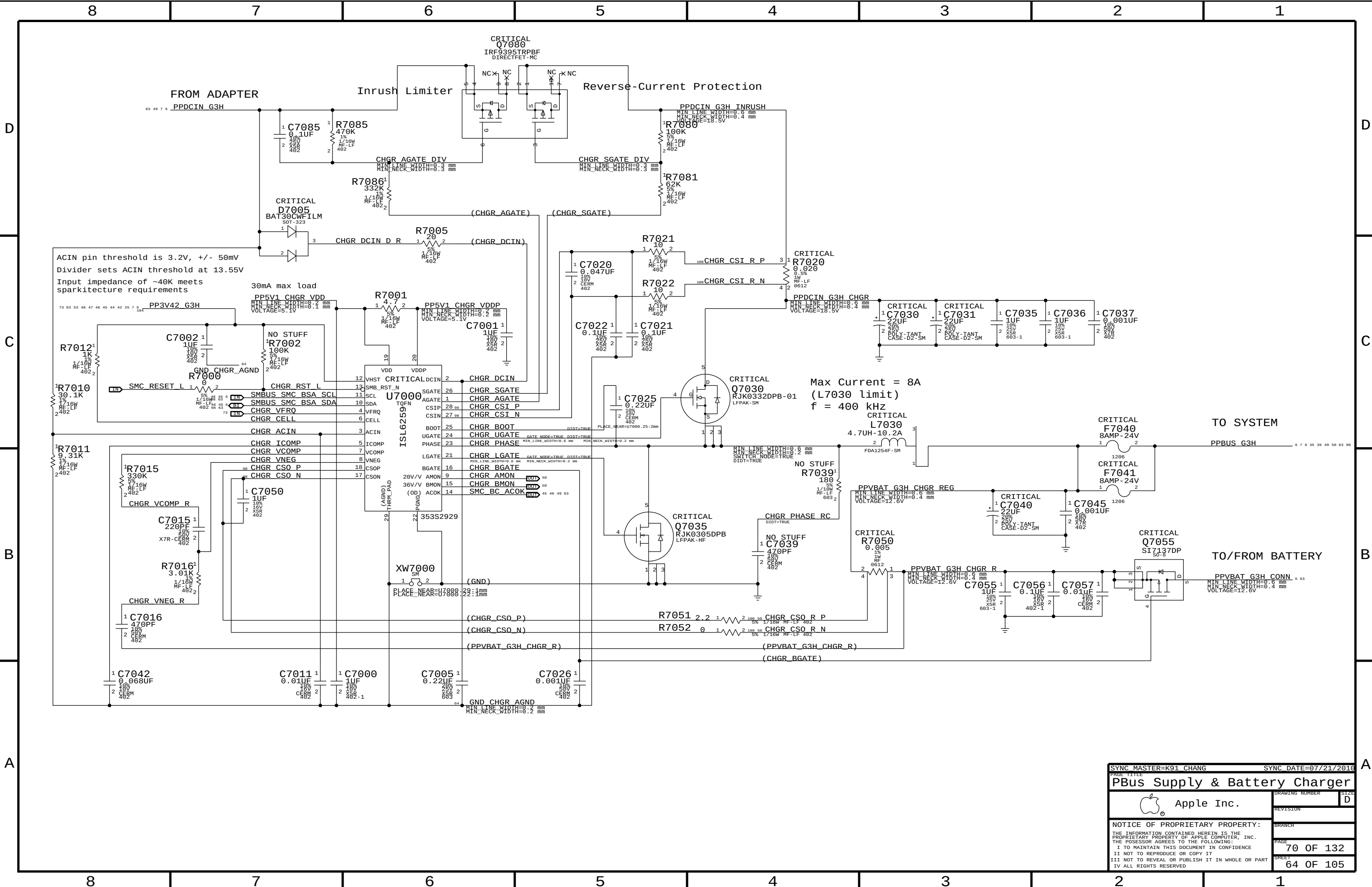


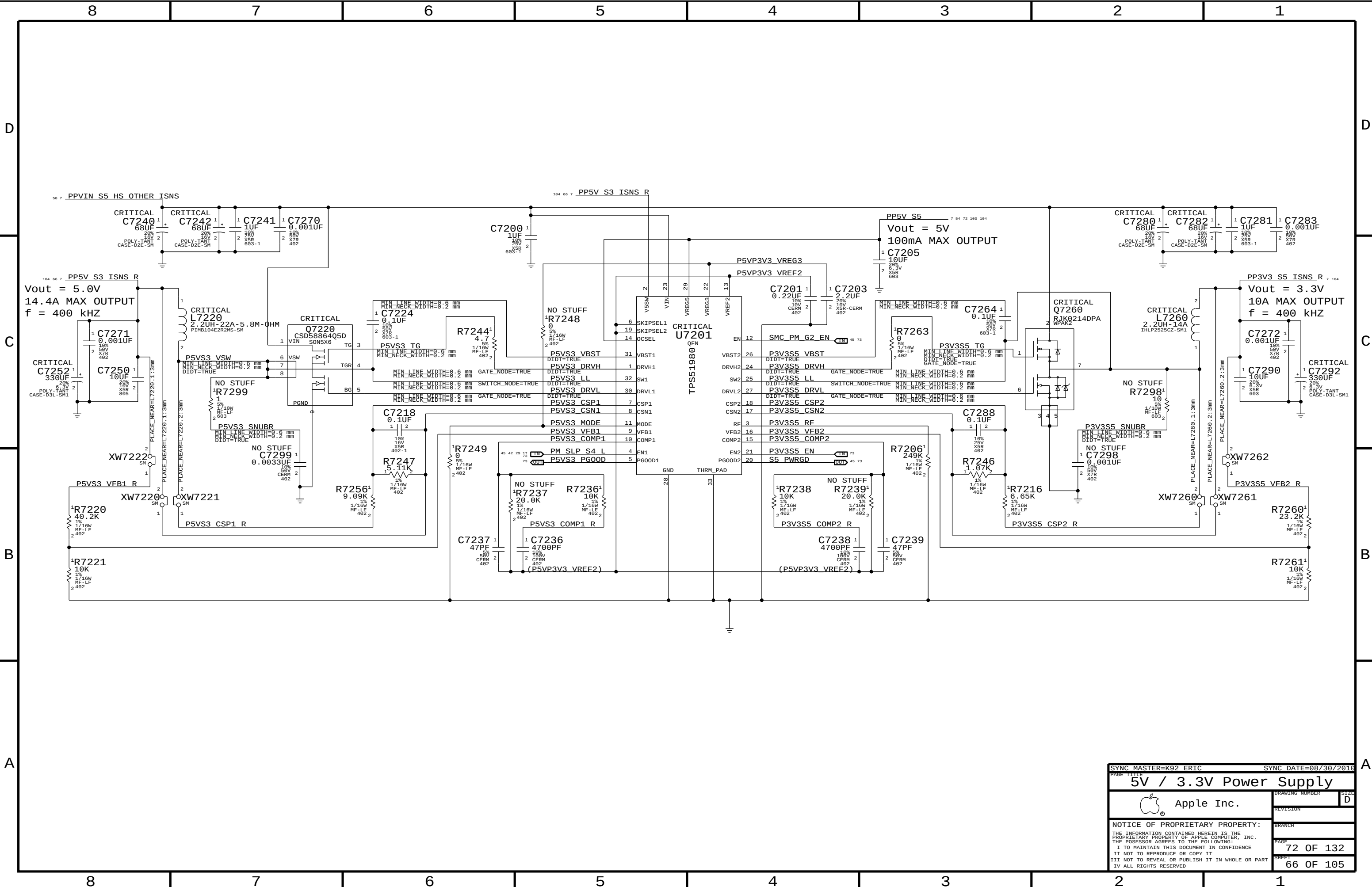
SYNC MASTER=K92_KAVITHA		SYNC DATE=11/02/2016	
PAGE TITLE		AUDIO: JACKS	
Apple Inc.		DRAWING NUMBER	SIZE D
NOTICE OF PROPRIETARY PROPERTY:		REVISION	
THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE COMPUTER, INC. THE POSSESSOR AGREES TO THE FOLLOWING:		BRANCH	
I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE		PAGE	67 OF 132
II NOT TO REPRODUCE OR COPY IT		SHEET	61 OF 105
III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART			
IV ALL RIGHTS RESERVED			

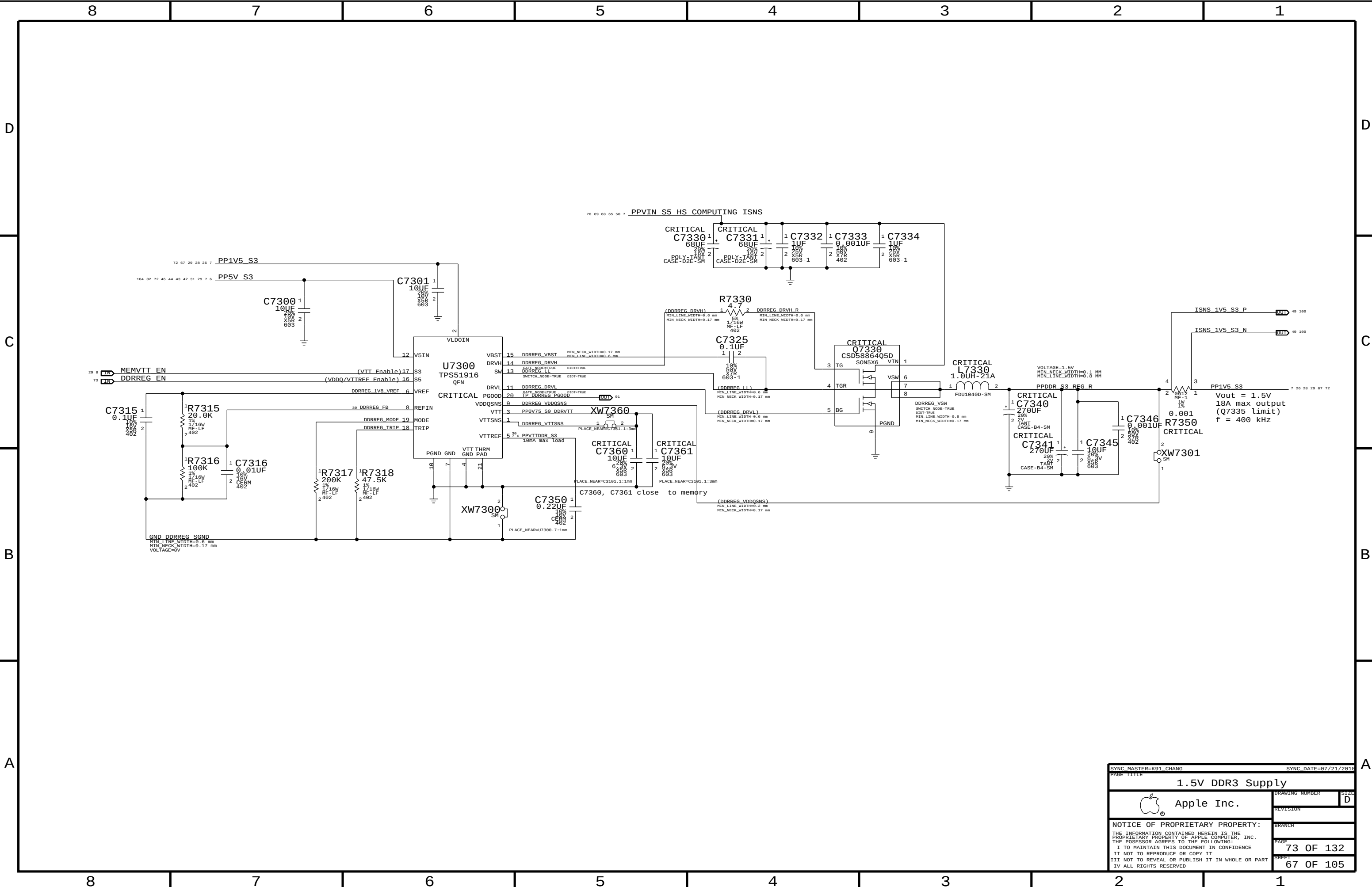


PAGE TITLE		PAGE TITLE	
AUDIO: JACK TRANSLATORS		DRAWING NUMBER	
Apple Inc.		REVISION	
NOTICE OF PROPRIETARY PROPERTY:		BRANCH	
THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE COMPUTER, INC. THE POSSESSOR AGREES TO THE FOLLOWING:		PAGE	
I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE		68 OF 132	
II NOT TO REPRODUCE OR COPY IT		SHEET	
III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART		62 OF 105	
IV ALL RIGHTS RESERVED			

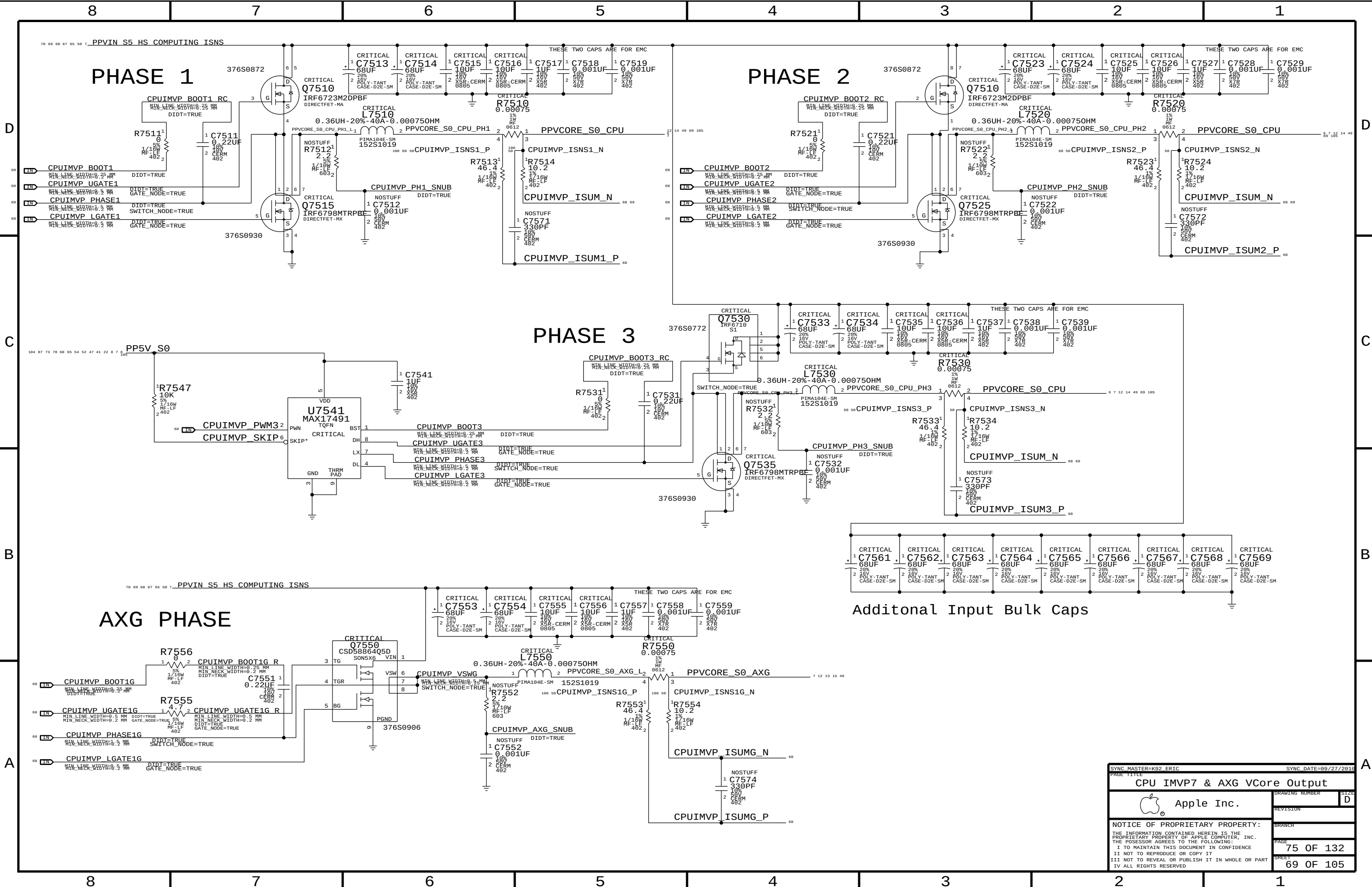








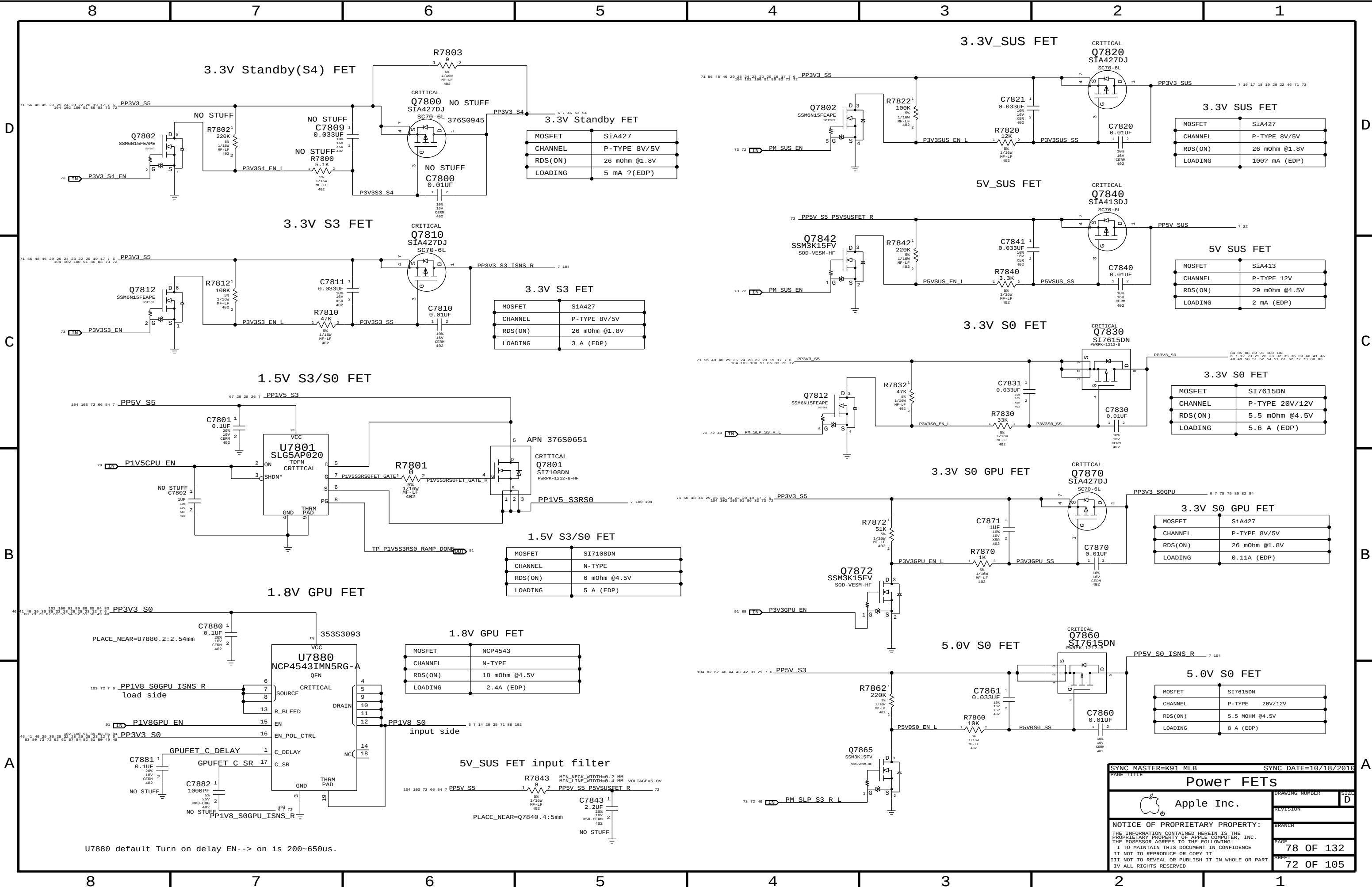
SYNC MASTER=K91 CHANG		SYNC DATE=07/21/2010	
PAGE TITLE		1.5V DDR3 Supply	
Apple Inc.		DRAWING NUMBER	SIZE
		REVISION	D
NOTICE OF PROPRIETARY PROPERTY:		BRANCH	
THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE COMPUTER, INC. THE POSSESSOR AGREES TO THE FOLLOWING:		PAGE	73 OF 132
I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE		SHEET	67 OF 105
II NOT TO REPRODUCE OR COPY IT			
III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART			
IV ALL RIGHTS RESERVED			

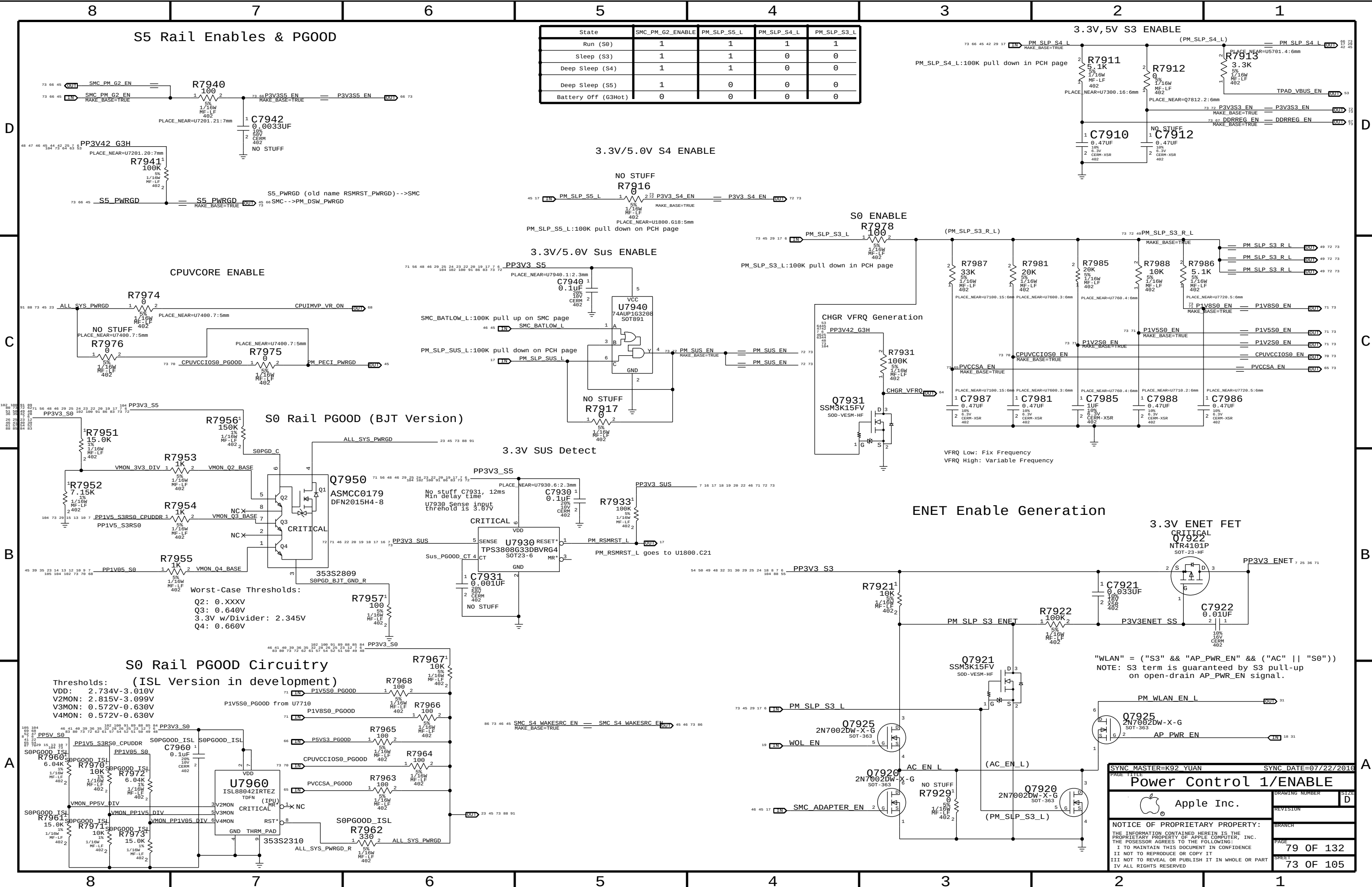


SYNC MASTER=K92 ERIC		SYNC DATE=89/27/2810	
PAGE TITLE		DRAWING NUMBER	
CPU IMVP7 & AXG VCore Output		SIZE	
Apple Inc.		REVISION	
NOTICE OF PROPRIETARY PROPERTY: THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE COMPUTER, INC. THE POSSESSOR AGREES TO THE FOLLOWING: I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE II NOT TO REPRODUCE OR COPY IT III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART IV ALL RIGHTS RESERVED		BRANCH	
		PAGE	
		75 OF 132	
		SHEET	
		69 OF 105	

D

A





State	SMC_PM_G2_ENABLE	PM_SLP_S5_L	PM_SLP_S4_L	PM_SLP_S3_L
Run (S0)	1	1	1	1
Sleep (S3)	1	1	0	0
Deep Sleep (S4)	1	1	0	0
Deep Sleep (S5)	1	0	0	0
Battery Off (G3Hot)	0	0	0	0

SYNC MASTER=K92_YUAN

SYNC DATE=07/22/2016

Power Control 1/ENABLE

Apple Inc.

NOTICE OF PROPRIETARY PROPERTY:

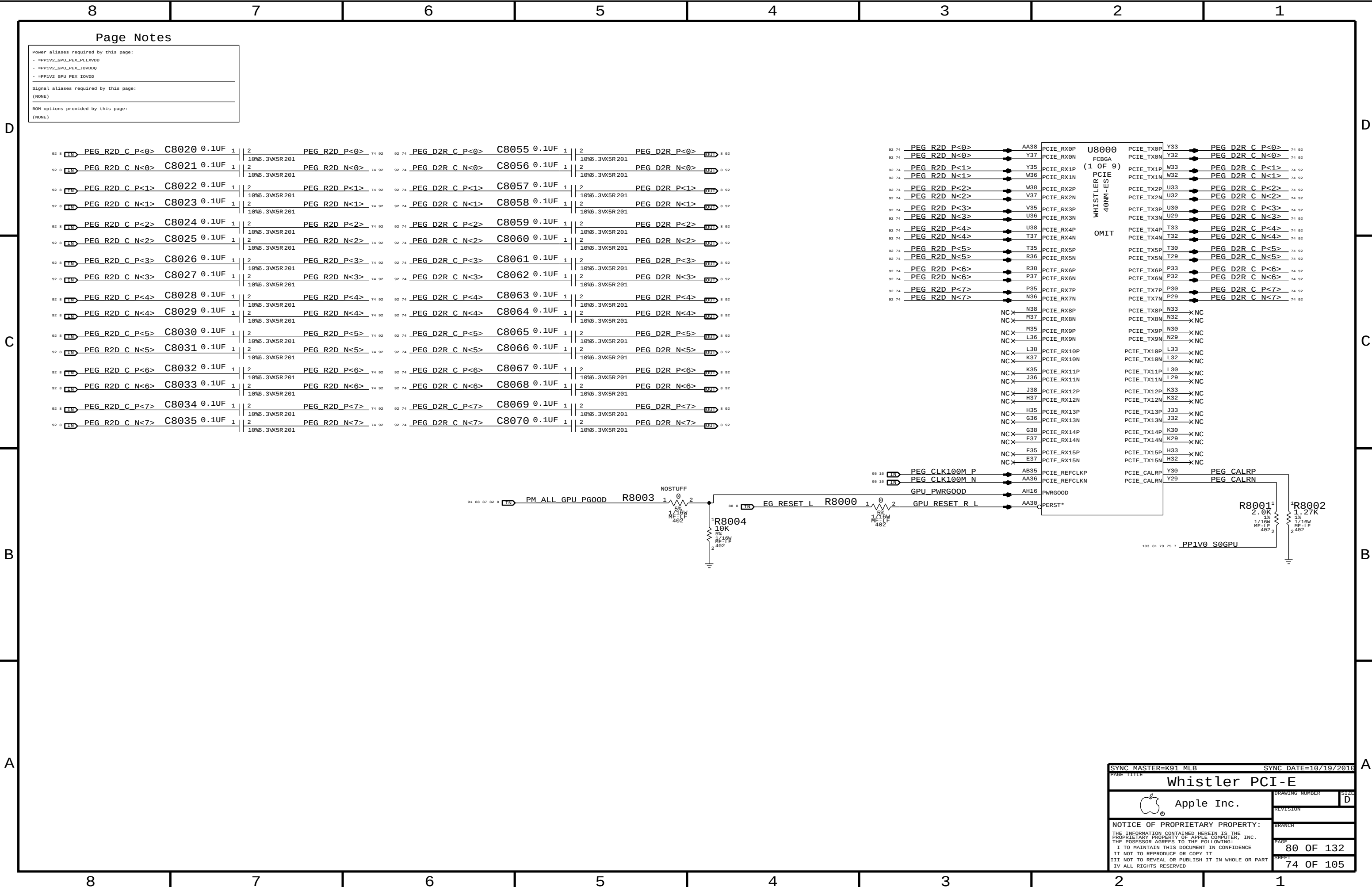
THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE COMPUTER, INC. THE POSSESSOR AGREES TO THE FOLLOWING:
I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE
II NOT TO REPRODUCE OR COPY IT
III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART
IV ALL RIGHTS RESERVED

DRAWING NUMBER

79 OF 132

REVISION

73 OF 105



Page Notes

Power aliases required by this page:

- =PP1V2_GPU_PEX_PLLXVDD
- =PP1V2_GPU_PEX_IOVDDQ
- =PP1V2_GPU_PEX_IOVDD

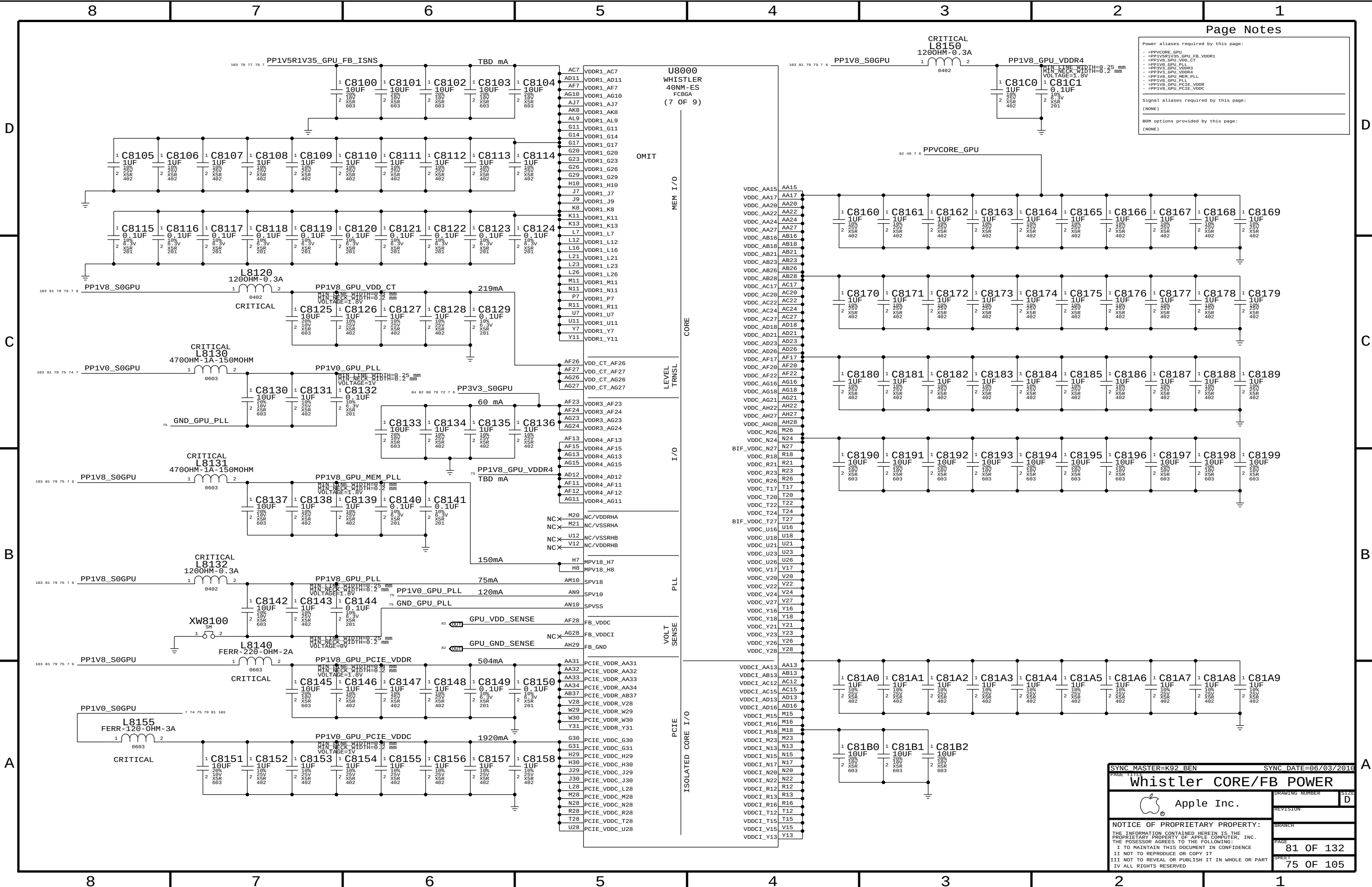
Signal aliases required by this page:

(NONE)

BOM options provided by this page:

(NONE)

SYNC MASTER=K91 MLB		SYNC DATE=10/19/2016	
PAGE TITLE			
whistler PCI-E			
Apple Inc.		DRAWING NUMBER	SIZE
		REVISION	D
NOTICE OF PROPRIETARY PROPERTY:		BRANCH	
THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE COMPUTER, INC. THE POSSESSOR AGREES TO THE FOLLOWING:			
I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE		PAGE	80 OF 132
II NOT TO REPRODUCE OR COPY IT		SHEET	74 OF 105
III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART			
IV ALL RIGHTS RESERVED			



Page Notes

Power aliases required by this page:

- #PPVCORE_GPU
- #PP1V5R1V35_GPU_FB_ISNS
- #PP1V8_GPU_VDD_CT
- #PP1V8_GPU_PLL
- #PP3V3_GPU_VDDR3
- #PP1V8_GPU_VDDR4
- #PP1V8_GPU_MEM_PLL
- #PP1V8_GPU_PCIE_VDDC
- #PP1V8_GPU_PCIE_VDDC

Signal aliases required by this page:

(NONE)

BOM options provided by this page:

(NONE)

SYNC MASTER=K92 BEN

SYNC DATE=06/03/2016

whistler CORE/FB POWER

Apple Inc.

NOTICE OF PROPRIETARY PROPERTY:

THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE COMPUTER, INC. THE POSSESSOR AGREES TO THE FOLLOWING:

I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE

II NOT TO REPRODUCE OR COPY IT

III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART

IV ALL RIGHTS RESERVED

DRAWING NUMBER

REVISION

BRANCH

PAGE

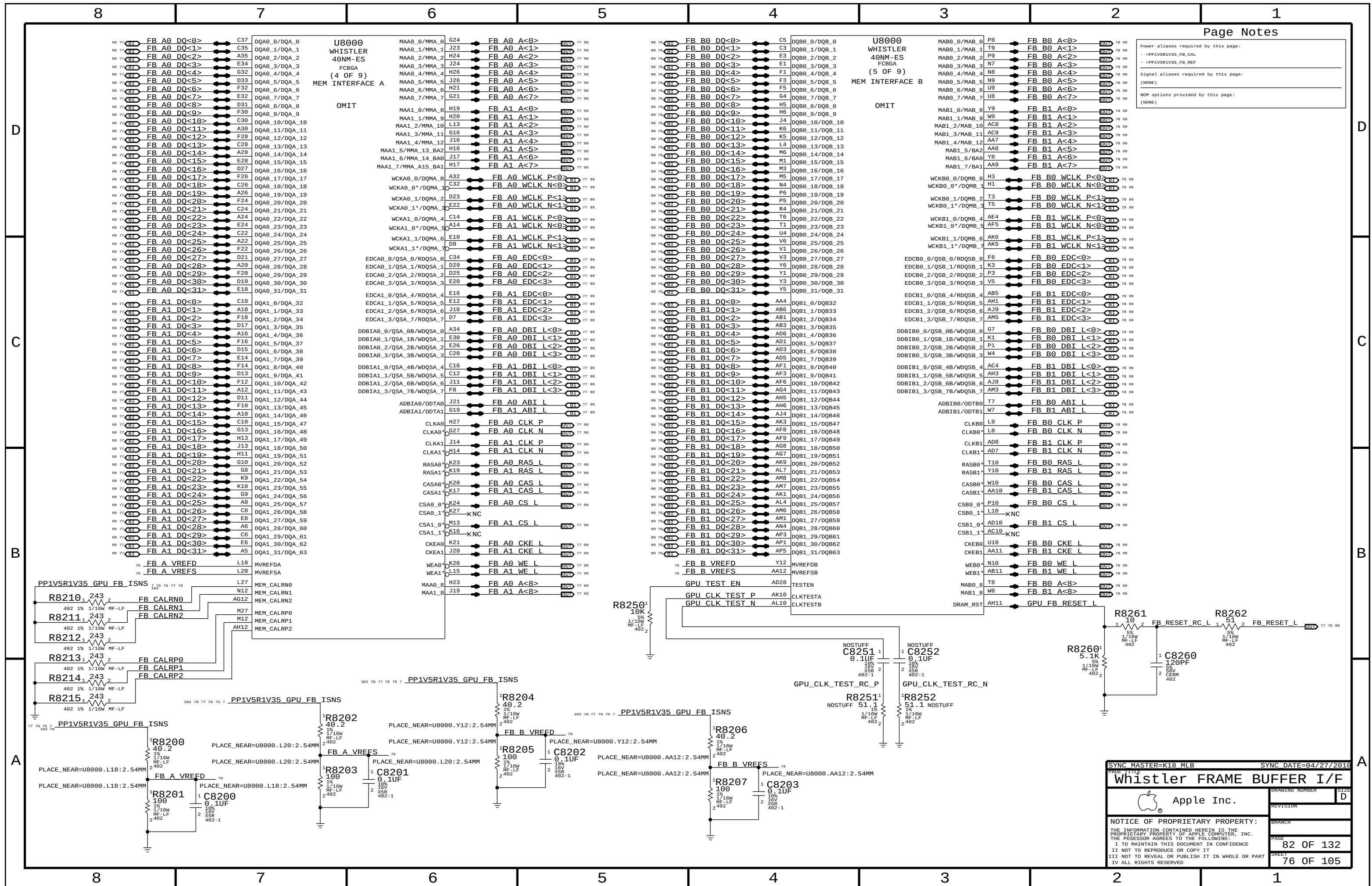
SHEET

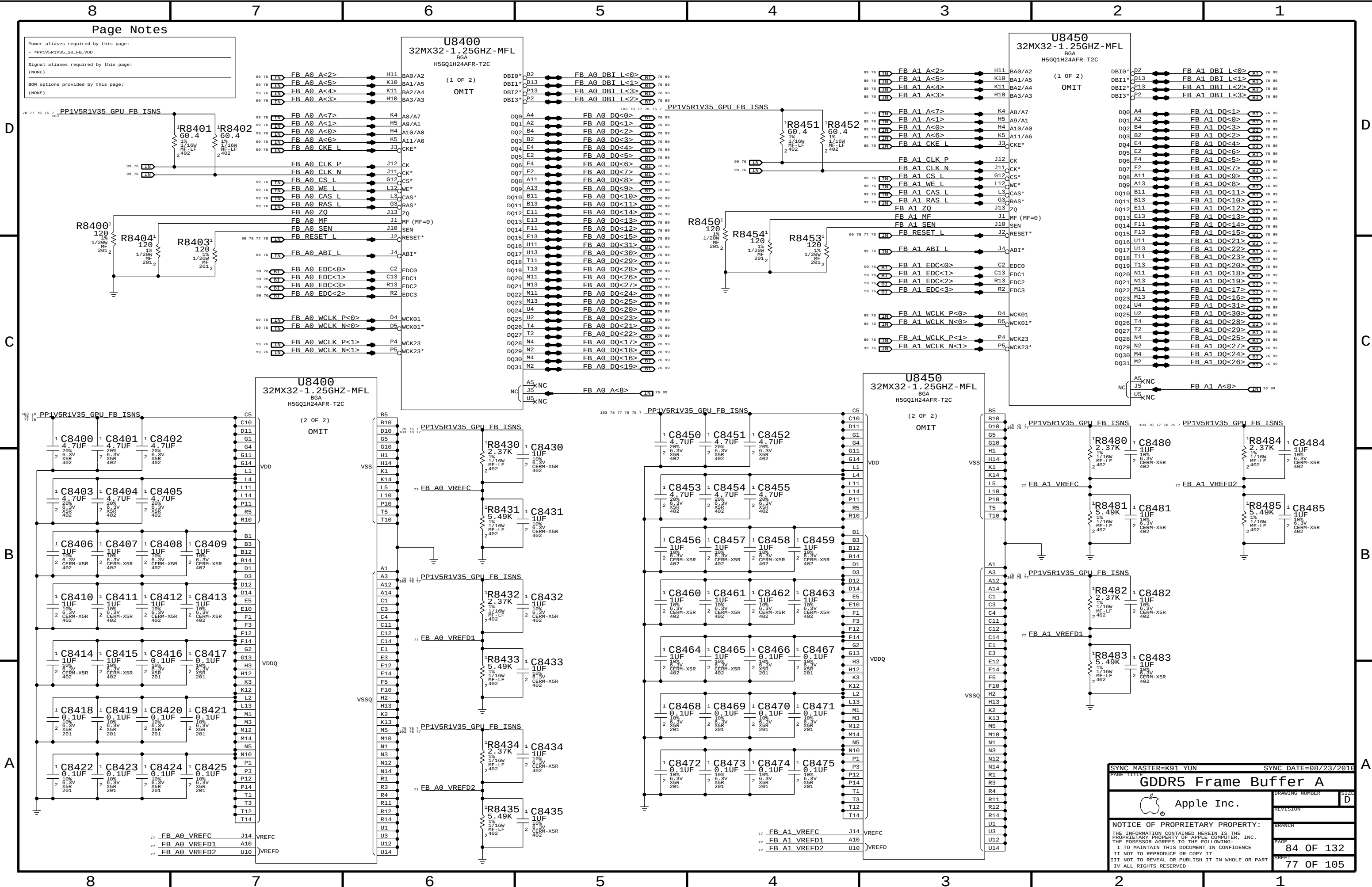
SIZE

D

81 OF 132

75 OF 105





Power aliases required by this page:
- #PP1V5R1V35_GPU_FB_ISNS

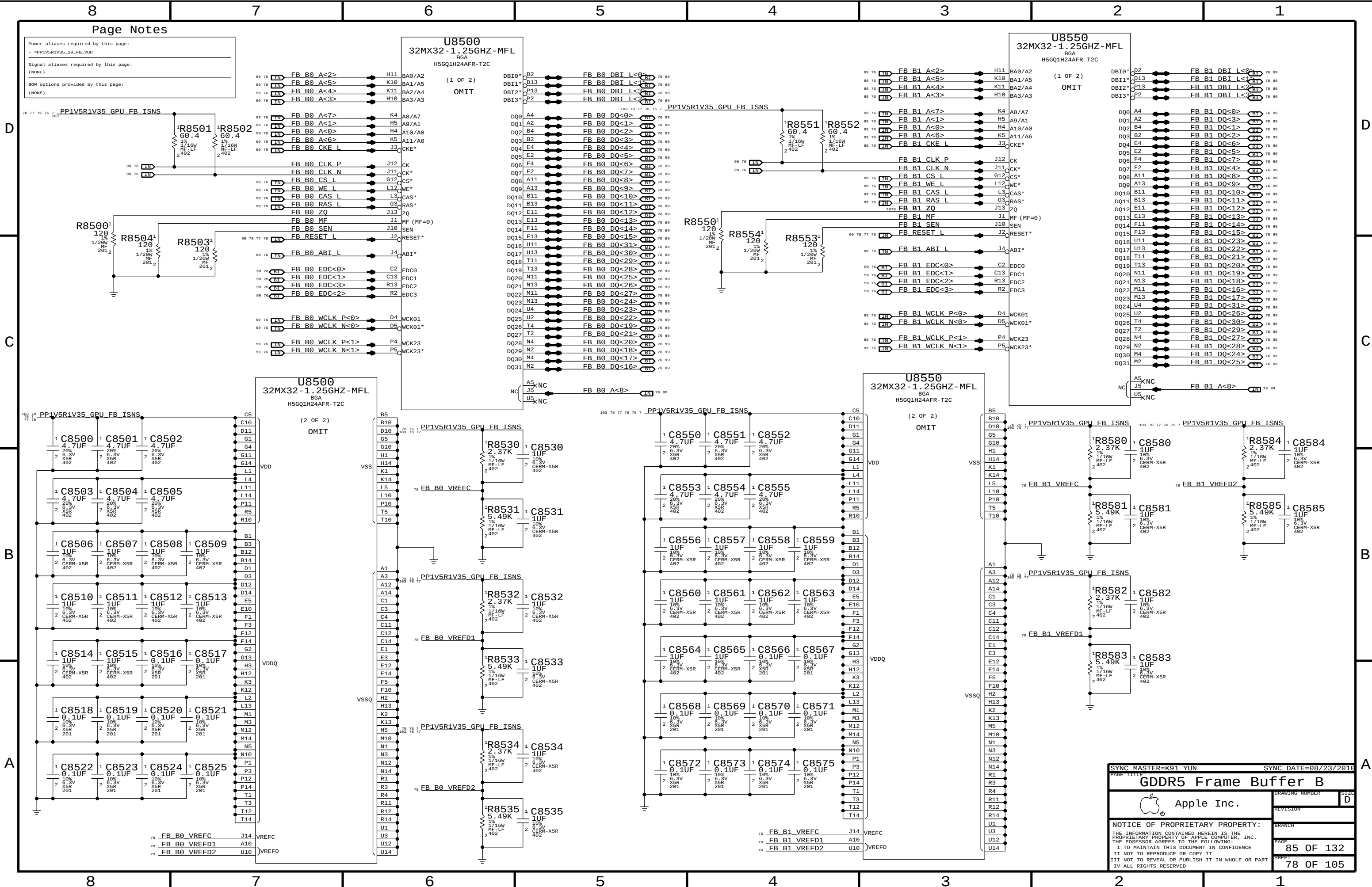
Signal aliases required by this page:
(NONE)

BOM options provided by this page:
(NONE)

U8400
32MX32-1.25GHZ-MFL
BGA
H5GQ1H24AFR-T2C
(1 OF 2)
OMIT

U8450
32MX32-1.25GHZ-MFL
BGA
H5GQ1H24AFR-T2C
(1 OF 2)
OMIT

PAGE TITLE		SYNC MASTER=K91 YUN		SYNC DATE=08/23/2016	
GDDR5 Frame Buffer A				DRAWING NUMBER	SIZE
Apple Inc.				REVISION	D
NOTICE OF PROPRIETARY PROPERTY: THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE COMPUTER, INC. THE POSSESSOR AGREES TO THE FOLLOWING: I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE II NOT TO REPRODUCE OR COPY IT III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART IV ALL RIGHTS RESERVED				BRANCH	
				PAGE	84 OF 132
				SHEET	77 OF 105



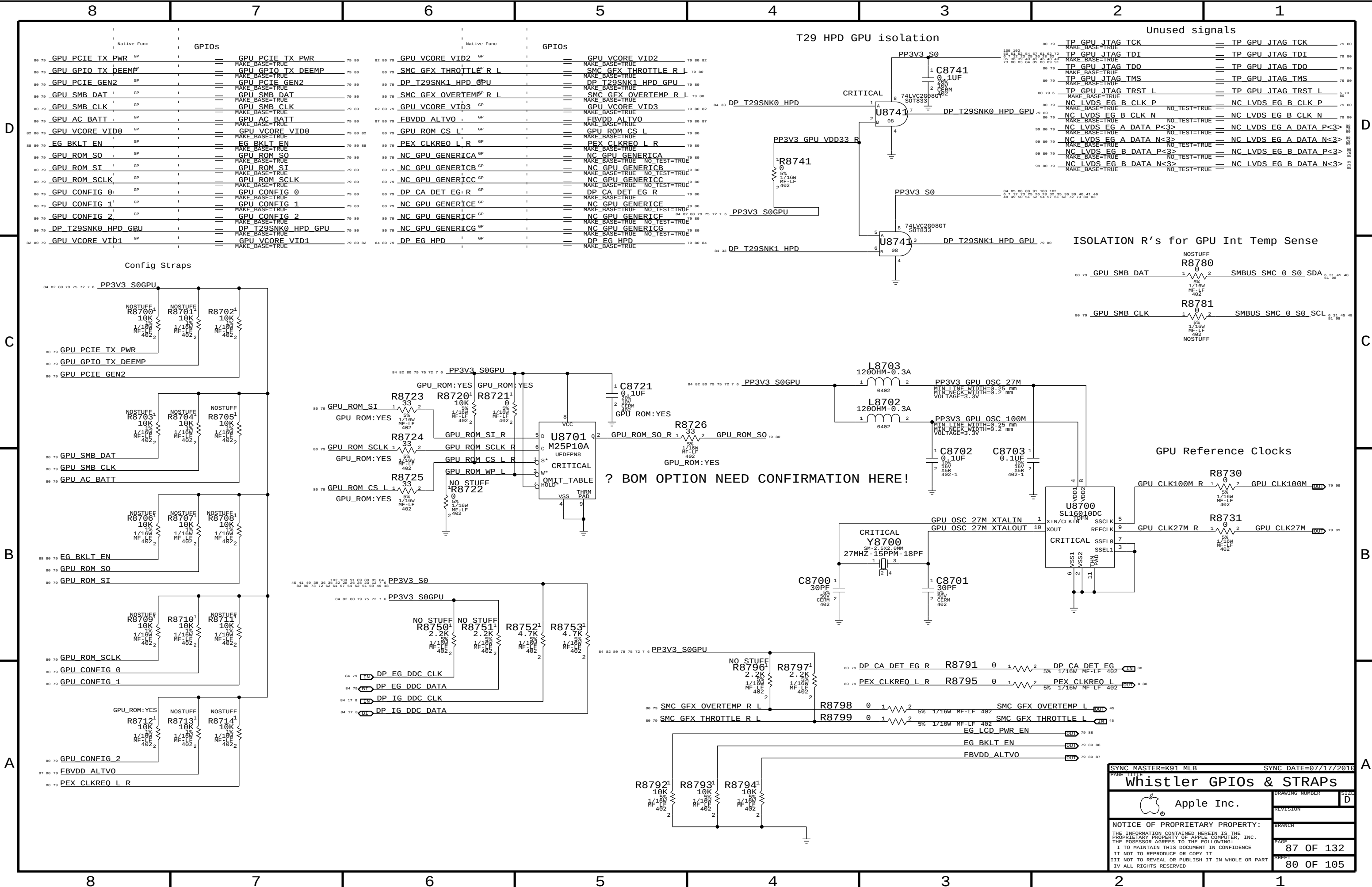
Page Notes

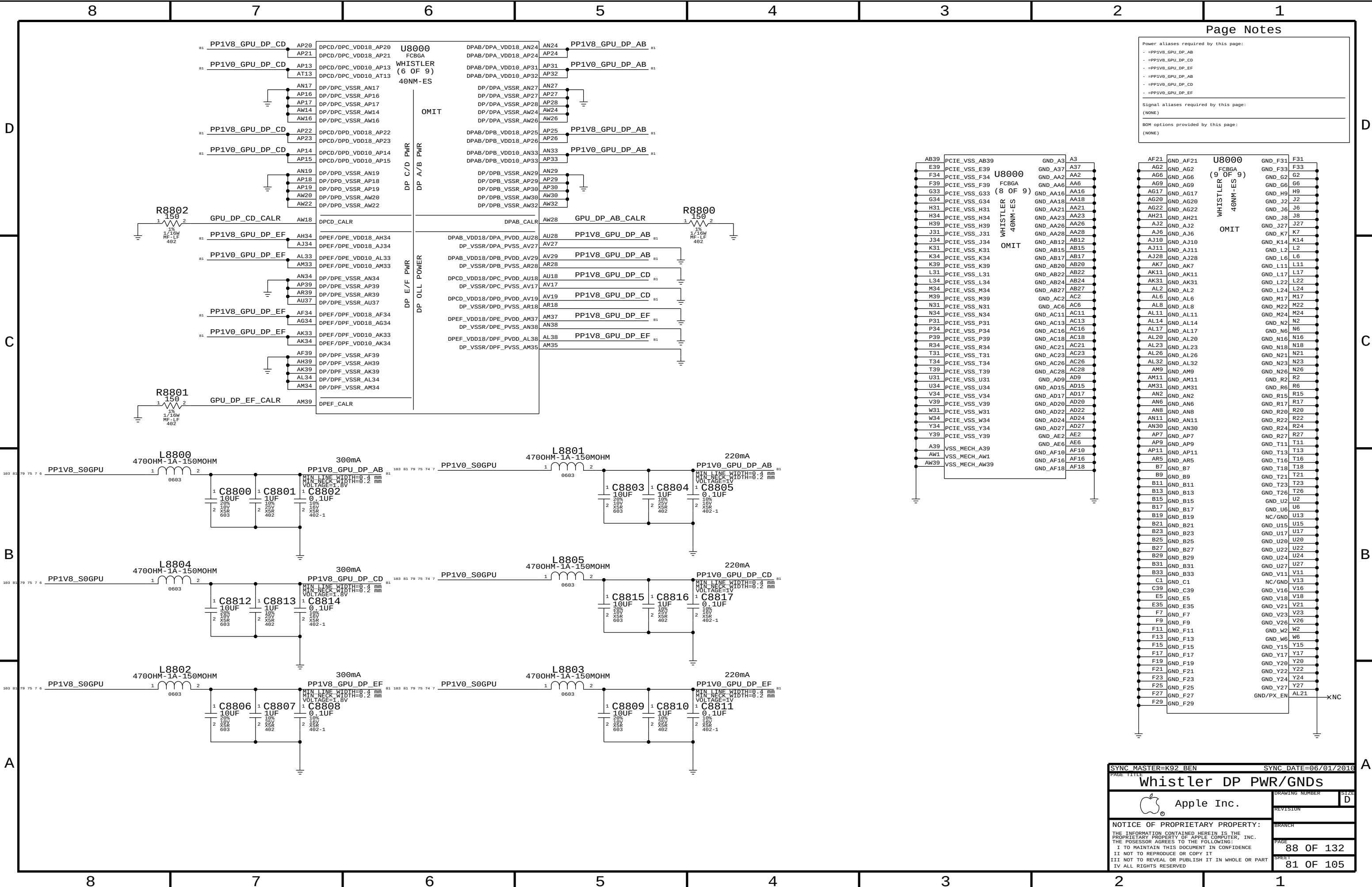
Power aliases required by this page:
- #PP1V5R1V35_50_FB_VDD

Signal aliases required by this page:
(NONE)

BOM options provided by this page:
(NONE)

SYNC MASTER=K91 YUN		SYNC DATE=08/23/2016	
PAGE TITLE			
GDDR5 Frame Buffer B			
 Apple Inc.		DRAWING NUMBER	SIZE
			D
NOTICE OF PROPRIETARY PROPERTY: THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE COMPUTER, INC. THE POSSESSOR AGREES TO THE FOLLOWING: I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE II NOT TO REPRODUCE OR COPY IT III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART IV ALL RIGHTS RESERVED		REVISION	
		BRANCH	
		PAGE	85 OF 132
		SHEET	78 OF 105





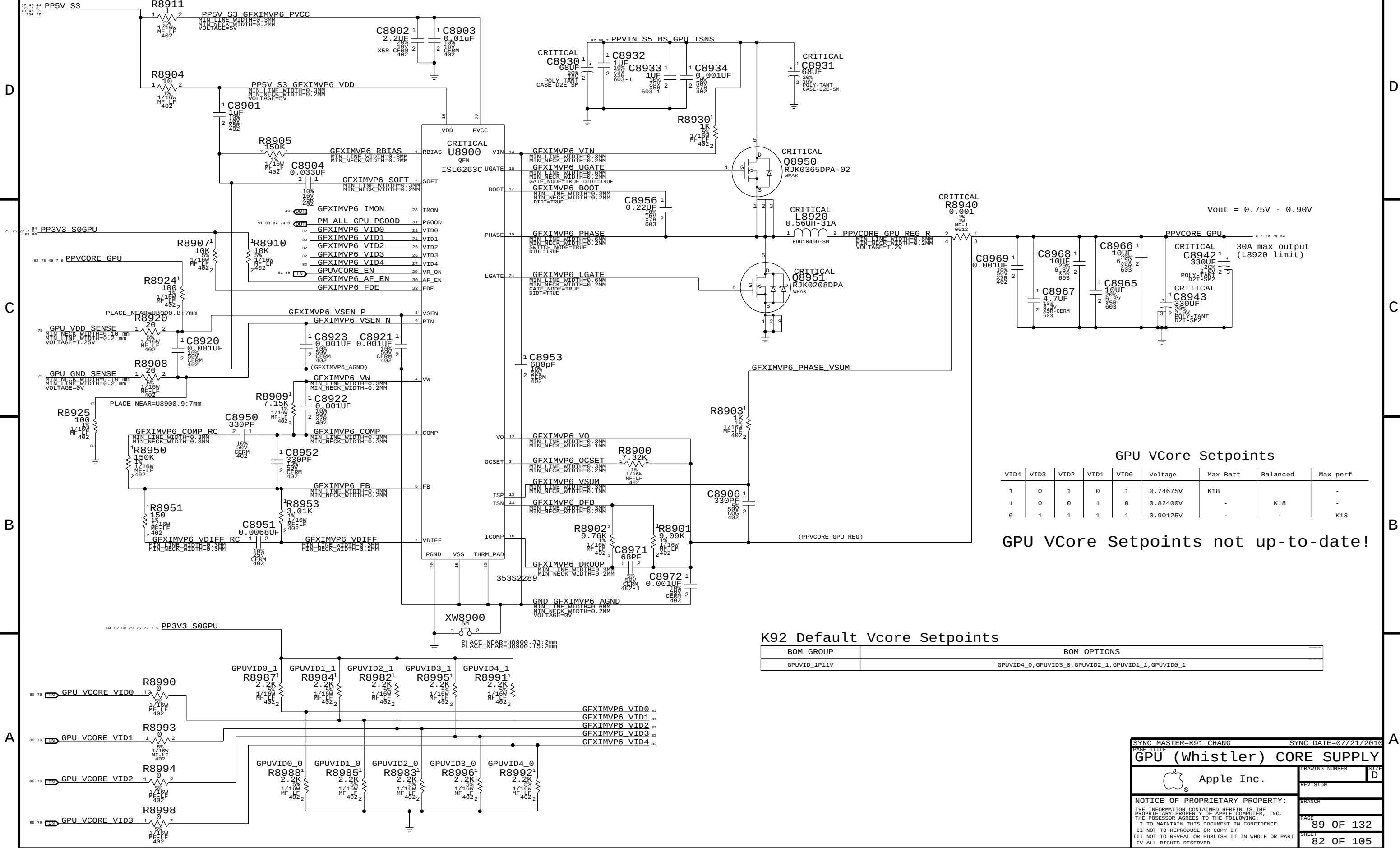
Page Notes	
Power aliases required by this page:	
- =PP1V8_GPU_DP_AB	
- =PP1V8_GPU_DP_CD	
- =PP1V8_GPU_DP_EF	
- =PP1V0_GPU_DP_AB	
- =PP1V0_GPU_DP_CD	
- =PP1V0_GPU_DP_EF	
Signal aliases required by this page:	
(NONE)	
BOM options provided by this page:	
(NONE)	

AB39	PCIE_VSS_AB39	GND_A3	A3
E39	PCIE_VSS_E39	GND_A37	A37
F34	PCIE_VSS_F34	GND_AA2	AA2
F39	PCIE_VSS_F39	GND_AA6	AA6
G33	PCIE_VSS_G33	GND_AA16	AA16
G34	PCIE_VSS_G34	GND_AA18	AA18
H31	PCIE_VSS_H31	GND_AA21	AA21
H34	PCIE_VSS_H34	GND_AA23	AA23
H39	PCIE_VSS_H39	GND_AA26	AA26
J31	PCIE_VSS_J31	GND_AA28	AA28
J34	PCIE_VSS_J34	GND_AB12	AB12
K31	PCIE_VSS_K31	GND_AB15	AB15
K34	PCIE_VSS_K34	GND_AB17	AB17
K39	PCIE_VSS_K39	GND_AB20	AB20
L31	PCIE_VSS_L31	GND_AB22	AB22
L34	PCIE_VSS_L34	GND_AB24	AB24
M34	PCIE_VSS_M34	GND_AB27	AB27
M39	PCIE_VSS_M39	GND_AC2	AC2
N31	PCIE_VSS_N31	GND_AC6	AC6
N34	PCIE_VSS_N34	GND_AC11	AC11
P31	PCIE_VSS_P31	GND_AC13	AC13
P34	PCIE_VSS_P34	GND_AC16	AC16
P39	PCIE_VSS_P39	GND_AC18	AC18
R34	PCIE_VSS_R34	GND_AC21	AC21
T31	PCIE_VSS_T31	GND_AC23	AC23
T34	PCIE_VSS_T34	GND_AC26	AC26
T39	PCIE_VSS_T39	GND_AC28	AC28
U31	PCIE_VSS_U31	GND_AD9	AD9
U34	PCIE_VSS_U34	GND_AD15	AD15
V34	PCIE_VSS_V34	GND_AD17	AD17
V39	PCIE_VSS_V39	GND_AD20	AD20
W31	PCIE_VSS_W31	GND_AD22	AD22
W34	PCIE_VSS_W34	GND_AD24	AD24
Y34	PCIE_VSS_Y34	GND_AD27	AD27
Y39	PCIE_VSS_Y39	GND_AE2	AE2
A39	VSS_MECH_A39	GND_AE6	AE6
AW1	VSS_MECH_AW1	GND_AF10	AF10
AW39	VSS_MECH_AW39	GND_AF16	AF16
		GND_AF18	AF18

AF21	GND_AF21	U8000	GND_F31	F31
AG2	GND_AG2	FCBGA	GND_F33	F33
AG6	GND_AG6	(9 OF 9)	GND_G2	G2
AG9	GND_AG9	WHISTLER	GND_G6	G6
AG17	GND_AG17	40NM-ES	GND_H9	H9
AG20	GND_AG20	OMIT	GND_J2	J2
AG22	GND_AG22		GND_J6	J6
AH21	GND_AH21		GND_J8	J8
AJ2	GND_AJ2		GND_J27	J27
AJ6	GND_AJ6		GND_K7	K7
AJ10	GND_AJ10		GND_K14	K14
AJ11	GND_AJ11		GND_L2	L2
AJ28	GND_AJ28		GND_L6	L6
AK7	GND_AK7		GND_L11	L11
AK11	GND_AK11		GND_L17	L17
AK31	GND_AK31		GND_L22	L22
AL2	GND_AL2		GND_L24	L24
AL6	GND_AL6		GND_M17	M17
AL8	GND_AL8		GND_M22	M22
AL11	GND_AL11		GND_M24	M24
AL14	GND_AL14		GND_N2	N2
AL17	GND_AL17		GND_N6	N6
AL20	GND_AL20		GND_N16	N16
AL23	GND_AL23		GND_N18	N18
AL26	GND_AL26		GND_N21	N21
AL32	GND_AL32		GND_N23	N23
AM9	GND_AM9		GND_N26	N26
AM11	GND_AM11		GND_R2	R2
AM31	GND_AM31		GND_R6	R6
AN2	GND_AN2		GND_R15	R15
AN6	GND_AN6		GND_R17	R17
AN8	GND_AN8		GND_R20	R20
AN11	GND_AN11		GND_R22	R22
AN30	GND_AN30		GND_R24	R24
AP7	GND_AP7		GND_R27	R27
AP9	GND_AP9		GND_T11	T11
AP11	GND_AP11		GND_T13	T13
AR5	GND_AR5		GND_T16	T16
B7	GND_B7		GND_T18	T18
B9	GND_B9		GND_T21	T21
B11	GND_B11		GND_T23	T23
B13	GND_B13		GND_T26	T26
B15	GND_B15		GND_U2	U2
B17	GND_B17		GND_U6	U6
B19	GND_B19		NC/GND	U13
B21	GND_B21		GND_U15	U15
B23	GND_B23		GND_U17	U17
B25	GND_B25		GND_U20	U20
B27	GND_B27		GND_U22	U22
B29	GND_B29		GND_U24	U24
B31	GND_B31		GND_U27	U27
B33	GND_B33		GND_V11	V11
C1	GND_C1		GND_V13	V13
C39	GND_C39		GND_V16	V16
E5	GND_E5		GND_V18	V18
E35	GND_E35		GND_V21	V21
F7	GND_F7		GND_V23	V23
F9	GND_F9		GND_V26	V26
F11	GND_F11		GND_W2	W2
F13	GND_F13		GND_W6	W6
F15	GND_F15		GND_Y15	Y15
F17	GND_F17		GND_Y17	Y17
F19	GND_F19		GND_Y20	Y20
F21	GND_F21		GND_Y22	Y22
F23	GND_F23		GND_Y24	Y24
F25	GND_F25		GND_Y27	Y27
F27	GND_F27		GND/PX_EN	AL21
F29	GND_F29			

SYNC MASTER=K92 BEN		SYNC DATE=06/01/2016	
PAGE TITLE		Whistler DP PWR/GNDs	
Apple Inc.		DRAWING NUMBER	SIZE D
NOTICE OF PROPRIETARY PROPERTY:		REVISION	
THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE COMPUTER, INC. THE POSSESSOR AGREES TO THE FOLLOWING:		BRANCH	
I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE		PAGE	88 OF 132
II NOT TO REPRODUCE OR COPY IT		SHEET	81 OF 105
III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART			
IV ALL RIGHTS RESERVED			

GPU VCore Regulator



GPU VCore Setpoints									
VID4	VID3	VID2	VID1	VID0	Voltage	Max Batt	Balanced	Max perf	
1	0	1	0	1	0.74675V	K18		-	
1	0	0	1	0	0.82400V	-	K18	-	
0	1	1	1	1	0.90125V	-	-	K18	

GPU VCore Setpoints not up-to-date!

K92 Default Vcore Setpoints

BOM GROUP	BOM OPTIONS
GPUVID_1P11V	GPUVID4_0, GPUVID3_0, GPUVID2_1, GPUVID1_1, GPUVID0_1

SYNC MASTER=K91 CHANG

SYNC DATE=07/21/2016

GPU (Whistler) CORE SUPPLY

Apple Inc.

NOTICE OF PROPRIETARY PROPERTY:

THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE COMPUTER, INC. THE POSSESSOR AGREES TO THE FOLLOWING:
I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE
II NOT TO REPRODUCE OR COPY IT
III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART
IV ALL RIGHTS RESERVED

DRAWING NUMBER

REVISION

BRANCH

PAGE

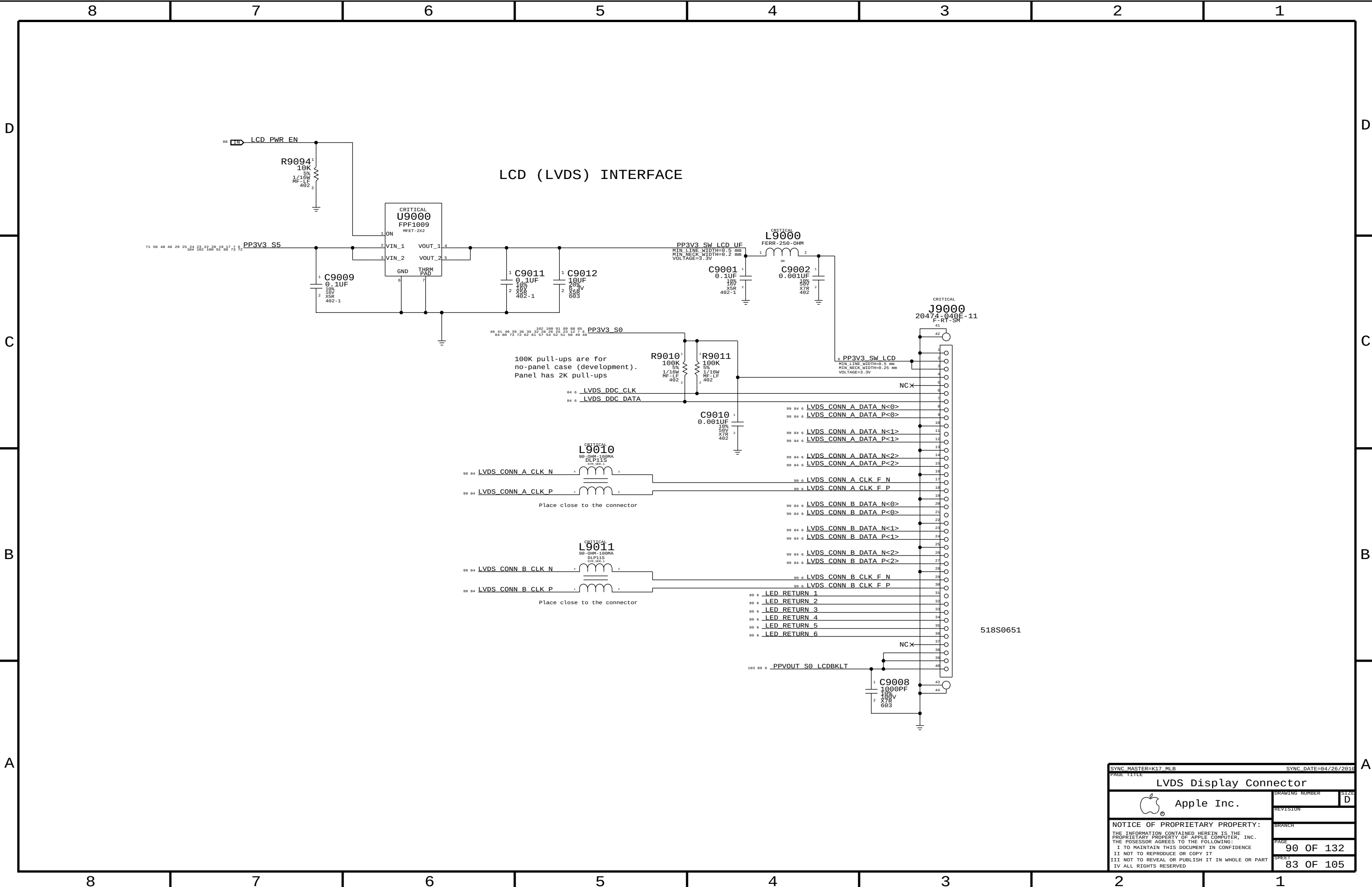
SHEET

SIZE

D

89 OF 132

82 OF 105



LCD (LVDS) INTERFACE

100K pull-ups are for
no-panel case (development).
Panel has 2K pull-ups

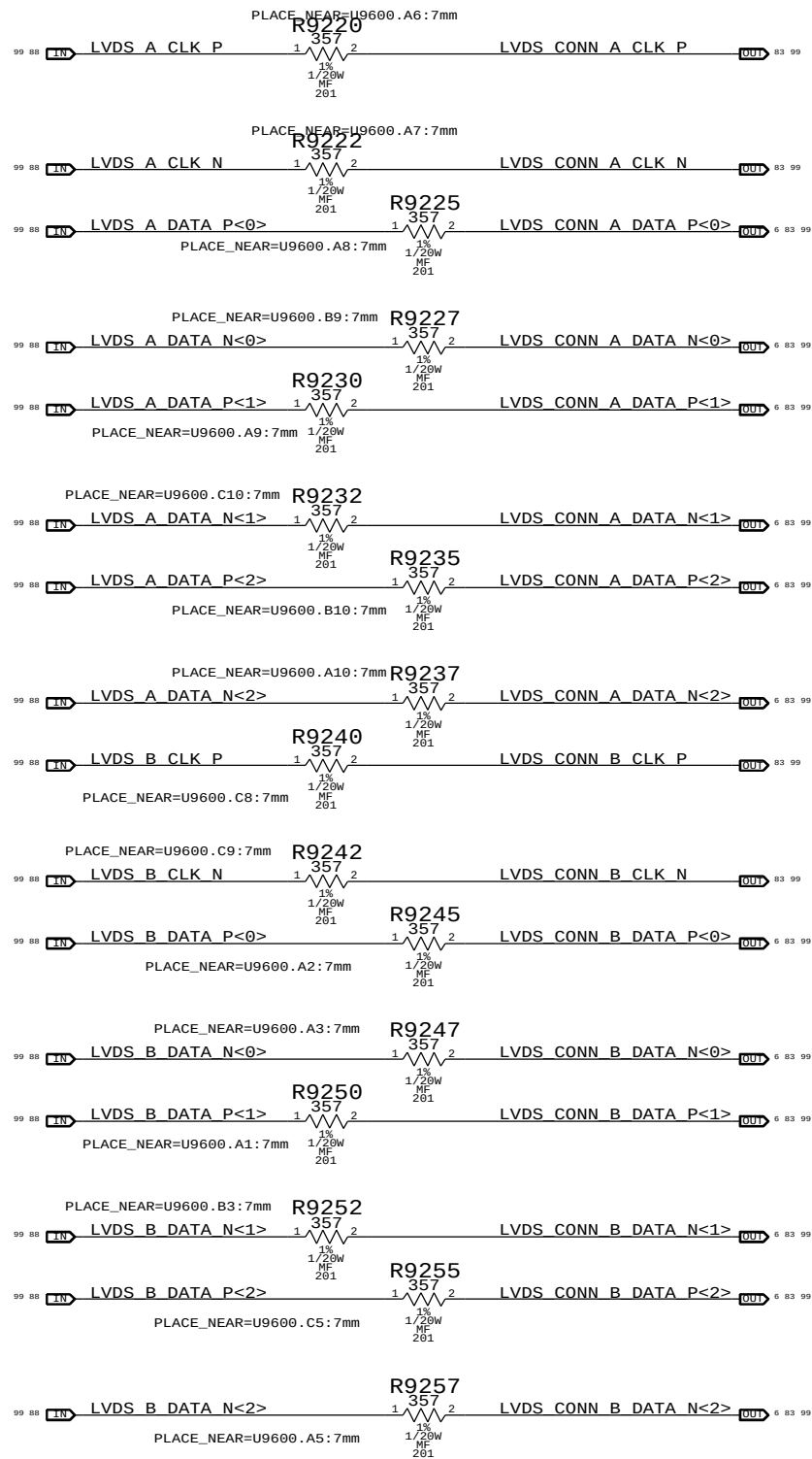
Place close to the connector

Place close to the connector

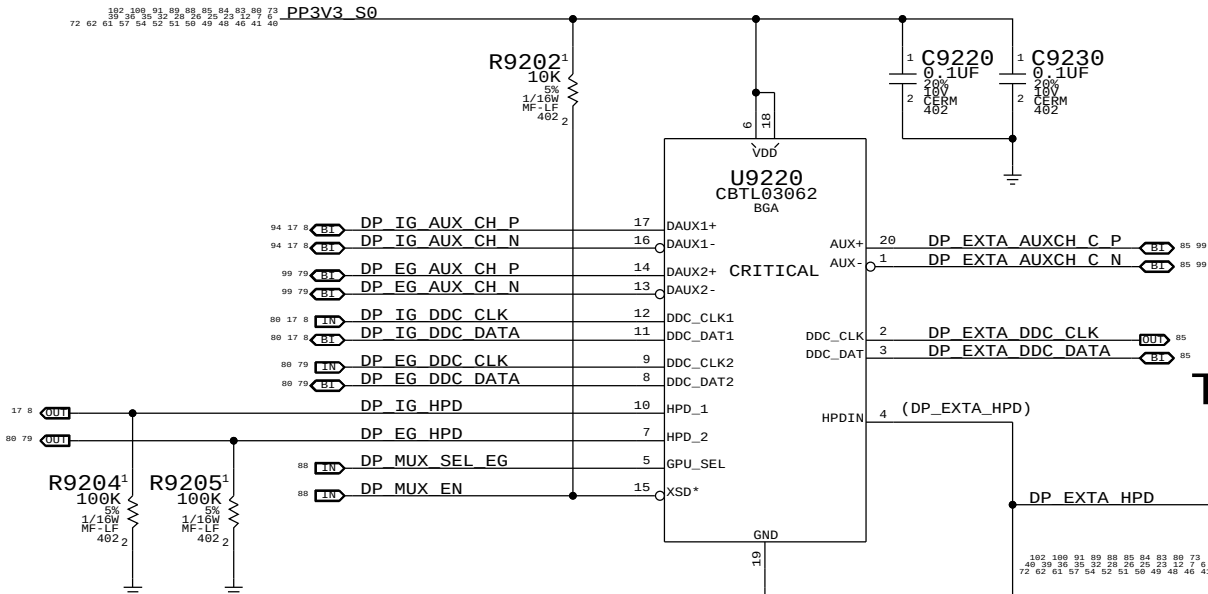
518S0651

LVDS Transmitter Termination

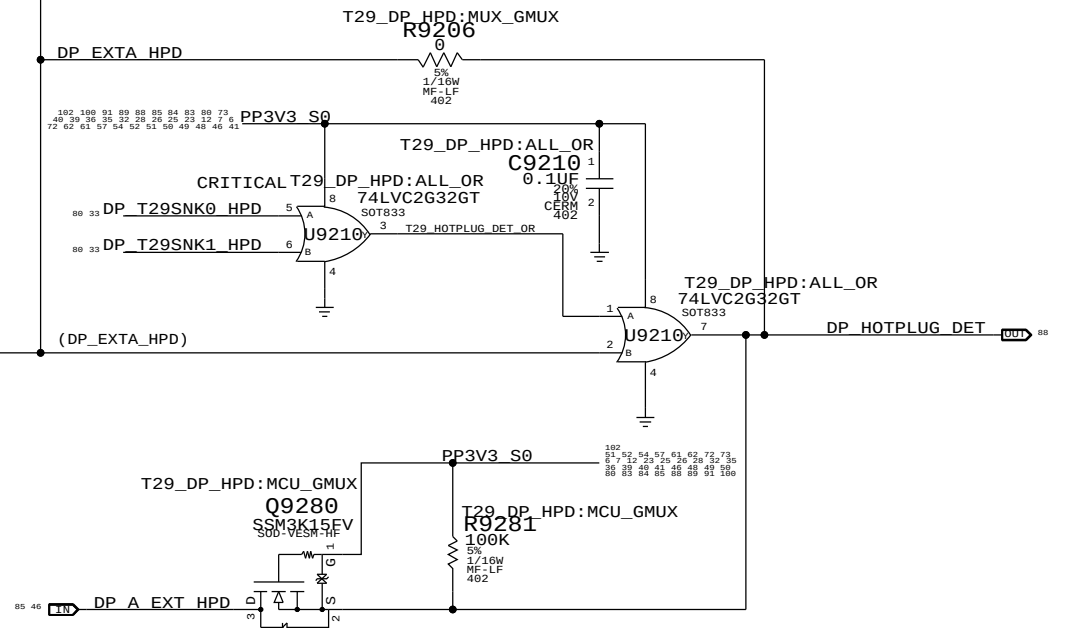
All emulated LVDS outputs require this termination



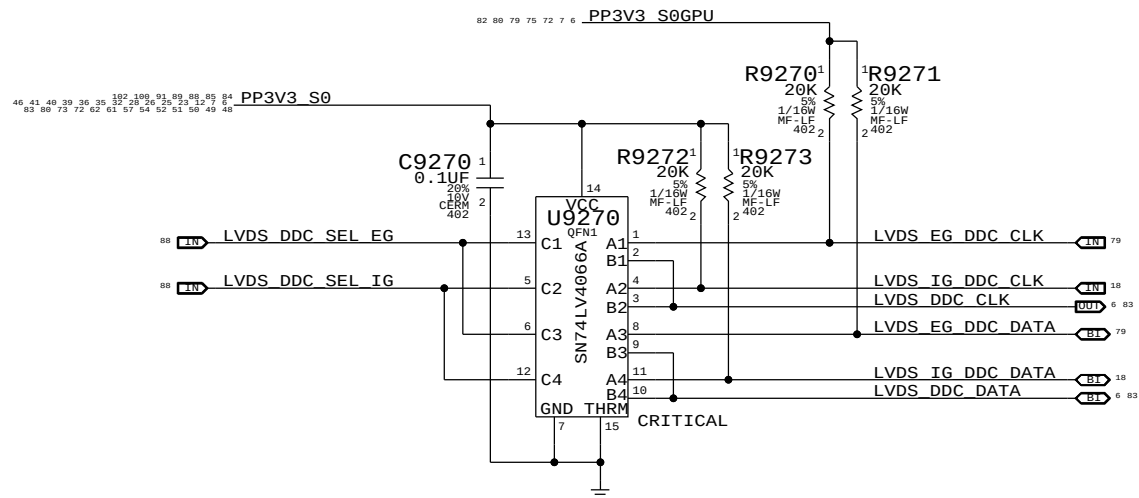
DP AUX, DDC, & HPD muxing to IG/EG



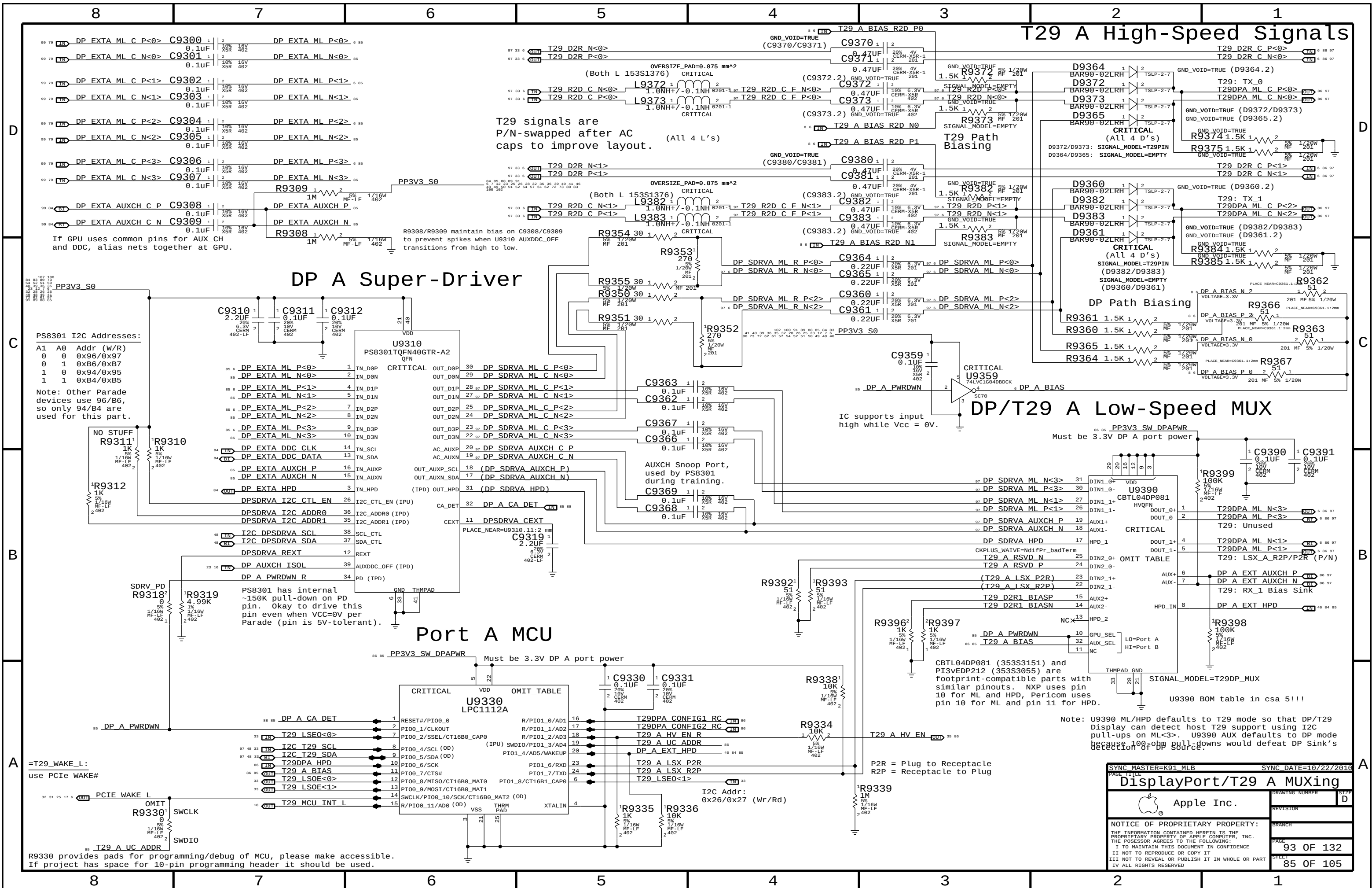
T29/DP HOT PLUG IN



LVDS DDC MUX



PAGE TITLE		PAGE TITLE	
Muxed Graphics Support		Muxed Graphics Support	
Apple Inc.		Apple Inc.	
NOTICE OF PROPRIETARY PROPERTY:		NOTICE OF PROPRIETARY PROPERTY:	
THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE COMPUTER, INC. THE POSSESSOR AGREES TO THE FOLLOWING:		THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE COMPUTER, INC. THE POSSESSOR AGREES TO THE FOLLOWING:	
I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE		I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE	
II NOT TO REPRODUCE OR COPY IT		II NOT TO REPRODUCE OR COPY IT	
III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART		III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART	
IV ALL RIGHTS RESERVED		IV ALL RIGHTS RESERVED	
DRAWING NUMBER		DRAWING NUMBER	
REVISION		REVISION	
BRANCH		BRANCH	
PAGE		PAGE	
92 OF 132		92 OF 132	
SHEET		SHEET	
84 OF 105		84 OF 105	



VQFN25

A

B

C

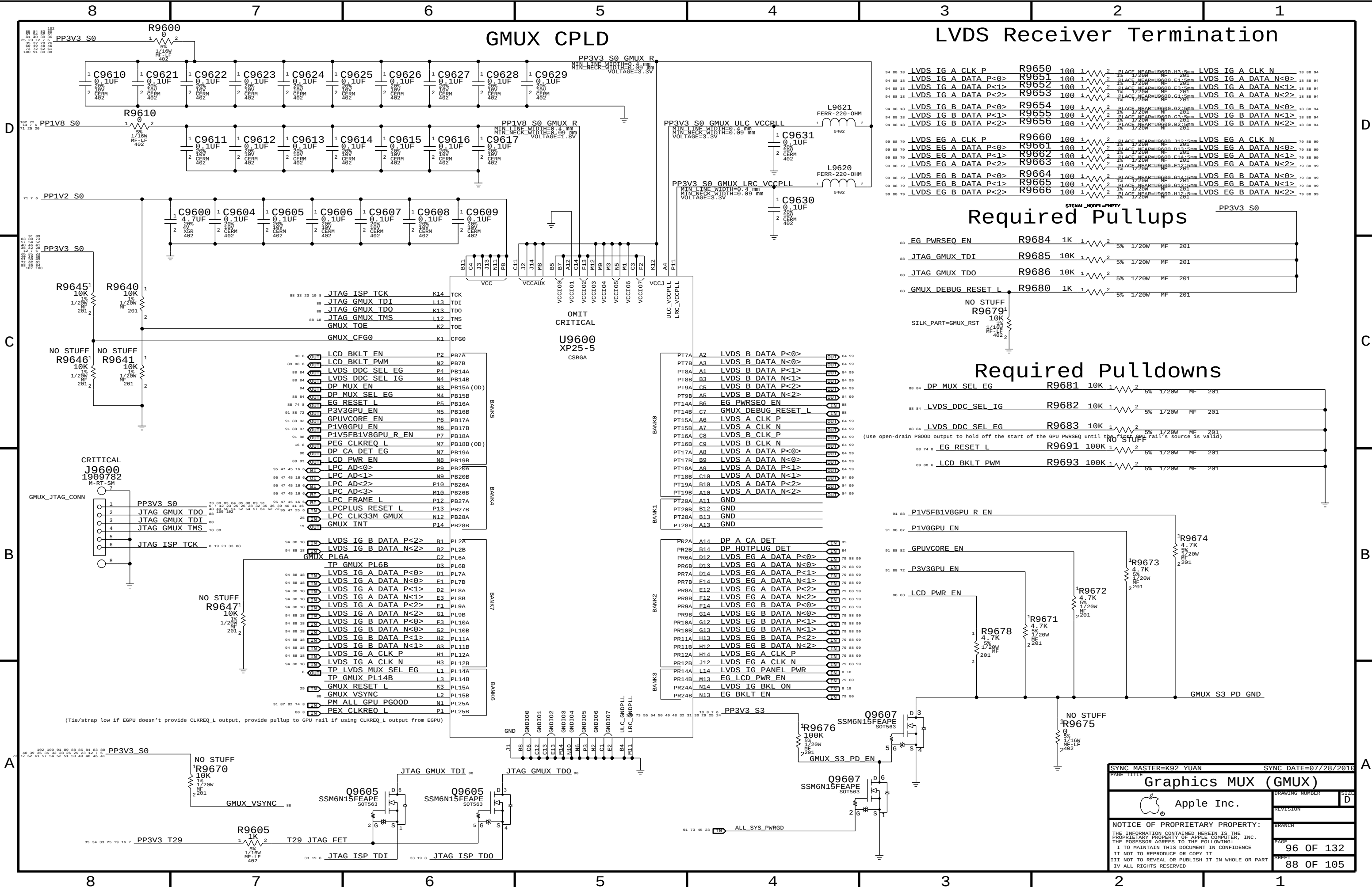
D

A

B

C

D



LVDS Receiver Termination

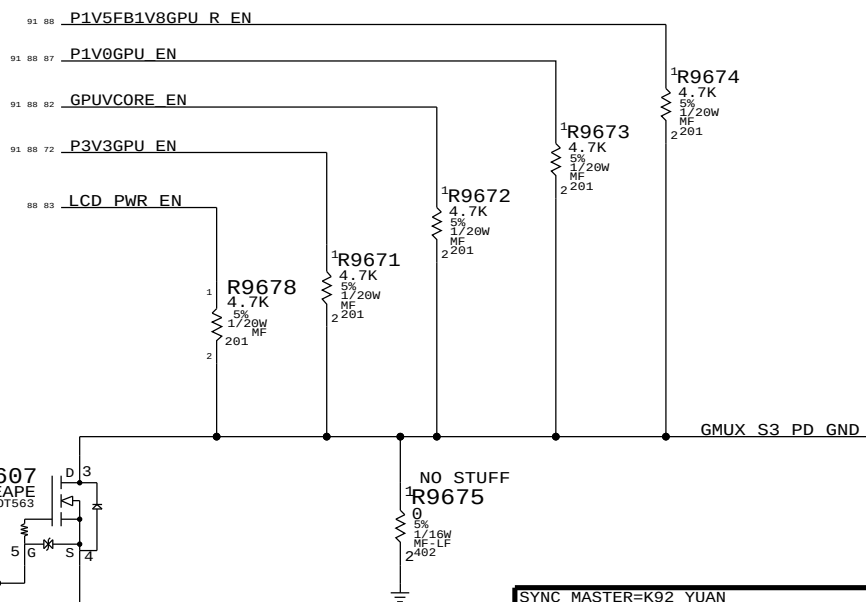
LVDS IG A CLK P	R9650	100	1	2	PLACE NEAR=U9600.H3:5mm	LVDS IG A CLK N	10	88	94
LVDS IG A DATA P<0>	R9651	100	1	2	PLACE NEAR=U9600.E1:5mm	LVDS IG A DATA N<0>	10	88	94
LVDS IG A DATA P<1>	R9652	100	1	2	PLACE NEAR=U9600.E3:5mm	LVDS IG A DATA N<1>	10	88	94
LVDS IG A DATA P<2>	R9653	100	1	2	PLACE NEAR=U9600.G1:5mm	LVDS IG A DATA N<2>	10	88	94
LVDS IG B DATA P<0>	R9654	100	1	2	PLACE NEAR=U9600.G2:5mm	LVDS IG B DATA N<0>	10	88	94
LVDS IG B DATA P<1>	R9655	100	1	2	PLACE NEAR=U9600.G3:5mm	LVDS IG B DATA N<1>	10	88	94
LVDS IG B DATA P<2>	R9656	100	1	2	PLACE NEAR=U9600.B2:5mm	LVDS IG B DATA N<2>	10	88	94
LVDS EG A CLK P	R9660	100	1	2	PLACE NEAR=U9600.H12:5mm	LVDS EG A CLK N	79	88	99
LVDS EG A DATA P<0>	R9661	100	1	2	PLACE NEAR=U9600.D13:5mm	LVDS EG A DATA N<0>	79	88	99
LVDS EG A DATA P<1>	R9662	100	1	2	PLACE NEAR=U9600.E14:5mm	LVDS EG A DATA N<1>	79	88	99
LVDS EG A DATA P<2>	R9663	100	1	2	PLACE NEAR=U9600.F12:5mm	LVDS EG A DATA N<2>	79	88	99
LVDS EG B DATA P<0>	R9664	100	1	2	PLACE NEAR=U9600.G14:5mm	LVDS EG B DATA N<0>	79	88	99
LVDS EG B DATA P<1>	R9665	100	1	2	PLACE NEAR=U9600.G13:5mm	LVDS EG B DATA N<1>	79	88	99
LVDS EG B DATA P<2>	R9666	100	1	2	PLACE NEAR=U9600.H12:5mm	LVDS EG B DATA N<2>	79	88	99

Required Pullups

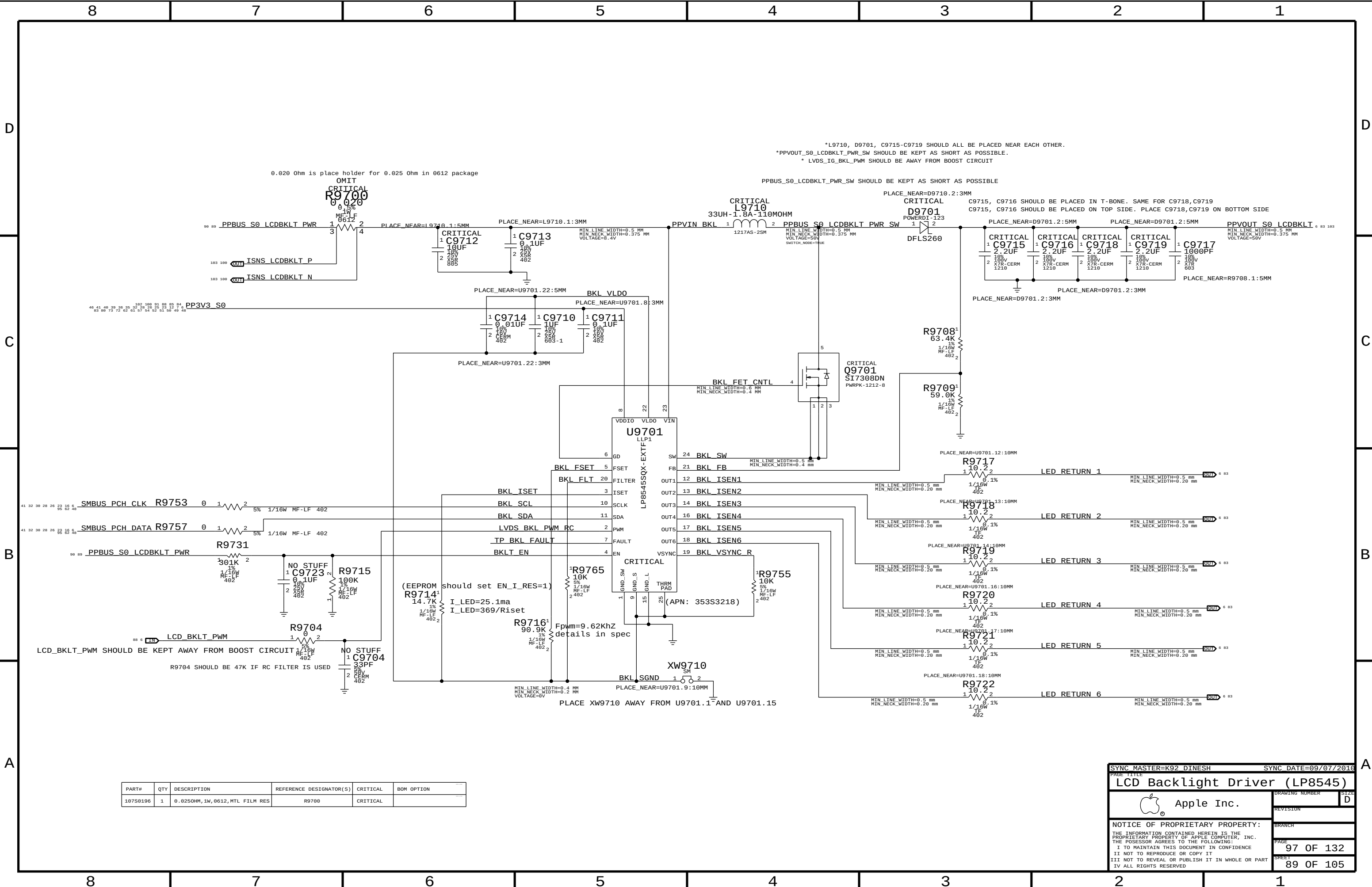
EG_PWRSEQ_EN	R9684	1K	1	2	5%	1/20W	MF	201
JTAG_GMUX_TDI	R9685	10K	1	2	5%	1/20W	MF	201
JTAG_GMUX_TDO	R9686	10K	1	2	5%	1/20W	MF	201
GMUX_DEBUG_RESET_L	R9680	1K	1	2	5%	1/20W	MF	201

Required Pulldowns

DP_MUX_SEL_EG	R9681	10K	1	2	5%	1/20W	MF	201
LVDS_DDC_SEL_IG	R9682	10K	1	2	5%	1/20W	MF	201
LVDS_DDC_SEL_EG	R9683	10K	1	2	5%	1/20W	MF	201
EG_RESET_L	R9691	100K	1	2	5%	1/20W	MF	201
LCD_BKLT_PWM	R9693	100K	1	2	5%	1/20W	MF	201



SYNC MASTER=K92 YUAN		SYNC DATE=07/28/2016	
PAGE TITLE			
Graphics MUX (GMUX)		DRAWING NUMBER	SIZE
Apple Inc.		REVISION	D
NOTICE OF PROPRIETARY PROPERTY:		BRANCH	
THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE COMPUTER, INC. THE POSSESSOR AGREES TO THE FOLLOWING:		PAGE	
I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE		96 OF 132	
II NOT TO REPRODUCE OR COPY IT		SHEET	
III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART		88 OF 105	
IV ALL RIGHTS RESERVED			



PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
10750196	1	0.0250HM,1W,0612,MTL FILM RES	R9700	CRITICAL	

SYNC MASTER=K92 DINESH

SYNC DATE=09/07/2016

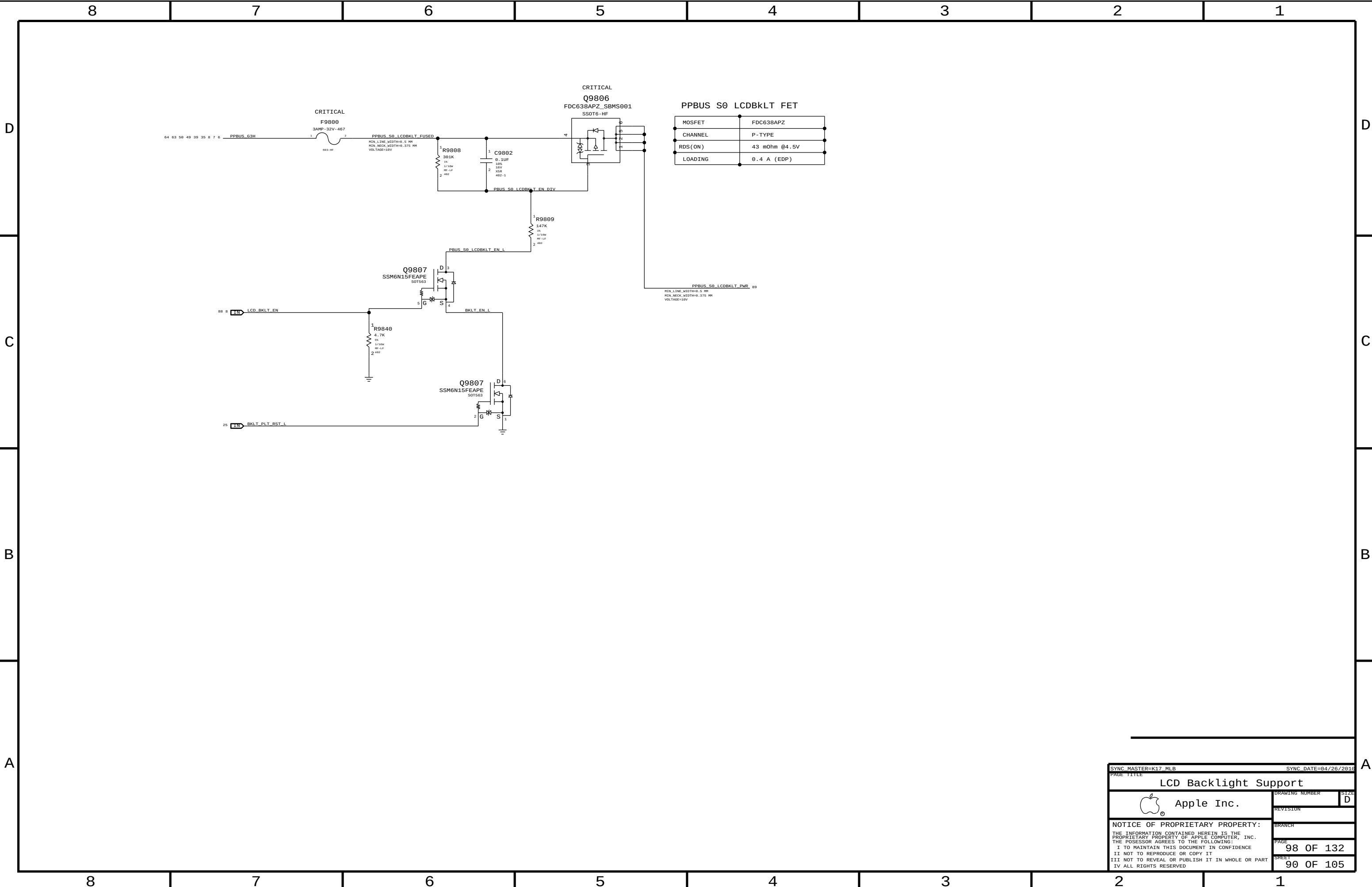
LCD Backlight Driver (LP8545)

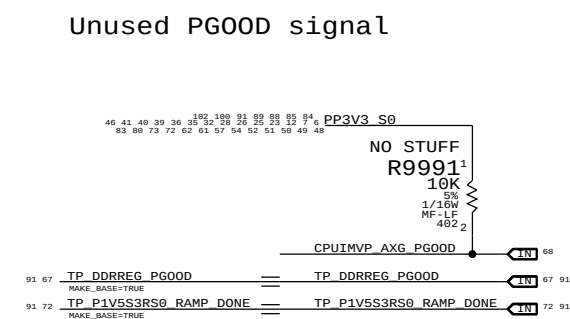
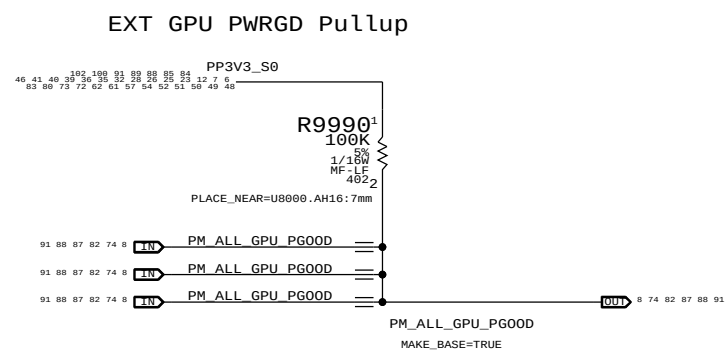
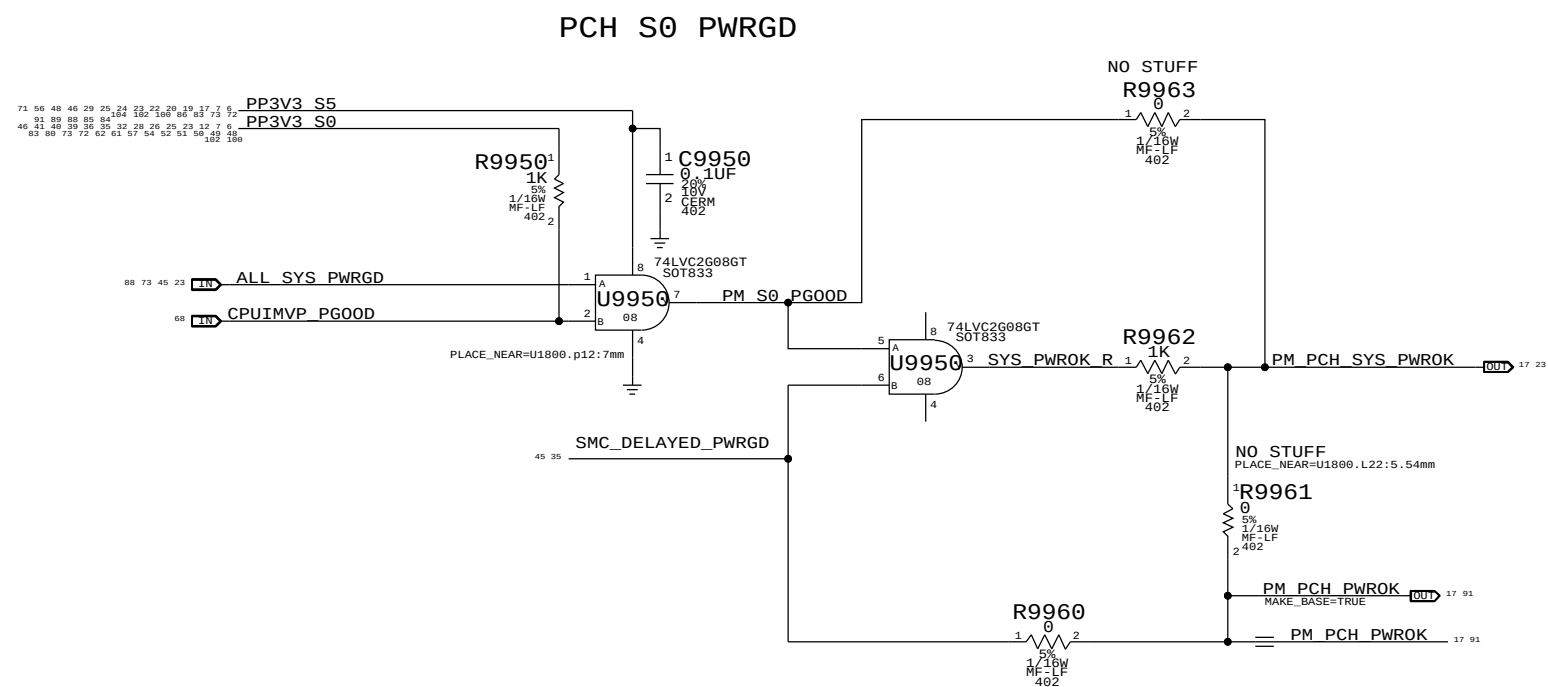
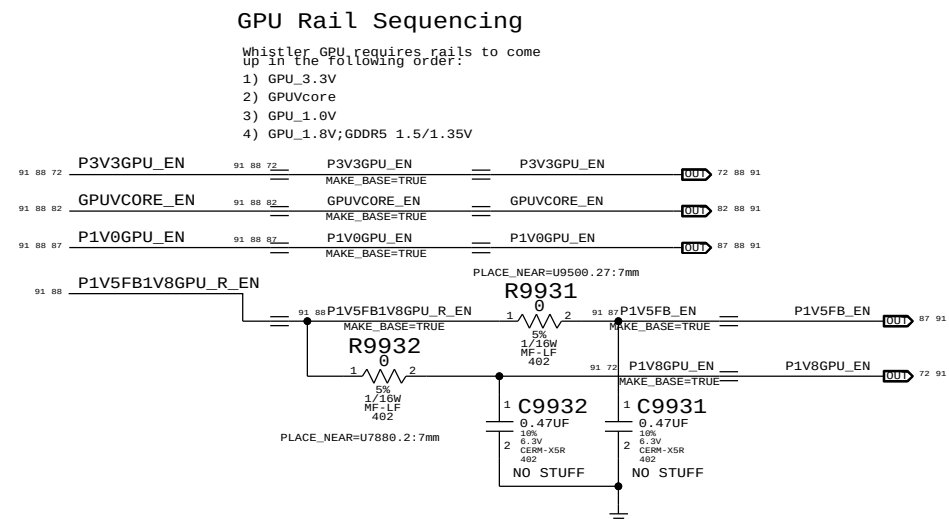
 Apple Inc.

DRAWING NUMBER
SIZE
D

NOTICE OF PROPRIETARY PROPERTY:
THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE COMPUTER, INC. THE POSSESSOR AGREES TO THE FOLLOWING:
I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE
II NOT TO REPRODUCE OR COPY IT
III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART
IV ALL RIGHTS RESERVED

REVISION
BRANCH
PAGE
97 OF 132
SHEET
89 OF 105





SYNC MASTER=K92 YUAN		SYNC DATE=07/30/2016	
PAGE 1		PAGE 1	
Power Sequencing		EG/PCH S0	
 Apple Inc.		DRAWING NUMBER D	
		REVISION	
NOTICE OF PROPRIETARY PROPERTY:		BRANCH	
THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE COMPUTER, INC. THE POSSESSOR AGREES TO THE FOLLOWING: I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE II NOT TO REPRODUCE OR COPY IT III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART IV ALL RIGHTS RESERVED		PAGE 99 OF 132	
		SHEET 91 OF 105	

CPU Signal Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
CPU_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD
CPU_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD
CPU_27P4S	*	=27P4_OHM_SE	=27P4_OHM_SE	=27P4_OHM_SE	=27P4_OHM_SE	7 MIL	7 MIL

NOTE: 7 mil gap is for VCCSense pair, which Intel says to route with 7 mil spacing without specifying a target impedance.

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT	SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
CPU_AGTL	*	=STANDARD	?	CPU_AGTL	TOP, BOTTOM	=2X_DIELECTRIC	?
CPU_8MIL	*	8 MIL	?	CPU_VID	*	0.457 MM	?
CPU_COMP	*	20 MIL	?				
CPU_ITP	*	=2:1_SPACING	?				
CPU_VCCSENSE	*	25 MIL	?				

Most CPU signals with impedance requirements are 50-ohm single-ended. Some signals require 27.4-ohm single-ended impedance.

SOURCE: Calpella SFF DG (DG-407364_v1.5), Section 2.8

PCI-Express

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
PCIE_85D	*	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF
CLK_PCIE_90D	*	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT	SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
PCIE	*	=3X_DIELECTRIC	?	PCIE	TOP, BOTTOM	=4X_DIELECTRIC	?
CLK_PCIE	*	20 MIL	?				

CPU Net Properties

		NET_TYPE		
	ELECTRICAL_CONSTRAINT_SET	PHYSICAL	SPACING	
	DMI_S2N	PCIE_85D	PCIE	DMI S2N P<3:0> 6 9 17
	DMI_S2N	PCIE_85D	PCIE	DMI S2N N<3:0> 6 9 17
	DMI_N2S	PCIE_85D	PCIE	DMI N2S P<3:0> 6 9 17
	DMI_N2S	PCIE_85D	PCIE	DMI N2S N<3:0> 6 9 17
	FDI_DATA	PCIE_85D	PCIE	FDI DATA P<7:0> 9 17
	FDI_DATA	PCIE_85D	PCIE	FDI DATA N<7:0> 9 17
		CPU_50S	CPU_AGTL	FDI FSYNC<1..0> 9 17
		CPU_50S	CPU_AGTL	FDI LSYNC<1..0> 9 17
112D	DMI_CLK100M	CLK_PCIE_90D	CLK_PCIE	DMI CLK100M CPU P 10 16
112D		CLK_PCIE_90D	CLK_PCIE	DMI CLK100M CPU N 10 16
		CPU_50S	CPU_AGTL	FDI INT 9 17
	CPU_PECT	CPU_50S	PCIE	CPU PECT 10 19 45
	PM_SYNC	CPU_50S	CPU_AGTL	PM SYNC 10 17
	PM_MEM_PWRGD	CPU_50S	CPU_AGTL	PM MEM_PWRGD 10 17 29
	XDP_CPU_PWRG00D	CPU_50S	CPU_ITP	XDP CPU_PWRGD 23
	XDP_DBRESET_I	CPU_50S	CPU_ITP	XDP DBRESET L 10 23 25
	XDP_PRDY_I	CPU_50S	CPU_ITP	XDP CPU_PRDY L 10 23
	XDP_PREQ_L	CPU_50S	CPU_ITP	XDP CPU_PREQ L 10 23
	CPU_SM_RCOMP	CPU_27P4S	CPU_COMP	CPU SM_RCOMP0
	CPU_SM_RCOMP	CPU_27P4S	CPU_COMP	CPU SM_RCOMP1
	CPU_SM_RCOMP	CPU_27P4S	CPU_COMP	CPU SM_RCOMP2
	CPU_CFG	CPU_50S	CPU_ITP	CPU_CFG<11..0> 9 23
112D	CPU_CFG	CPU_50S	CPU_ITP	CPU_CFG<17..16> 9 23
112D	CPU_CATERR_I	CPU_50S	CPU_AGTL	CPU_CATERR L 10
112D		CPU_50S	CPU_AGTL	CPU_PROC_SEL L 10 17
		CPU_50S	CPU_AGTL	TP_CPU_VTT_SELECT 8
	CPU_PROCHOT_L	CPU_50S	CPU_AGTL	CPU_PROCHOT L 10 46 68
	CPU_PWRGD	CPU_50S	CPU_AGTL	CPU_PWRGD 10 19 23
	PM_THRMTRIP_L	CPU_50S	CPU_8MIL	PM_THRMTRIP L 10 19
	XDP_CLK_CPU	CLK_PCIE_90D	CLK_PCIE	ITPCPU_CLK100M P 10 16
	XDP_CLK_CPU	CLK_PCIE_90D	CLK_PCIE	ITPCPU_CLK100M N 10 16
	XDP_CLK_PCH	CLK_PCIE_90D	CLK_PCIE	ITPXDPCLK100M P 16 23
	XDP_CLK_PCH	CLK_PCIE_90D	CLK_PCIE	ITPXDPCLK100M N 16 23
	XDP_CLK_ITP	CLK_PCIE_90D	CLK_PCIE	XDPCPU_CLK100M_P 23
	XDP_CLK_ITP	CLK_PCIE_90D	CLK_PCIE	XDPCPU_CLK100M_N 23
	PM_DPRSPLVR	CPU_55S	CPU_8MTI	CPU_PSI L
		CPU_50S	CPU_AGTL	PM_DPRSPLVR
		CPU_27P4S	CPU_COMP	CPU_PEG_COMP 9
		CPU_27P4S	CPU_COMP	CPU_PEG_RBIAS
	CPU_COMP	CPU_27P4S	CPU_COMP	CPU_COMP3
	CPU_COMP	CPU_27P4S	CPU_COMP	CPU_COMP2
	CPU_COMP	CPU_27P4S	CPU_COMP	CPU_COMP1
	CPU_COMP	CPU_27P4S	CPU_COMP	CPU_COMP0
	XDP_TDI	CPU_50S	CPU_ITP	XDPCPU_TDI 10 23
	XDP_TDO	CPU_50S	CPU_ITP	XDPCPU_TDO 10 23
	XDP_TMS	CPU_50S	CPU_ITP	XDPCPU_TMS 10 23
	XDP_TCK	CPU_50S	CPU_ITP	XDPCPU_TCK 10 23
	XDP_TRST_I	CPU_50S	CPU_ITP	XDPCPU_TRST L 10 23
	XDP_BPM	CPU_50S	CPU_ITP	XDPCPU_BPM L<3..0> 10 23
	XDP_BPM_L	CPU_50S	CPU_ITP	XDPCPU_BPM L<7..4> 10 23
	(FSB_CPURST_L)	CPU_50S	CPU_ITP	XDPCPURST L 23
		CPU_55S	CPU_8MTI	CPU_VID<6..0> 8
		CPU_50S	CPU_AGTL	CPUIMVP_IMON
	CPU_VCCSENSE	CPU_27P4S	CPU_VCCSENSE	CPU_VCCSENSE_P 12 68
	CPU_VCCSENSE	CPU_27P4S	CPU_VCCSENSE	CPU_VCCSENSE_N 12 68
	CPU_VCCSENSE	CPU_27P4S	CPU_VCCSENSE	CPU_VCCIOSENSE_P 12 70
	CPU_VCCSENSE	CPU_27P4S	CPU_VCCSENSE	CPU_VCCIOSENSE_N 12 70
	CPU_VCCSENSE	CPU_27P4S	CPU_VCCSENSE	CPU_AXG_SENSE_P 12 68
	CPU_VCCSENSE	CPU_27P4S	CPU_VCCSENSE	CPU_AXG_SENSE_N 12 68
112D		CPU_27P4S	CPU_VCCSENSE	CPU_VCC_VALSENSE_P 12
112D		CPU_27P4S	CPU_VCCSENSE	CPU_VCC_VALSENSE_N 12
112D		CPU_27P4S	CPU_VCCSENSE	CPU_AXG_VALSENSE_P 12
112D		CPU_27P4S	CPU_VCCSENSE	CPU_AXG_VALSENSE_N 12
	PM_DPRSPLVR	CPU_55S	CPU_8MTI	GFX_VID<6..0> 8
		CPU_50S	CPU_AGTL	GFX_DPRSPLVR
		CPU_50S	CPU_AGTL	GFX_VR_EN
		CPU_50S	CPU_AGTL	GFXIMVP_IMON
		PCIE_85D	PCIE	PEG_R2D_P<7..0> 74
		PCIE_85D	PCIE	PEG_R2D_N<7..0> 74
	PEG_R2D	PCIE_85D	PCIE	PEG_R2D_C_P<7..0> 8 74
		PCIE_85D	PCIE	PEG_R2D_C_N<7..0> 8 74
	PEG_D2R	PCIE_85D	PCIE	PEG_D2R_P<7..0> 8 74
		PCIE_85D	PCIE	PEG_D2R_N<7..0> 8 74
		PCIE_85D	PCIE	PEG_D2R_C_P<7..0> 74
		PCIE_85D	PCIE	PEG_D2R_C_N<7..0> 74
		CPU_50S	CPU_VID	CPU_VIDSOUT 12 68
		CPU_50S	CPU_VID	CPU_VIDCLK 12 68
		CPU_50S	CPU_VID	CPU_VIDALERT_L 12 68

SYNC_MASTER=K91_MLB

SYNC_DATE=07/22/2016

CPU Constraints

Apple Inc.

NOTICE OF PROPRIETARY PROPERTY:
THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE COMPUTER, INC. THE POSSESSOR AGREES TO THE FOLLOWING:
I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE
II NOT TO REPRODUCE OR COPY IT
III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART
IV ALL RIGHTS RESERVED

DRAWING NUMBER

SIZE

REVISION

BRANCH

PAGE

SHEET

100 OF 132

92 OF 105

D

C

B

A

Memory Bus Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
MEM_37S	*	=37_OHM_SE	=37_OHM_SE	=37_OHM_SE	=37_OHM_SE	=STANDARD	=STANDARD
MEM_40S	*	=40_OHM_SE	=40_OHM_SE	=40_OHM_SE	=40_OHM_SE	=STANDARD	=STANDARD
MEM_72D	*	=72_OHM_DIFF	=72_OHM_DIFF	=72_OHM_DIFF	=72_OHM_DIFF	=72_OHM_DIFF	=72_OHM_DIFF
MEM_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD
MEM_85D	*	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
MEM_CLK2MEM	*	=4:1_SPACING	?
MEM_CTRL2CTRL	*	=3:1_SPACING	?
MEM_CTRL2MEM	*	=2.5:1_SPACING	?
MEM_CMD2CMD	*	=1.5:1_SPACING	?
MEM_CMD2MEM	*	=3:1_SPACING	?
MEM_DATA2DATA	*	=1.5:1_SPACING	?
MEM_DATA2MEM	*	=3:1_SPACING	?
MEM_DQS2MEM	*	=3:1_SPACING	?
MEM_20THER	*	25 MILS	?

Memory Bus Spacing Group Assignments

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_CLK	MEM_CLK	*	MEM_CLK2MEM
MEM_CLK	MEM_CTRL	*	MEM_CLK2MEM
MEM_CLK	MEM_CMD	*	MEM_CLK2MEM
MEM_CLK	MEM_DATA	*	MEM_CLK2MEM
MEM_CLK	MEM_DQS	*	MEM_CLK2MEM

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_CMD	MEM_CLK	*	MEM_CMD2MEM
MEM_CMD	MEM_CTRL	*	MEM_CMD2MEM
MEM_CMD	MEM_CMD	*	MEM_CMD2CMD
MEM_CMD	MEM_DATA	*	MEM_CMD2MEM
MEM_CMD	MEM_DQS	*	MEM_CMD2MEM

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_CTRL	MEM_CLK	*	MEM_CTRL2MEM
MEM_CTRL	MEM_CTRL	*	MEM_CTRL2CTRL
MEM_CTRL	MEM_CMD	*	MEM_CTRL2MEM
MEM_CTRL	MEM_DATA	*	MEM_CTRL2MEM
MEM_CTRL	MEM_DQS	*	MEM_CTRL2MEM

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_DATA	MEM_CLK	*	MEM_DATA2MEM
MEM_DATA	MEM_CTRL	*	MEM_DATA2MEM
MEM_DATA	MEM_CMD	*	MEM_DATA2MEM
MEM_DATA	MEM_DATA	*	MEM_DATA2DATA
MEM_DATA	MEM_DQS	*	MEM_DATA2MEM

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_DQS	MEM_CLK	*	MEM_DQS2MEM
MEM_DQS	MEM_CTRL	*	MEM_DQS2MEM
MEM_DQS	MEM_CMD	*	MEM_DQS2MEM
MEM_DQS	MEM_DATA	*	MEM_DQS2MEM
MEM_DQS	MEM_DQS	*	MEM_DQS2MEM

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_CLK	*	*	MEM_20THER
MEM_CTRL	*	*	MEM_20THER
MEM_CMD	*	*	MEM_20THER
MEM_DATA	*	*	MEM_20THER
MEM_DQS	*	*	MEM_20THER

DDR3:

DQ/DM signals should be matched within 0.508mm of associated DQS pair.

DQS intra-pair matching should be within 0.127mm, no inter-pair matching requirement.

DQS to clock matching should be within [CLK-12.7mm] and [CLK+25.4mm].

CLK intra-pair matching should be within 0.127mm, inter-pair matching should be within 0.508mm.

CONTROL signals should be matched within [CLK-12.7mm] to [CLK+0.0mm] of CLK pairs.

A/BA/CMD signals should be matched within [CLK-12.7mm] to [CLK+12.7mm] of CLK pairs.

DQ/DQS/A/BA/cmd signal spacing is 4x dielectric, CLK is 5x dielectric.

Maximum length of any signal from die pad to SODIMM pad is 139.7mm, from procesor ball to SODIMM pad is 114.3mm.

SOURCE: Calpella SFF Platform DG, Rev 1.5 (#407364), Section 2.2

Memory Net Properties

ELECTRICAL_CONSTRAINT_SET	NET_TYPE			
	PHYSICAL	SPACING		
MEM_A_CLK	MEM_72D	MEM_CLK	MEM A CLK P<5..0>	11 26
MEM_A_CLK	MEM_72D	MEM_CLK	MEM A CLK N<5..0>	11 26
MEM_A_CNTL	MEM_37S	MEM_CTRL	MEM A CKE<3..0>	11 26
MEM_A_CNTL	MEM_37S	MEM_CTRL	MEM A CS L<3..0>	11 26
MEM_A_CNTL	MEM_37S	MEM_CTRL	MEM A ODT<3..0>	11 26
MEM_A_CMD	MEM_40S	MEM_CMD	MEM A A<15..0>	11 26
MEM_A_CMD	MEM_40S	MEM_CMD	MEM A BA<2..0>	11 26
MEM_A_CMD	MEM_40S	MEM_CMD	MEM A RAS L	11 26
MEM_A_CMD	MEM_40S	MEM_CMD	MEM A CAS L	11 26
MEM_A_CMD	MEM_40S	MEM_CMD	MEM A WE L	11 26
MEM_A_DQ_BYTE0	MEM_50S	MEM_DATA	MEM A DQ<7..0>	11 27
MEM_A_DQ_BYTE1	MEM_50S	MEM_DATA	MEM A DQ<15..8>	11 27
MEM_A_DQ_BYTE2	MEM_50S	MEM_DATA	MEM A DQ<23..16>	11 27
MEM_A_DQ_BYTE3	MEM_50S	MEM_DATA	MEM A DQ<31..24>	11 27
MEM_A_DQ_BYTE4	MEM_50S	MEM_DATA	MEM A DQ<39..32>	11 26 27
MEM_A_DQ_BYTE5	MEM_50S	MEM_DATA	MEM A DQ<47..40>	11 27
MEM_A_DQ_BYTE6	MEM_50S	MEM_DATA	MEM A DQ<55..48>	11 27
MEM_A_DQ_BYTE7	MEM_50S	MEM_DATA	MEM A DQ<63..56>	11 27
MEM_A_DQS0	MEM_85D	MEM_DQS	MEM A DQS P<0>	11 27
MEM_A_DQS0	MEM_85D	MEM_DQS	MEM A DQS N<0>	11 26 27
MEM_A_DQS1	MEM_85D	MEM_DQS	MEM A DQS P<1>	11 27
MEM_A_DQS1	MEM_85D	MEM_DQS	MEM A DQS N<1>	11 27
MEM_A_DQS2	MEM_85D	MEM_DQS	MEM A DQS P<2>	11 27
MEM_A_DQS2	MEM_85D	MEM_DQS	MEM A DQS N<2>	11 27
MEM_A_DQS3	MEM_85D	MEM_DQS	MEM A DQS P<3>	11 27
MEM_A_DQS3	MEM_85D	MEM_DQS	MEM A DQS N<3>	11 27
MEM_A_DQS4	MEM_85D	MEM_DQS	MEM A DQS P<4>	11 27
MEM_A_DQS4	MEM_85D	MEM_DQS	MEM A DQS N<4>	11 27
MEM_A_DQS5	MEM_85D	MEM_DQS	MEM A DQS P<5>	11 27
MEM_A_DQS5	MEM_85D	MEM_DQS	MEM A DQS N<5>	11 27
MEM_A_DQS6	MEM_85D	MEM_DQS	MEM A DQS P<6>	11 27
MEM_A_DQS6	MEM_85D	MEM_DQS	MEM A DQS N<6>	11 27
MEM_A_DQS7	MEM_85D	MEM_DQS	MEM A DQS P<7>	11 27
MEM_A_DQS7	MEM_85D	MEM_DQS	MEM A DQS N<7>	11 27
MEM_B_CLK	MEM_72D	MEM_CLK	MEM B CLK P<5..0>	11 28
MEM_B_CLK	MEM_72D	MEM_CLK	MEM B CLK N<5..0>	11 28
MEM_B_CNTL	MEM_37S	MEM_CTRL	MEM B CKE<3..0>	11 28
MEM_B_CNTL	MEM_37S	MEM_CTRL	MEM B CS L<3..0>	11 28
MEM_B_CNTL	MEM_37S	MEM_CTRL	MEM B ODT<3..0>	11 28
MEM_B_CMD	MEM_40S	MEM_CMD	MEM B A<15..0>	11 28
MEM_B_CMD	MEM_40S	MEM_CMD	MEM B BA<2..0>	11 28
MEM_B_CMD	MEM_40S	MEM_CMD	MEM B RAS L	11 28
MEM_B_CMD	MEM_40S	MEM_CMD	MEM B CAS L	11 28
MEM_B_CMD	MEM_40S	MEM_CMD	MEM B WE L	11 28
MEM_B_DQ_BYTE0	MEM_50S	MEM_DATA	MEM B DQ<7..0>	11 27
MEM_B_DQ_BYTE1	MEM_50S	MEM_DATA	MEM B DQ<15..8>	11 27
MEM_B_DQ_BYTE2	MEM_50S	MEM_DATA	MEM B DQ<23..16>	11 27
MEM_B_DQ_BYTE3	MEM_50S	MEM_DATA	MEM B DQ<31..24>	11 27
MEM_B_DQ_BYTE4	MEM_50S	MEM_DATA	MEM B DQ<39..32>	11 27 28
MEM_B_DQ_BYTE5	MEM_50S	MEM_DATA	MEM B DQ<47..40>	11 27
MEM_B_DQ_BYTE6	MEM_50S	MEM_DATA	MEM B DQ<55..48>	11 27
MEM_B_DQ_BYTE7	MEM_50S	MEM_DATA	MEM B DQ<63..56>	11 27
MEM_B_DQS0	MEM_85D	MEM_DQS	MEM B DQS P<0>	11 27 28
MEM_B_DQS0	MEM_85D	MEM_DQS	MEM B DQS N<0>	11 27 28
MEM_B_DQS1	MEM_85D	MEM_DQS	MEM B DQS P<1>	11 27
MEM_B_DQS1	MEM_85D	MEM_DQS	MEM B DQS N<1>	11 27
MEM_B_DQS2	MEM_85D	MEM_DQS	MEM B DQS P<2>	11 27
MEM_B_DQS2	MEM_85D	MEM_DQS	MEM B DQS N<2>	11 27
MEM_B_DQS3	MEM_85D	MEM_DQS	MEM B DQS P<3>	11 27
MEM_B_DQS3	MEM_85D	MEM_DQS	MEM B DQS N<3>	11 27
MEM_B_DQS4	MEM_85D	MEM_DQS	MEM B DQS P<4>	11 27
MEM_B_DQS4	MEM_85D	MEM_DQS	MEM B DQS N<4>	11 27
MEM_B_DQS5	MEM_85D	MEM_DQS	MEM B DQS P<5>	11 27
MEM_B_DQS5	MEM_85D	MEM_DQS	MEM B DQS N<5>	11 27
MEM_B_DQS6	MEM_85D	MEM_DQS	MEM B DQS P<6>	11 27
MEM_B_DQS6	MEM_85D	MEM_DQS	MEM B DQS N<6>	11 27
MEM_B_DQS7	MEM_85D	MEM_DQS	MEM B DQS P<7>	11 27
MEM_B_DQS7	MEM_85D	MEM_DQS	MEM B DQS N<7>	11 27

SYNC_MASTER=K17_MLB

SYNC_DATE=05/14/2016

PAGE TITLE

Memory Constraints

 Apple Inc.

DRAWING NUMBER

SIZE

D

NOTICE OF PROPRIETARY PROPERTY:

THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE COMPUTER, INC. THE POSSESSOR AGREES TO THE FOLLOWING:

I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE

II NOT TO REPRODUCE OR COPY IT

III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART

IV ALL RIGHTS RESERVED

REVISION

BRANCH

PAGE

101 OF 132

SHEET

93 OF 105

D

CB

A

B

CB

A

B

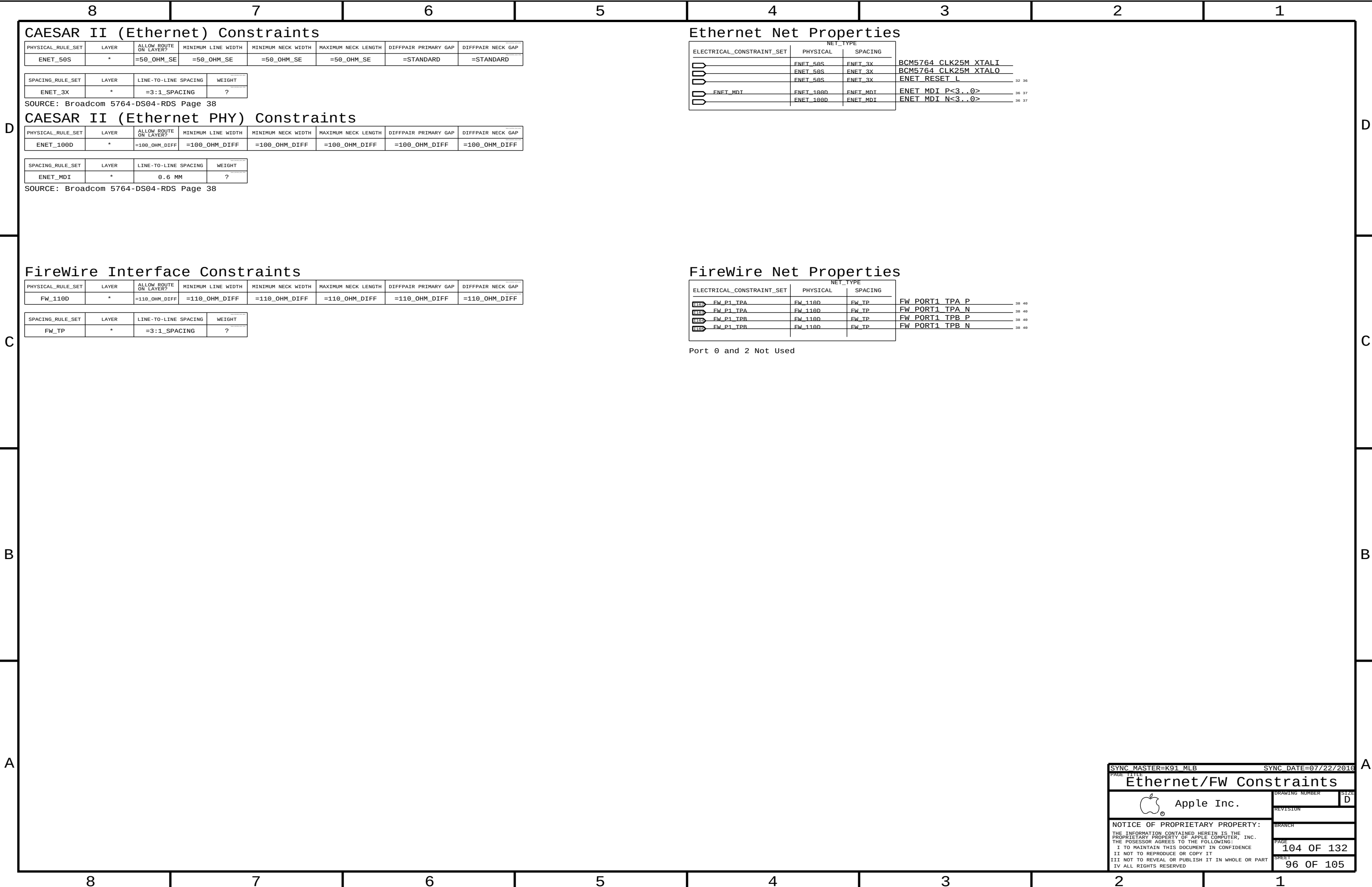
A

B

A

D

CB



D

C

B

A

D

B

AD

B

AD

A

B

A

A

D

C

B

A

A

D

C

B

A

GDDR5 Frame Buffer Signal Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
GDDR5_45R50SE	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	12.7 MM	=STANDARD	=STANDARD
GDDR5_45SE	*	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=STANDARD	=STANDARD
GDDR5_80D	*	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
GDDR5_CLK	*	=5x_DIELECTRIC	?
GDDR5_CMD	*	=2x_DIELECTRIC	?
GDDR5_DATA	*	=3x_DIELECTRIC	?
GDDR5_EDC	*	=7x_DIELECTRIC	?

Digital Video Signal Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
DP_85D	*	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF
LVDS_85D	*	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
DISPLAYPORT	*	=3x_DIELECTRIC	?
LVDS	*	=3x_DIELECTRIC	?

LVDS intra-pair matching should be 0.127 mm. Pairs should be within 0.508mm of entire channel.
DisplayPort/TMDS intra-pair matching should be 0.127mm. Inter-pair matching should be within 2.54cm. Max Length 241.3mm.
DisplayPort AUX CH intra-pair matching should be 0.127mm. Max length 330.2mm.
Max length of LVDS/DisplayPort/TMDS traces: 13 inches.
SOURCE: Calpella SFF DG Rev 1.5 (407364) and Family GPU DG-04202-001-v04.

GDDR5 FB A Net Properties

ELECTRICAL_CONSTRAINT_SET		NET_TYPE		
		PHYSICAL	SPACING	
	FB_A0_CLK	GDDR5_80D	GDDR5_CLK	FB A0 CLK P
	FB_A0_CLK	GDDR5_80D	GDDR5_CLK	FB A0 CLK N
	FB_A1_CLK	GDDR5_80D	GDDR5_CLK	FB A1 CLK P
	FB_A1_CLK	GDDR5_80D	GDDR5_CLK	FB A1 CLK N
	FB_A0_CMD	GDDR5_45R50SE	GDDR5_CMD	FB A0 A<8..0>
	FB_A1_CMD	GDDR5_45R50SE	GDDR5_CMD	FB A1 A<8..0>
	FB_A0_CMD	GDDR5_45R50SE	GDDR5_CMD	FB A0 ABI L
	FB_A1_CMD	GDDR5_45R50SE	GDDR5_CMD	FB A1 ABI L
	FB_A0_CMD	GDDR5_45R50SE	GDDR5_CMD	FB A0 RAS L
	FB_A1_CMD	GDDR5_45R50SE	GDDR5_CMD	FB A1 RAS L
	FB_A0_CMD	GDDR5_45R50SE	GDDR5_CMD	FB A0 CAS L
	FB_A1_CMD	GDDR5_45R50SE	GDDR5_CMD	FB A1 CAS L
	FB_A0_CMD	GDDR5_45R50SE	GDDR5_CMD	FB A0 WE L
	FB_A1_CMD	GDDR5_45R50SE	GDDR5_CMD	FB A1 WE L
	FB_A0_CMD	GDDR5_45R50SE	GDDR5_CMD	FB A0 CKE L
	FB_A1_CMD	GDDR5_45R50SE	GDDR5_CMD	FB A1 CKE L
	FB_A0_CMD	GDDR5_45R50SE	GDDR5_CMD	FB A0 CS L
	FB_A1_CMD	GDDR5_45R50SE	GDDR5_CMD	FB A1 CS L
	FB_A0_EDC0	GDDR5_45SE	GDDR5_EDC	FB A0 EDC<0>
1P20	FB_A0_EDC1	GDDR5_45SE	GDDR5_EDC	FB A0 EDC<1>
1P20	FB_A0_EDC2	GDDR5_45SE	GDDR5_EDC	FB A0 EDC<2>
1P20	FB_A0_EDC3	GDDR5_45SE	GDDR5_EDC	FB A0 EDC<3>
1P20	FB_A1_EDC0	GDDR5_45SE	GDDR5_EDC	FB A1 EDC<0>
1P20	FB_A1_EDC1	GDDR5_45SE	GDDR5_EDC	FB A1 EDC<1>
1P20	FB_A1_EDC2	GDDR5_45SE	GDDR5_EDC	FB A1 EDC<2>
1P20	FB_A1_EDC3	GDDR5_45SE	GDDR5_EDC	FB A1 EDC<3>
	FB_A0_DBI_L0	GDDR5_45SE	GDDR5_DATA	FB A0 DBI L<0>
1P20	FB_A0_DBI_L1	GDDR5_45SE	GDDR5_DATA	FB A0 DBI L<1>
1P20	FB_A0_DBI_L2	GDDR5_45SE	GDDR5_DATA	FB A0 DBI L<2>
1P20	FB_A0_DBI_L3	GDDR5_45SE	GDDR5_DATA	FB A0 DBI L<3>
1P20	FB_A1_DBI_L0	GDDR5_45SE	GDDR5_DATA	FB A1 DBI L<0>
1P20	FB_A1_DBI_L1	GDDR5_45SE	GDDR5_DATA	FB A1 DBI L<1>
1P20	FB_A1_DBI_L2	GDDR5_45SE	GDDR5_DATA	FB A1 DBI L<2>
1P20	FB_A1_DBI_L3	GDDR5_45SE	GDDR5_DATA	FB A1 DBI L<3>
	FB_A0_WCLK0	GDDR5_80D	GDDR5_CMD	FB A0 WCLK P<0>
	FB_A0_WCLK0	GDDR5_80D	GDDR5_CMD	FB A0 WCLK N<0>
	FB_A0_WCLK1	GDDR5_80D	GDDR5_CMD	FB A0 WCLK P<1>
	FB_A0_WCLK1	GDDR5_80D	GDDR5_CMD	FB A0 WCLK N<1>
	FB_A1_WCLK0	GDDR5_80D	GDDR5_CMD	FB A1 WCLK P<0>
	FB_A1_WCLK0	GDDR5_80D	GDDR5_CMD	FB A1 WCLK N<0>
	FB_A1_WCLK1	GDDR5_80D	GDDR5_CMD	FB A1 WCLK P<1>
	FB_A1_WCLK1	GDDR5_80D	GDDR5_CMD	FB A1 WCLK N<1>
	FB_A0_DQ_BYTE0	GDDR5_45SE	GDDR5_DATA	FB A0 DQ<7..0>
	FB_A0_DQ_BYTE1	GDDR5_45SE	GDDR5_DATA	FB A0 DQ<15..8>
	FB_A0_DQ_BYTE2	GDDR5_45SE	GDDR5_DATA	FB A0 DQ<23..16>
	FB_A0_DQ_BYTE3	GDDR5_45SE	GDDR5_DATA	FB A0 DQ<31..24>
	FB_A1_DQ_BYTE0	GDDR5_45SE	GDDR5_DATA	FB A1 DQ<7..0>
	FB_A1_DQ_BYTE1	GDDR5_45SE	GDDR5_DATA	FB A1 DQ<15..8>
	FB_A1_DQ_BYTE2	GDDR5_45SE	GDDR5_DATA	FB A1 DQ<23..16>
	FB_A1_DQ_BYTE3	GDDR5_45SE	GDDR5_DATA	FB A1 DQ<31..24>
	FB_AB_RESET	GDDR5_45R50SE	GDDR5_CMD	FB RESET L

GDDR5 FB B Net Properties

ELECTRICAL_CONSTRAINT_SET		NET_TYPE	
		PHYSICAL	SPACING
	FB_B0_CLK	GDDR5_80D	GDDR5_CLK
	FB_B0_CLK	GDDR5_80D	GDDR5_CLK
	FB_B1_CLK	GDDR5_80D	GDDR5_CLK
	FB_B1_CLK	GDDR5_80D	GDDR5_CLK
	FB_B0_CMD	GDDR5_45R50SE	GDDR5_CMD
	FB_B1_CMD	GDDR5_45R50SE	GDDR5_CMD
	FB_B0_CMD	GDDR5_45R50SE	GDDR5_CMD
	FB_B1_CMD	GDDR5_45R50SE	GDDR5_CMD
	FB_B0_CMD	GDDR5_45R50SE	GDDR5_CMD
	FB_B1_CMD	GDDR5_45R50SE	GDDR5_CMD
	FB_B0_CMD	GDDR5_45R50SE	GDDR5_CMD
	FB_B1_CMD	GDDR5_45R50SE	GDDR5_CMD
	FB_B0_CMD	GDDR5_45R50SE	GDDR5_CMD
	FB_B1_CMD	GDDR5_45R50SE	GDDR5_CMD
	FB_B0_CMD	GDDR5_45R50SE	GDDR5_CMD
	FB_B1_CMD	GDDR5_45R50SE	GDDR5_CMD
	FB_B0_CMD	GDDR5_45R50SE	GDDR5_CMD
	FB_B1_CMD	GDDR5_45R50SE	GDDR5_CMD
	FB_B0_CMD	GDDR5_45R50SE	GDDR5_CMD
	FB_B1_CMD	GDDR5_45R50SE	GDDR5_CMD
	FB_B0_CMD	GDDR5_45R50SE	GDDR5_CMD
	FB_B1_CMD	GDDR5_45R50SE	GDDR5_CMD
	FB_B0_CMD	GDDR5_45R50SE	GDDR5_CMD
	FB_B1_CMD	GDDR5_45R50SE	GDDR5_CMD
	FB_B0_CMD	GDDR5_45R50SE	GDDR5_CMD
	FB_B1_CMD	GDDR5_45R50SE	GDDR5_CMD
	FB_B0_CMD	GDDR5_45R50SE	GDDR5_CMD
	FB_B1_CMD	GDDR5_45R50SE	GDDR5_CMD
	FB_B0_CMD	GDDR5_45R50SE	GDDR5_CMD
	FB_B1_CMD	GDDR5_45R50SE	GDDR5_CMD
	FB_B0_CMD	GDDR5_45R50SE	GDDR5_CMD
	FB_B1_CMD	GDDR5_45R50SE	GDDR5_CMD
	FB_B0_CMD	GDDR5_45R50SE	GDDR5_CMD
	FB_B1_CMD	GDDR5_45R50SE	GDDR5_CMD
	FB_B0_CMD	GDDR5_45R50SE	GDDR5_CMD
	FB_B1_CMD	GDDR5_45R50SE	GDDR5_CMD
	FB_B0_CMD	GDDR5_45R50SE	GDDR5_CMD
	FB_B1_CMD	GDDR5_45R50SE	GDDR5_CMD
	FB_B0_CMD	GDDR5_45R50SE	GDDR5_CMD
	FB_B1_CMD	GDDR5_45R50SE	GDDR5_CMD
	FB_B0_CMD	GDDR5_45R50SE	GDDR5_CMD
	FB_B1_CMD	GDDR5_45R50SE	GDDR5_CMD
	FB_B0_CMD	GDDR5_45R50SE	GDDR5_CMD
	FB_B1_CMD	GDDR5_45R50SE	GDDR5_CMD
	FB_B0_CMD	GDDR5_45R50SE	GDDR5_CMD
	FB_B1_CMD	GDDR5_45R50SE	GDDR5_CMD
	FB_B0_CMD	GDDR5_45R50SE	GDDR5_CMD
	FB_B1_CMD	GDDR5_45R50SE	GDDR5_CMD
	FB_B0_CMD	GDDR5_45R50SE	GDDR5_CMD
	FB_B1_CMD	GDDR5_45R50SE	GDDR5_CMD
	FB_B0_CMD	GDDR5_45R50SE	GDDR5_CMD
	FB_B1_CMD	GDDR5_45R50SE	GDDR5_CMD
	FB_B0_CMD	GDDR5_45R50SE	GDDR5_CMD
	FB_B1_CMD	GDDR5_45R50SE	GDDR5_CMD
	FB_B0_CMD	GDDR5_45R50SE	GDDR5_CMD
	FB_B1_CMD	GDDR5_45R50SE	GDDR5_CMD
	FB_B0_CMD	GDDR5_45R50SE	GDDR5_CMD
	FB_B1_CMD	GDDR5_45R50SE	GDDR5_CMD
	FB_B0_CMD	GDDR5_45R50SE	GDDR5_CMD
	FB_B1_CMD	GDDR5_45R50SE	GDDR5_CMD
	FB_B0_CMD	GDDR5_45R50SE	GDDR5_CMD
	FB_B1_CMD	GDDR5_45R50SE	GDDR5_CMD
	FB_B0_CMD	GDDR5_45R50SE	GDDR5_CMD
	FB_B1_CMD	GDDR5_45R50SE	GDDR5_CMD
	FB_B0_CMD	GDDR5_45R50SE	GDDR5_CMD
	FB_B1_CMD	GDDR5_45R50SE	GDDR5_CMD
	FB_B0_CMD	GDDR5_45R50SE	GDDR5_CMD
	FB_B1_CMD	GDDR5_45R50SE	GDDR5_CMD
	FB_B0_CMD	GDDR5_45R50SE	GDDR5_CMD
	FB_B1_CMD	GDDR5_45R50SE	GDDR5_CMD
	FB_B0_CMD	GDDR5_45R50SE	GDDR5_CMD
	FB_B1_CMD	GDDR5_45R50SE	GDDR5_CMD
	FB_B0_CMD	GDDR5_45R50SE	GDDR5_CMD
	FB_B1_CMD	GDDR5_45R50SE	GDDR5_CMD
	FB_B0_CMD	GDDR5_45R50SE	GDDR5_CMD
	FB_B1_CMD	GDDR5_45R50SE	GDDR5_CMD
	FB_B0_CMD	GDDR5_45R50SE	GDDR5_CMD
	FB_B1_CMD	GDDR5_45R50SE	GDDR5_CMD
	FB_B0_CMD	GDDR5_45R50SE	GDDR5_CMD
	FB_B1_CMD	GDDR5_45R50SE	GDDR5_CMD
	FB_B0_CMD	GDDR5_45R50SE	GDDR5_CMD
	FB_B1_CMD	GDDR5_45R50SE	GDDR5_CMD
	FB_B0_CMD	GDDR5_45R50SE	GDDR5_CMD
	FB_B1_CMD	GDDR5_45R50SE	GDDR5_CMD
	FB_B0_CMD	GDDR5_45R50SE	GDDR5_CMD
	FB_B1_CMD	GDDR5_45R50SE	GDDR5_CMD
	FB_B0_CMD	GDDR5_45R50SE	GDDR5_CMD
	FB_B1_CMD	GDDR5_45R50SE	GDDR5_CMD
	FB_B0_CMD	GDDR5_45R50SE	GDDR5_CMD
	FB_B1_CMD	GDDR5_45R50SE	GDDR5_CMD
	FB_B0_CMD	GDDR5_45R50SE	GDDR5_CMD
	FB_B1_CMD	GDDR5_45R50SE	GDDR5_CMD
	FB_B0_CMD	GDDR5_45R50SE	GDDR5_CMD
	FB_B1_CMD	GDDR5_45R50SE	GDDR5_CMD
	FB_B0_CMD	GDDR5_45R50SE	GDDR5_CMD
	FB_B1_CMD	GDDR5_45R50SE	GDDR5_CMD
	FB_B0_CMD	GDDR5_45R50SE	GDDR5_CMD
	FB_B1_CMD	GDDR5_45R50SE	GDDR5_CMD
	FB_B0_CMD	GDDR5_45R50SE	GDDR5_CMD
	FB_B1_CMD	GDDR5_45R50SE	GDDR5_CMD
	FB_B0_CMD	GDDR5_45R50SE	GDDR5_CMD
	FB_B1_CMD	GDDR5_45R50SE	GDDR5_CMD
	FB_B0_CMD	GDDR5_45R50SE	GDDR5_CMD
	FB_B1_CMD	GDDR5_45R50SE	GDDR5_CMD
	FB_B0_CMD	GDDR5_45R50SE	GDDR5_CMD
	FB_B1_CMD	GDDR5_45R50SE	GDDR5_CMD
	FB_B0_CMD	GDDR5_45R50SE	GDDR5_CMD
	FB_B1_CMD	GDDR5_45R50SE	GDDR5_CMD
	FB_B0_CMD	GDDR5_45R50SE	GDDR5_CMD
	FB_B1_CMD	GDDR5_45R50SE	GDDR5_CMD
	FB_B0_CMD	GDDR5_45R50SE	GDDR5_CMD
	FB_B1_CMD	GDDR5_45R50SE	GDDR5_CMD
	FB_B0_CMD	GDDR5_45R50SE	GDDR5_CMD
	FB_B1_CMD	GDDR5_45R50SE	GDDR5_CMD
	FB_B0_CMD	GDDR5_45R50SE	GDDR5_CMD
	FB_B1_CMD	GDDR5_45R50SE	GDDR5_CMD
	FB_B0_CMD	GDDR5_45R50SE	GDDR5_CMD
	FB_B1_CMD	GDDR5_45R50SE	GDDR5_CMD
	FB_B0_CMD	GDDR5_45R50SE	GDDR5_CMD
	FB_B1_CMD	GDDR5_45R50SE	GDDR5_CMD
	FB_B0_CMD	GDDR5_45R50SE	GDDR5_CMD
	FB_B1_CMD	GDDR5_45R50SE	GDDR5_CMD
	FB_B0_CMD	GDDR5_45R50SE	GDDR5_CMD
	FB_B1_CMD	GDDR5_45R50SE	GDDR5_CMD
	FB_B0_CMD	GDDR5_45R50SE	GDDR5_CMD
	FB_B1_CMD	GDDR5_45R50SE	GDDR5_CMD
	FB_B0_CMD	GDDR5_45R50SE	GDDR5_CMD
	FB_B1_CMD	GDDR5_45R50SE	GDDR5_CMD
	FB_B0_CMD	GDDR5_45R50SE	GDDR5_CMD
	FB_B1_CMD	GDDR5_45R50SE	GDDR5_CMD
	FB_B0_CMD	GDDR5_45R50SE	GDDR5_CMD
	FB_B1_CMD	GDDR5_45R50SE	GDDR5_CMD
	FB_B0_CMD	GDDR5_45R50SE	GDDR5_CMD
	FB_B1_CMD	GDDR5_45R50SE	GDDR5_CMD
	FB_B0_CMD	GDDR5_45R50SE	GDDR5_CMD
	FB_B1_CMD	GDDR5_45R50SE	GDDR5_CMD
	FB_B0_CMD	GDDR5_45R50SE	GDDR5_CMD
	FB_B1_CMD	GDDR5_45R50SE	GDDR5_CMD
	FB_B0_CMD	GDDR5_45R50SE	GDDR5_CMD
	FB_B1_CMD	GDDR5_45R50SE	GDDR5_CMD
	FB_B0_CMD	GDDR5_45R50SE	GDDR5_CMD
	FB_B1_CMD	GDDR5_45R50SE	GDDR5_CMD
	FB_B0_CMD	GDDR5_45R50SE	GDDR5_CMD
	FB_B1_CMD	GDDR5_45R50SE	GDDR5_CMD
	FB_B0_CMD	GDDR5_45R50SE	GDDR5_CMD
	FB_B1_CMD	GDDR5_45R50SE	GDDR5_CMD
	FB_B0_CMD	GDDR5_45R50SE	GDDR5_CMD
	FB_B1_CMD	GDDR5_45R50SE	GDDR5_CMD
	FB_B0_CMD	GDDR5_45R50SE	GDDR5_CMD
	FB_B1_CMD	GDDR5_45R50SE	GDDR5_CMD
	FB_B0_CMD	GDDR5_45R50SE	GDDR5_CMD
	FB_B1_CMD	GDDR5_45R50SE	GDDR5_CMD
	FB_B0_CMD	GDDR5_45R50SE	GDDR5_CMD
	FB_B1_CMD	GDDR5_45R50SE	GDDR5_CMD
	FB_B0_CMD	GDDR5_45R50SE	GDDR5_CMD
	FB_B1_CMD	GDDR5_45R50SE	GDDR5_CMD
	FB_B0_CMD	GDDR5_45R50SE	GDDR5_CMD
	FB_B1_CMD	GDDR5_45R50SE	GDDR5_CMD
	FB_B0_CMD	GDDR5_45R50SE	GDDR5_CMD
	FB_B1_CMD	GDDR5_45R50SE	GDDR5_CMD

MUXGFX Net Properties

ELECTRICAL_CONSTRAINT_SET		NET_TYPE		
		PHYSICAL	SPACING	
□	LVDS_A_CLK	LVDS_850	LVDS	LVDS A CLK P 84 88
□	LVDS_A_CLK	LVDS_850	LVDS	LVDS A CLK N 84 88
□	LVDS_A_DATA	LVDS_850	LVDS	LVDS A DATA P<2..0> 84 88
□	LVDS_A_DATA	LVDS_850	LVDS	LVDS A DATA N<2..0> 84 88
□	LVDS_B_CLK	LVDS_850	LVDS	LVDS B CLK P 84 88
□	LVDS_B_CLK	LVDS_850	LVDS	LVDS B CLK N 84 88
□	LVDS_B_DATA	LVDS_850	LVDS	LVDS B DATA P<2..0> 84 88
□	LVDS_B_DATA	LVDS_850	LVDS	LVDS B DATA N<2..0> 84 88
□		LVDS_850	LVDS	LVDS CONN A CLK F P 6 83
□		LVDS_850	LVDS	LVDS CONN A CLK F N 6 83
□		LVDS_850	LVDS	LVDS CONN B CLK F P 6 83
□		LVDS_850	LVDS	LVDS CONN B CLK F N 6 83
□		LVDS_850	LVDS	LVDS CONN A CLK P 83 84
□		LVDS_850	LVDS	LVDS CONN A CLK N 83 84
□		LVDS_850	LVDS	LVDS CONN A DATA P<2..0> 6 83 84
□		LVDS_850	LVDS	LVDS CONN A DATA N<2..0> 6 83 84
□		LVDS_850	LVDS	LVDS CONN B CLK P 83 84
□		LVDS_850	LVDS	LVDS CONN B CLK N 83 84
□		LVDS_850	LVDS	LVDS CONN B DATA P<2..0> 6 83 84
□		LVDS_850	LVDS	LVDS CONN B DATA N<2..0> 6 83 84

Whistler Net Properties

ELECTRICAL_CONSTRAINT_SET		NET_TYPE		
		PHYSICAL	SPACING	
	GPU_CLK27M	CLK_SLOW_55S	CLK_SLOW	GPU_CLK27M 79 80
	GPU_CLK100M	CLK_SLOW_55S	CLK_SLOW	GPU_CLK100M 79 80
	LVDS_EG_A_CLK	LVDS_85D	LVDS	LVDS_EG_A_CLK_P 79 88
	LVDS_EG_A_CLK	LVDS_85D	LVDS	LVDS_EG_A_CLK_N 79 88
	LVDS_EG_A_DATA	LVDS_85D	LVDS	LVDS_EG_A_DATA P<2...0> 79 88
	LVDS_EG_A_DATA	LVDS_85D	LVDS	LVDS_EG_A_DATA N<2...0> 79 88
	LVDS_EG_A_DATA3	LVDS_85D	LVDS	NC LVDS_EG_A_DATA P<3> 79 80
	LVDS_EG_A_DATA3	LVDS_85D	LVDS	NC LVDS_EG_A_DATA N<3> 79 80
	LVDS_EG_B_DATA	LVDS_85D	LVDS	LVDS_EG_B_DATA P<2...0> 79 88
	LVDS_EG_B_DATA	LVDS_85D	LVDS	LVDS_EG_B_DATA N<2...0> 79 88
	LVDS_EG_B_DATA3	LVDS_85D	LVDS	NC LVDS_EG_B_DATA P<3> 79 80
	LVDS_EG_B_DATA3	LVDS_85D	LVDS	NC LVDS_EG_B_DATA N<3> 79 80
	DP_MI	DP_85D	DISPLAYPORT	DP_EXT_A_ML_C_P<3...0> 79 80
		DP_85D	DISPLAYPORT	DP_EXT_A_ML_C_N<3...0> 79 80
	DP_AUX_CH	DP_85D	DISPLAYPORT	DP_EXT_AUXCH_C_P 84 85
		DP_85D	DISPLAYPORT	DP_EXT_AUXCH_C_N 84 85
	DP_AUX_CH	DP_85D	DISPLAYPORT	DP_EG_AUX_CH_P 79 84
		DP_85D	DISPLAYPORT	DP_EG_AUX_CH_N 79 84

K92 Board-Specific Spacing & Physical Constraints

BOARD LAYERS	BOARD AREAS	BOARD UNITS (MIL or MM)	ALLEGRO VERSION
TOP, ISL2, ISL3, ISL4, ISL5, ISL6, ISL7, ISL8, ISL9, ISL10, ISL11, BOTTOM	NO_TYPE, BGA	MM	15.5.1

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
DEFAULT	*	Y	=50_OHM_SE	=50_OHM_SE	10 MM	0 MM	0 MM
STANDARD	*	Y	=DEFAULT	=DEFAULT	10 MM	=DEFAULT	=DEFAULT

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
27P4_OHM_SE	TOP,BOTTOM	Y	0.310 MM	0.095 MM			
27P4_OHM_SE	*	Y	0.250 MM	0.1 MM	=STANDARD	=STANDARD	=STANDARD

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
37_0HM_SE	TOP,BOTTOM	Y	0.185 MM	0.095 MM			
37_0HM_SE	*	Y	0.155 MM	0.090 MM	=STANDARD	=STANDARD	=STANDARD

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
40_OHM_SE	TOP,BOTTOM	Y	0.165 MM	0.095 MM			
40_OHM_SE	*	Y	0.135 MM	0.090 MM	=STANDARD	=STANDARD	=STANDARD

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
45_OHM_SE	TOP,BOTTOM	Y	0.13 MM	0.13 MM			
45_OHM_SE	*	Y	0.099 MM	0.099 MM	=STANDARD	=STANDARD	=STANDARD

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
50_OHM_SE	TOP,BOTTOM	Y	0.110 MM	0.095 MM			
50_OHM_SE	*	Y	0.090 MM	0.090 MM	=STANDARD	=STANDARD	=STANDARD

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
55_OHM_SE	TOP,BOTTOM	Y	0.090 MM	0.090 MM			
55_OHM_SE	*	Y	0.076 MM	0.076 MM	=STANDARD	=STANDARD	=STANDARD

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
72_OHM_DIFF	*	N	=STANDARD	=STANDARD	=STANDARD	=STANDARD	=STANDARD
72_OHM_DIFF	ISL3, ISL4	Y	0.154 MM	0.154 MM		0.200 MM	0.200 MM
72_OHM_DIFF	ISL0, ISL10	Y	0.154 MM	0.154 MM		0.200 MM	0.200 MM
72_OHM_DIFF	ISL2, ISL11	Y	0.154 MM	0.154 MM		0.200 MM	0.200 MM
72_OHM_DIFF	TOP, BOTTOM	Y	0.175 MM	0.175 MM		0.200 MM	0.200 MM

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
80_OHM_DIFF	*	N	=STANDARD	=STANDARD	=STANDARD	=STANDARD	=STANDARD
80_OHM_DIFF	ISL3, ISL4, ISL9, ISL10	Y	0.105 MM	0.105 MM		0.120 MM	0.120 MM
80_OHM_DIFF	ISL2, ISL11	Y	0.105 MM	0.105 MM		0.120 MM	0.120 MM
80_OHM_DIFF	TOP, BOTTOM	Y	0.135 MM	0.135 MM		0.160 MM	0.160 MM

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
85_OHM_DIFF	*	N	=STANDARD	=STANDARD	=STANDARD	=STANDARD	=STANDARD
85_OHM_DIFF	ISL3, ISL4	Y	0.110 MM	0.090 MM		0.180 MM	0.180 MM
85_OHM_DIFF	ISL9, ISL10	Y	0.110 MM	0.090 MM		0.180 MM	0.180 MM
85_OHM_DIFF	ISL2, ISL11	Y	0.110 MM	0.090 MM		0.180 MM	0.180 MM
85_OHM_DIFF	TOP, BOTTOM	Y	0.125 MM	0.090 MM		0.190 MM	0.190 MM

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
90_OHM_DIFF	*	N	=STANDARD	=STANDARD	=STANDARD	=STANDARD	=STANDARD
90_OHM_DIFF	ISL3, ISL4	Y	0.102 MM	0.090 MM		0.220 MM	0.220 MM
90_OHM_DIFF	ISL9, ISL10	Y	0.102 MM	0.090 MM		0.220 MM	0.220 MM
90_OHM_DIFF	ISL2, ISL11	Y	0.102 MM	0.090 MM		0.220 MM	0.220 MM
90_OHM_DIFF	TOP, BOTTOM	Y	0.115 MM	0.090 MM		0.230 MM	0.230 MM

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
100_OHM_DIFF	*	N	=STANDARD	=STANDARD	=STANDARD	=STANDARD=STANDARD	
100_OHM_DIFF	ISL3, ISL4	Y	0.080 MM	0.080 MM		0.200 MM	0.200 MM
100_OHM_DIFF	ISL9, ISL10	Y	0.080 MM	0.080 MM		0.200 MM	0.200 MM
100_OHM_DIFF	ISL2, ISL11	Y	0.080 MM	0.080 MM		0.200 MM	0.200 MM
100_OHM_DIFF	TOP, BOTTOM	Y	0.089 MM	0.089 MM		0.220 MM	0.220 MM

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
110_OHM_DIFF	*	N	=STANDARD	=STANDARD	=STANDARD	=STANDARD	=STANDARD
110_OHM_DIFF	ISL3, ISL4	Y	0.065 MM	0.065 MM		0.200 MM	0.200 MM
110_OHM_DIFF	ISL9, ISL10	Y	0.065 MM	0.065 MM		0.200 MM	0.200 MM
110_OHM_DIFF	ISL2, ISL11	Y	0.065 MM	0.065 MM		0.200 MM	0.200 MM
110_OHM_DIFF	TOP, BOTTOM	Y	0.075 MM	0.075 MM		0.330 MM	0.330 MM

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
DEFAULT	*	0.1 MM	?
STANDARD	*	=DEFAULT	?
BGA_P1MM	*	=DEFAULT	?
BGA_P2MM	*	=DEFAULT	?
P072_SPACE	*	0.071 MM	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
1:5:1_SPACING	*	0.15 MM	?
2:1_SPACING	*	0.2 MM	?
2.5:1_SPACING	*	0.25 MM	?
3:1_SPACING	*	0.3 MM	?
4:1_SPACING	*	0.4 MM	?
5:1_SPACING	*	0.5 MM	?

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
1:1_DIFFPAIR	*	Y	=STANDARD	=STANDARD	=STANDARD	0.1 MM	0.1 MM

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
100_DIFF_BGA	*	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF
100_DIFF_BGA	ISL3, ISL4	Y	0.075 MM	0.075 MM		0.125 MM	0.125 MM
100_DIFF_BGA	ISL9, ISL10	Y	0.075 MM	0.075 MM		0.125 MM	0.125 MM

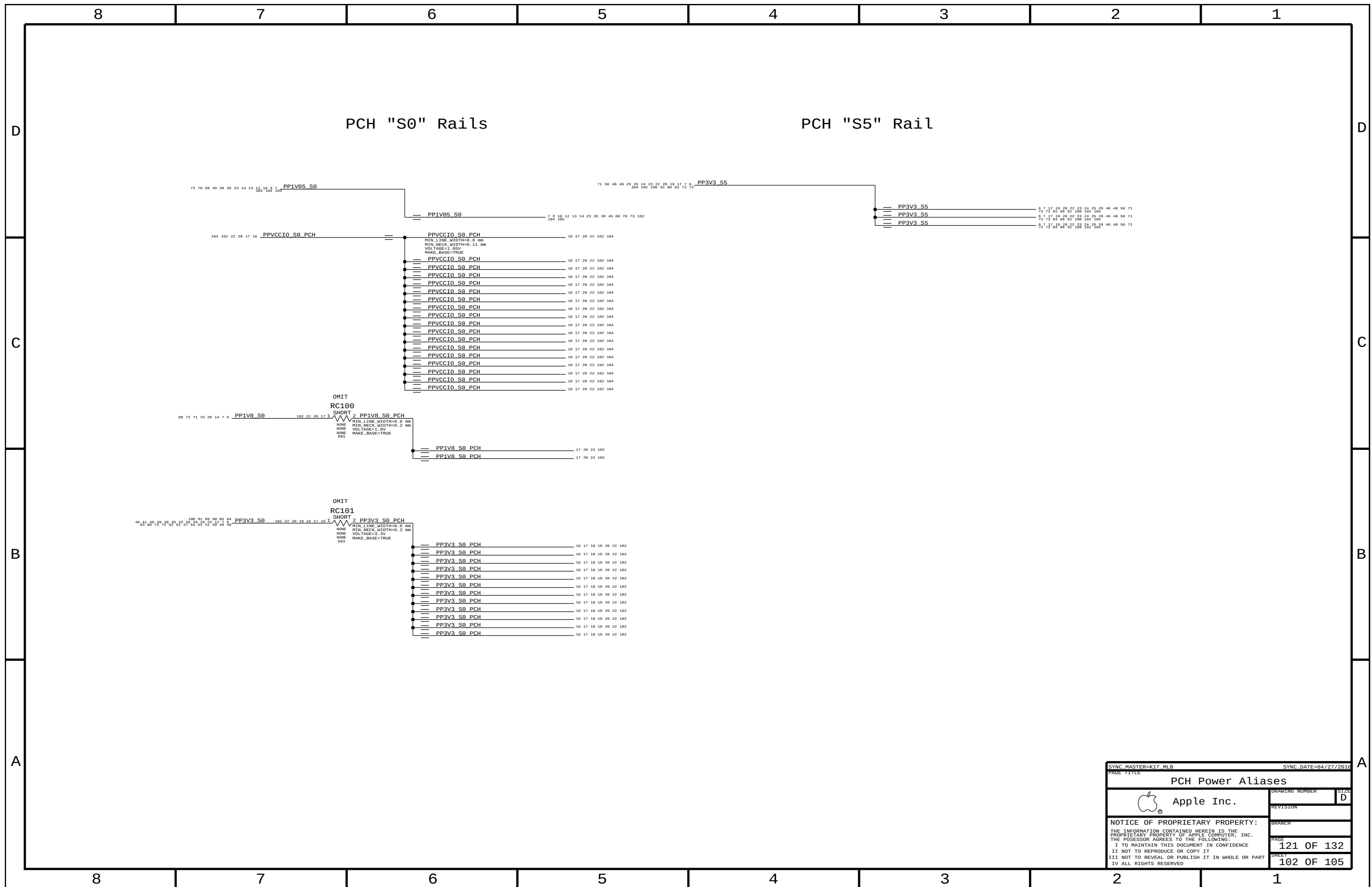
NOTE: 100_DIFF_BGA is 100-ohms differential impedance on outer layers and 95-ohms on inner layers.

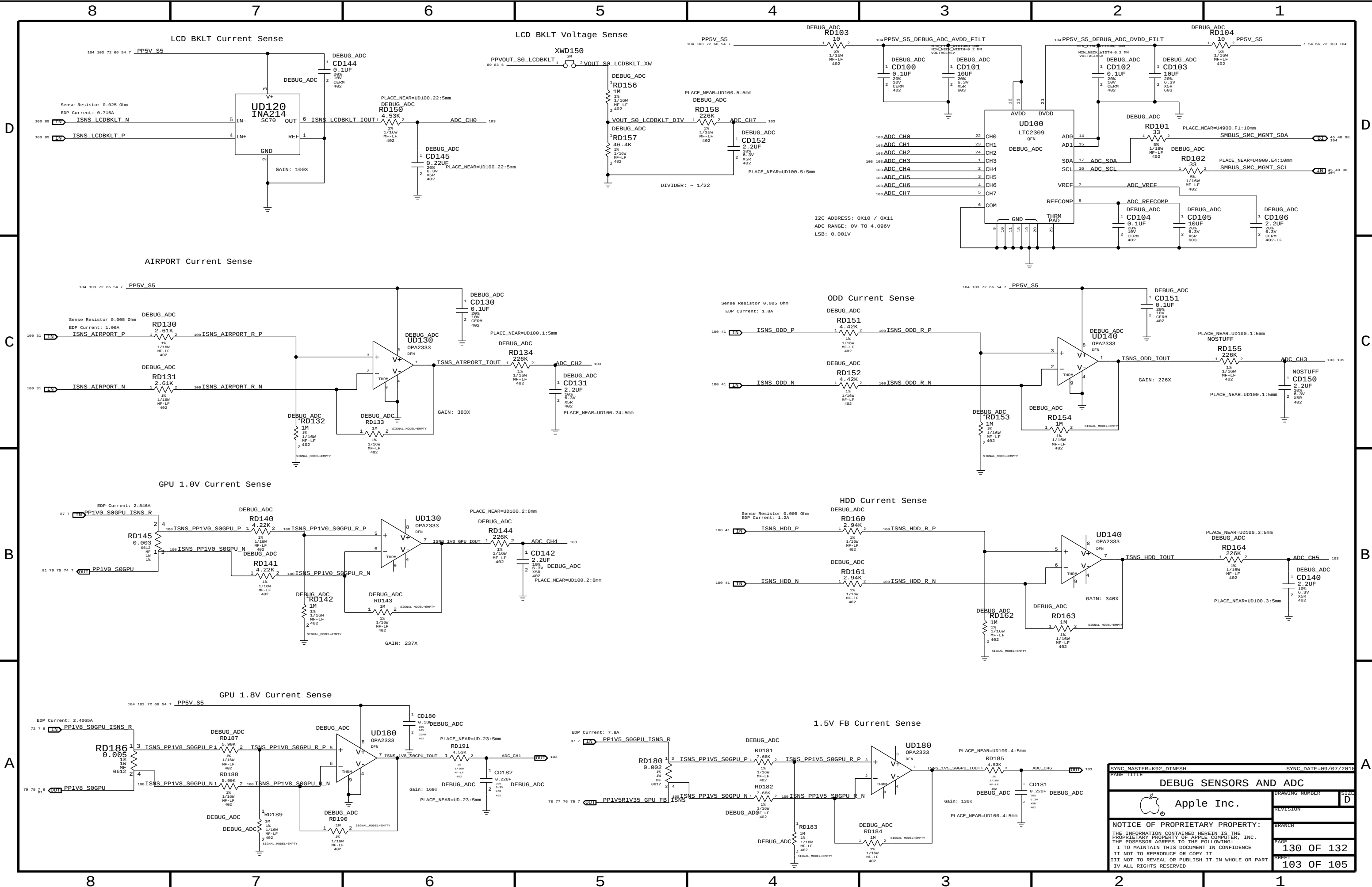
NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
*	*	BGA	P072_SPACE

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
2X_DIELECTRIC	*	0.140 MM	?
3X_DIELECTRIC	*	0.210 MM	?
4X_DIELECTRIC	*	0.280 MM	?
5X_DIELECTRIC	*	0.350 MM	?
7X_DIELECTRIC	*	0.490 MM	?

NOTE: Based on K92 mlb stackup.

SYNC MASTER=K17 MLB	SYNC DATE=05/14/2016																
PAGE TITLE																	
PCB Rule Definitions																	
 Apple Inc.	<table border="1" style="width: 100%; border-collapse: collapse;"> <tr> <td style="width: 80%; padding: 5px;">DRAWING NUMBER</td> <td style="width: 20%; padding: 5px; text-align: center;">SIZE</td> </tr> <tr> <td style="padding: 5px;"></td> <td style="text-align: center; padding: 5px;">D</td> </tr> <tr> <td colspan="2" style="padding: 5px;">REVISION</td> </tr> <tr> <td colspan="2" style="padding: 5px;">BRANCH</td> </tr> <tr> <td colspan="2" style="padding: 5px;">PAGE</td> </tr> <tr> <td colspan="2" style="text-align: center; padding: 5px;">109 OF 132</td> </tr> <tr> <td colspan="2" style="padding: 5px;">SHEET</td> </tr> <tr> <td colspan="2" style="text-align: center; padding: 5px;">101 OF 105</td> </tr> </table>	DRAWING NUMBER	SIZE		D	REVISION		BRANCH		PAGE		109 OF 132		SHEET		101 OF 105	
DRAWING NUMBER	SIZE																
	D																
REVISION																	
BRANCH																	
PAGE																	
109 OF 132																	
SHEET																	
101 OF 105																	
NOTICE OF PROPRIETARY PROPERTY: THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE COMPUTER, INC. THE POSSESSOR AGREES TO THE FOLLOWING: I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE II NOT TO REPRODUCE OR COPY IT III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART IV ALL RIGHTS RESERVED																	





SYNC MASTER=K92 DINESH		SYNC DATE=89/07/2810	
PAGE TITLE		DRAWING NUMBER	
DEBUG SENSORS AND ADC		D	
Apple Inc.		REVISION	
NOTICE OF PROPRIETARY PROPERTY:		BRANCH	
THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE COMPUTER, INC. THE POSSESSOR AGREES TO THE FOLLOWING:		PAGE	
I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE		130 OF 132	
II NOT TO REPRODUCE OR COPY IT		SHEET	
III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART		103 OF 105	
IV ALL RIGHTS RESERVED			

